Engineering Mechanics 1 · Unit_04 · UNIT 4

# DRAFT — UE26ME142A · Unit_04 — UNIT 4 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents


1. 1 · UE26ME142A · Unit_04: UNIT 4
2. 2 · CLASS 46 Introduction to Moment of Inertia
3. 3 · CLASS 38 Moment of Inertia_ Numerical
4. 4 · CLASS 39 Moment of Inertia_ Numerical
5. 5 · CLASS 40 Moment of Inertia_ Numerical
6. 6 · CLASS 41 Moment of Inertia_ Numerical
7. 7 · CLASS 51.1 Moment of Inertia_ Numerical
8. 8 · CLASS 51.2 Moment of Inertia_ Numerical
9. 9 · CLASS 51.2 Moment of Inertia_ Numerical
10. 10 · CLASS 47 Friction_ Numerical
11. 11 · CLASS 48 Friction_ Numerical
12. 12 · CLASS 56.1 Friction_ Numerical
13. 13 · CLASS 56.2 Friction_ Numerical
14. 14 · Source slide deck: Moment of Inertia
15. 15 · Source slide deck: UE20CV101 UNIT 4 SESSION 6
16. 16 · Source slide deck: UE20CV101 UNIT 4 SESSION 11
17. 17 · Source slide deck: UE20CV101 UNIT 4 SESSION 12
18. 18 · Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT
19. 19 · Source slide deck: UE20CV101 UNIT 5 SESSION 2
20. 20 · Source slide deck: UE20CV101 UNIT 5 SESSION 3
21. 21 · Source slide deck: UE20CV101 UNIT 5 SESSION 6
22. Appendix A · Symbol glossary
23. Appendix B · Sources, coverage and gaps
24. Appendix C · Revision and retrieval practice

M1

# Unit_04 · UNIT 4

## 1 UE26ME142A · Unit_04: UNIT 4

This unit master organizes source material for Engineering Mechanics 1, Unit_04, officially titled UNIT 4. It includes 12 video transcripts and 15 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26ME142A, Unit_04 (UNIT 4). Bucket folder present: yes. Videos: 12; slide decks: 15.

EV0000001

## 2 CLASS 46 Introduction to Moment of Inertia

This lesson is indexed from the source video titled CLASS 46 Introduction to Moment of Inertia. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

## Graphs and diagrams

**Figure G000001.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 1.

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# Engineering Mechanics

# Statics

**EIGHTH Edition**
**SI Version**

J. L. Meriam and L. G. Kraige

## Unit 4
## Moment of Inertia & Friction
## Appendix A
Moment of inertia

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**Figure G000002.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

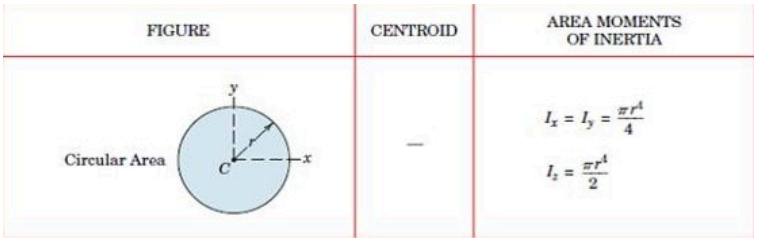




**Figure G000003.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 3.

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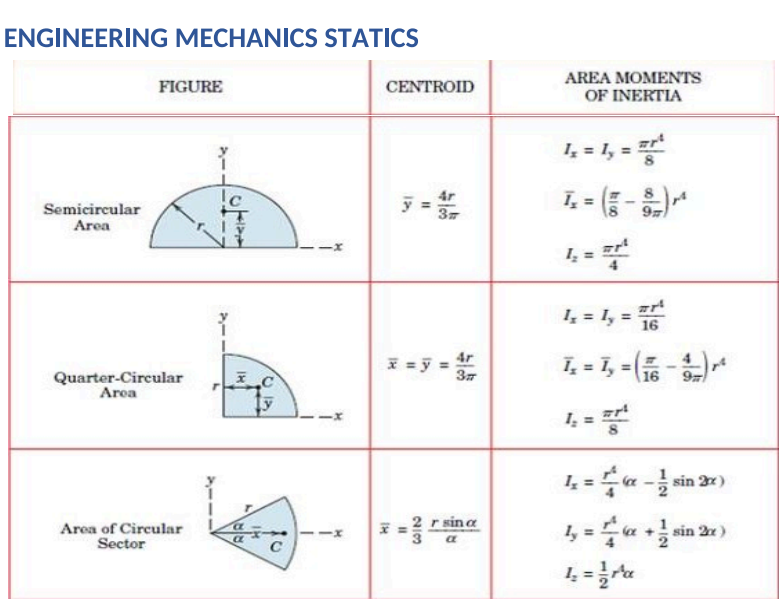


**Figure G000004.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 4.

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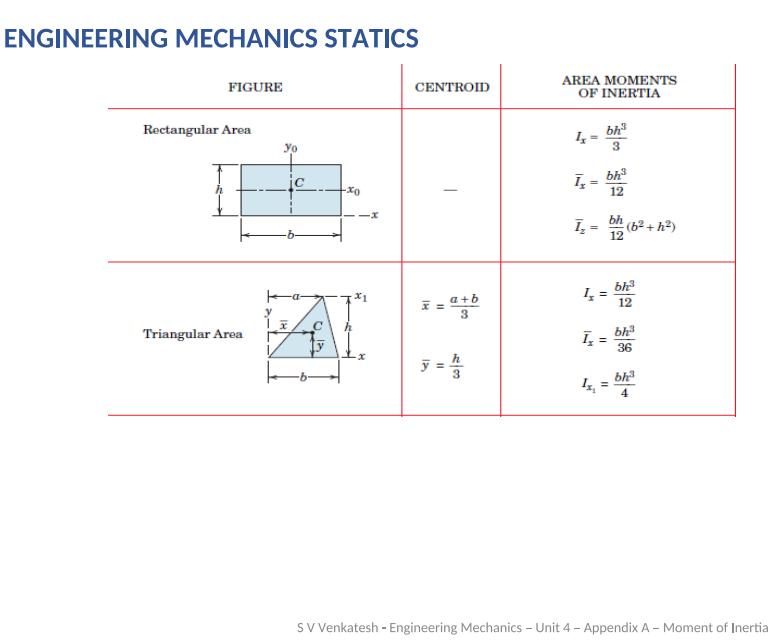


**Figure G000005.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 5.

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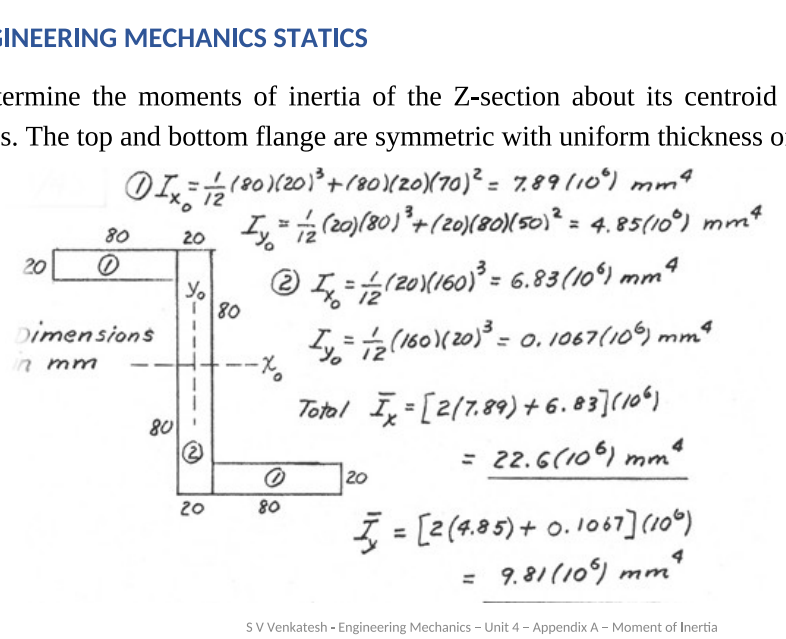


**Figure G000006.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 6.

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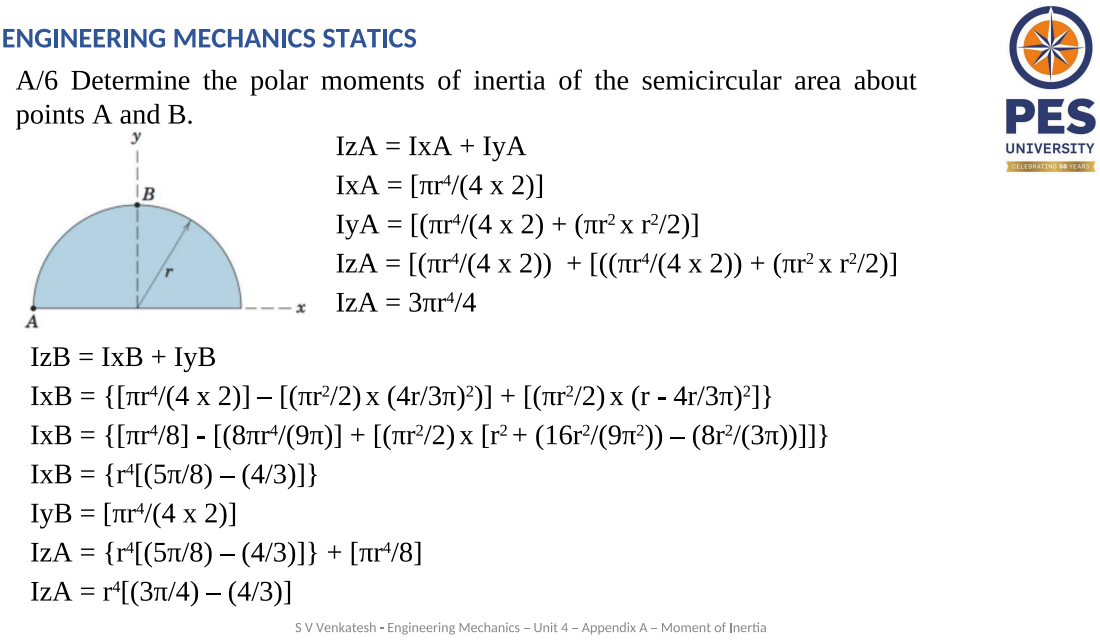


**Figure G000007.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 7.

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THANK YOU

**Dr. S V Venkatesh**
Civil Engineering
Department of S & H

**Figure G000008.** UE26ME142A Unit 4 Class 46 Lecture PPT · source slide page 8.

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### Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000190 · EV0000191 · EV0000192
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 4 Moment of Inertia & Friction Appendix A Moment of inertia Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia EV0000193 · EV0000194 · EV0000195
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia EV0000196 · EV0000197 · EV0000198
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia EV0000199 · EV0000200 · EV0000201
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia EV0000202 · EV0000203 · EV0000204
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS Determine the moments of inertia of the Z-section about its centroid x0- and y0-axes. The top and bottom flange are symmetric with uniform thickness of 20 mm. S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia EV0000205 · EV0000206 · EV0000207

7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS A/6 Determine the polar moments of inertia of the semicircular area about points A and B. IzA = IxA + IyA IxA = [πr4/(4 x 2)] IyA = [(πr4/(4 x 2) + (πr2 x r2/2)] IzA = [(πr4/(4 x 2)) + [((πr4/(4 x 2)) + (πr2 x r2/2)] IzA = 3πr4/4 IzB = IxB + IyB IxB = {[πr4/(4 x 2)] – [(πr2/2) x (4r/3π)2)] + [(πr2/2) x (r - 4r/3π)2]} IxB = {[πr4/8] - [(8πr4/(9π)] + [(πr2/2) x [r2 + (16r2/(9π2)) – (8r2/(3π))]]} IxB = {r4[(5π/8) – (4/3)]} IyB = [πr4/(4 x 2)] S V Venkatesh - Engineering Mechanics – Unit 4 – Appendix A – Moment of Inertia IzA = {r4[(5π/8) – (4/3)]} + [πr4/8] ## Page / slide 8 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000208
8. Source slide page 8: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000209

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.740] Hello, in today's session, let's discuss about the topic Area Moment of Inertia. Moment of inertia is an object's resistance to bending at the rotation, and it is also called as second moment of an area. [00:00:17.808] The concept of first moment of an area we discussed in locating the centroid of a lamina. So moment of inertia here is [00:00:27.024] It's nothing but second moment of an area. That's why moment of inertia I will be equal to distance square d square into area A. And it is also. [00:00:40.516] A purely a mathematical property of an area and itself doesn't have any physical significance. Now let's discuss about rectangular and polar movement of inertia. [00:00:54.400] Rectangular moment of inertia is nothing but moment of inertia phenamina about any two mutually perpendicular axes. Example like X and Y axis. Because X and Y we know they'll be always mutually perpendicular to each other. So moment of [00:01:12.112] inertia about x and y axis we call them as rectangular moment of inertia. Then moment of inertia about an axis which is perpendicular to the plane of an area or a lamina that we call it as a [00:01:26.112] Polar moment of inertia. As we know, z axis is always perpendicular to xy plane or a plane of a lamina. We consider moment of inertia about z axis as the polar moment of inertia. [00:01:42.032] To explain this, I am going to consider this figure. You can see here. Let us consider this is a lamina of total area capital A. It is in the XY plane. Now I want to find the moment of inertia of this lamina. [00:01:59.680] about the x axis then about the y axis. For that I am going to consider one smaller elemental area within this lamina. Let the area of this element let it be d. Now moment of inertia of this element about x and y axis [00:02:17.328] I want to calculate. So I am going to find the coordinates of the centroid of the smaller area. Since area is very very small, the centroid of this area will be the coordinates of the centroid are going to be x, y. [00:02:32.560] Now moment of inertia of this dA about x axis will be equal to we know moment of inertia is distance squaring area, so it is going to be [00:02:44.996] Distance is y, so that we know. So y square into area is dm. This is moment of inusia. Since we are talking about the smaller element, it [00:02:56.900] We go, we are going to take it as DIX. This is this stands for moment of inertia of the elemental area about x-axis. Similarly, moment of inertia of this elemental area about y-axis will be equal to. [00:03:11.344] Again it is distance square into area. Distance here is x. So it is x squaring to area is d and d iy stands for inertia of the smaller element above the [00:03:24.016] Y axis. Once we know moment of inertia of the element about x and about y axis. Now once we integrate these two, we are going to get inertia of the entire lamina about the x and about the y axis. That is ix is equals to integration. [00:03:41.648] integral of y

square into d a and iy is integral of x square into d a. i x stands for inertia of the lamina about x axis iy stands for inertia of the lamina about y axis since x and y are perpendicular to each other i x and i y [00:03:59.216] Why we call them as rectangular moment of inertia. [00:04:03.268] Next, let's get into polar moment of inertia. As I said before, moment of inertia about z axis we call them as polar moment of inertia. Now consider there is a z axis, passing to point one. Since there is perpendicular [00:04:19.936] to the plane of the lamina. Nothing but perpendicular to XY plane, we can't see that. [00:04:24.960] Just remember that is passing through point O. Then how to find inertia about the Z axis passing through O? We know moment of inertia is distance square into area. So here distance is R, so it is going to be R square into area is [00:04:41.328] D. This is the moment of the smaller element about the z axis. Next, I want to find the moment of inertia of this entire lamina about [00:04:52.144] Z-axis, so just differentiate integrate that. So iz will be equals to integral of R square into D. This is called as polar moment of inertia. Next.

EV0000002 · EV0000003 · EV0000004 · EV0000005

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.456] There is a relationship between polar moment of inertia and rectangular moment of inertia according to perpendicular axis theorem. According to perpendicular axis theorem, moment of inertia about a z-axis, which is perpendicular to the plane of the lamina, will be always equals to [00:05:21.072] Sum of the moment of inertia about the two mutually perpendicular axes passing through the same point O and lying in the same plane of the lamina. Nothing but IZ is equals to Ix plus I. The condition is [00:05:36.480] X, Y, Z should be passing to the same point. To prove that, I'm going to consider here. [00:05:44.464] If you look at this, there is a right angle triangle. Now I can write r square as x square plus y square. So r square is equals to x square plus y square. Once you substitute here, i z will be equals to integral of x square into d a plus integral of y square into d a. From here we know integral. [00:06:02.096] integral of x squaring to d is i y and integral of y squaring to d a is i x so iz will be equals to i x plus i y. So please remember this. [00:06:12.336] Once we know the rectangular moment of inertia about any point, like O R it can be anything, then we can find polar moment of inertia about the same point just by adding Ix and Iy. We are going to get this IZ. [00:06:30.020] Next, uh say there is a lamina like this, and this is the y axis given in the cushion. Now [00:06:38.628] An element whose coordinate is negative contributes as much as the moment of inertia as does an equal element with a positive coordinate of the same magnitude. To explain this, I am going to consider this rectangle. Since these two are identical, assume [00:06:55.168] The centroid will be some area. Now the moment of inertia of this right rectangle will be equal to distance square into area we know. Since this point centroid is on the right side of the axis of symmetry, it will be positive. So distance square into area will also be positive. [00:07:11.748] But when you consider moment of inertia of this left rectangle, the centrate of this left rectangle will be here. So X coordinate will be negative. [00:07:23.044] But since we are talking about the distance square, even though it is on the left side of the axis, even though it is minus x, distance square of the minus 6 will be positive. So it doesn't matter wherever the axis is. [00:07:40.420] It's a continuation of the same thing. Whereas the first moment of area can be positive, negative, or it can be zero, whereas the second moment of an area will be always positive. As I said before. [00:07:53.888] First moment of an area, we use this concept in locating the centroid. Because first moment of an area is area into distance, if the x coordinate is negative, then [00:08:08.400] The first moment of area can be negative. Few times it will be zero. When the coordinate is positive, first moment of area can be positive. Since here it is distance square, so it doesn't matter whether

that is positive or negative. [00:08:23.024] Negative. So second moment of an area will be always positive. Next is unit of moment of inertia is if the dimensions are given in millimeter, then the unit of moment of inertia is millimeter to the power of four. If the dimensions are given in [00:08:40.900] meter then it will be meter to the power of four. Next is radius of guidation. [00:08:48.580] To explain radius of gardenation, I am going to consider a lamina. Let us consider this a lamina. The total area of the lamina let it be A. Then Ix stands for moment of inertia of the lamina about x-axis. [00:09:04.736] Iy stands for moment of inertia of a lamina above y-axis. Now we know how to locate the centroid. X bar and y bar are the coordinates of the centroid. This is what we know in this particular case. We know area, we know Ix, Iy, X bar, and Y bar. [00:09:22.436] Next. We are going to visualize this area as a narrow rectangular strip. Just area remains the same area as [00:09:33.156] The same area here, and the moment of inertia of this narrow rectangular strip, which is parallel to x-axis, that is the same as that of the case one. The inertia in this case. [00:09:45.988] Is also Ix. Now the distance from this central to the x-axis is called as radius of guerrillo the x axis, and that is denoted by kx. Then how to find kx?

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.000] We know from the definition of moment of inertia, moment of inertia is distance square into area. So moment of inertia about x axis is Ix, that already we know. So Ix is equal to distance square is kx square into area is A. [00:10:18.320] Since we know Ix from here and a from this case, now kx can be written as under root of ix by area. Similarly, if we visualize this. [00:10:31.300] Area as a narrow rectangular strip which is parallel to y axis. Now the area remains the same as the base one. Then inertia of this narrow strip about y-axis is also the same as that of [00:10:47.680] This case i y. Now from the definition of moment of inertia about y axis that is i y is will be equal to distances ky, so ky square into area is a [00:11:00.864] Now, the distance from y-axis to here, this is called as radius of variation about the y axis. Ky will be equal to under root of iy by total area A. [00:11:14.148] Similarly, we can consider like a narrows a ring like this. And in this case also, momentum inertia of this area about [00:11:25.328] The point about the z-axis passing through O will be the same as that of the moment of finesse about the z axis passing through O in this case. Now, again from the definition, we know iz is equal to Kz square into area. [00:11:39.088] So the distance from the center are the z axis through uh this narrow state k is that this is called as radius of variation about the z axis. So kx is equal to under root of i x by e. [00:11:53.840] Ky is under root of IY by A, Kz is equals to under root of IZ by area. [00:11:59.876] When we know the value of moment of inertia about x-axis, then and we know total area, then we can find radius of variation about x-axis. Similarly, once we know i inertia about y-axis and total area, we can find radius of variation about y-axis. [00:12:16.048] Similarly, when we know inertia about z axis and area, we can find radius of variation about the z axis. And it is a measure of distribution of the area from the axis in the cushion. [00:12:31.428] And as we know here, I z is equals to Ix plus I. I z we know kz square into area, that will be equals to kx square into area plus ky square into area. Since area is [00:12:47.056] So you mean LHs and R Hs that will be get canceled each other. Now we get Kz square as uh K square plus KY square. [00:12:57.476] Thus, the square of the radius of guaration about the polar axis equals to the sum of the squares of the radii of variation about the two corresponding regular rectangular axis. Now, let's know about parallel axis theorem. It's also called as transfer of axis. [00:13:18.084] According to parallel axis theorem, moment of inertia about any axis will be always equal to sum of the moment of inertia about parallel centroidal axis plus

product of the area and square of the distance between the parallel. [00:13:31.552] Two lines. So I am going to write the statement of the parallel axis theorem here. If you consider moment of inertia about any horizontal axis, so I'm going to write it as ix will be always equals to moment of inertia about centroidal x axis. [00:13:46.544] So central x axis is I bar s plus area is capital A into distance square is the distance between the two parallel lines. This is the statement of the [00:13:57.824] Parallel axis theorem mark you can write it as dh. That is the distance between the two uh x axis. To prove this, I am going to consider there is a lamina like this. Like the area of this lamina let it be capital A. [00:14:16.004] A is the area of the lamina. Next, consider this is in the xy plane like this. This is x-axis and this is [00:14:26.340] Y axis. So the lamina is in xy plane. Now I'm I know how to locate the centroid of this. Let C be the centroid of this lamina. Then uh let it be at a distance of dx with respect to [00:14:45.540] X axis. We know how to find DX. Now if I draw a horizontal line passing to the centroid, this will be the centroidal x axis. So it is x0.

EV0000010 · EV0000011 · EV0000012 · EV0000013

Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.484] Next, I am going to consider one smaller elemental area within this lamina. The shaded region, let it be the smaller elemental area, let d be the area of this element. Let it be at a distance of 10 with respect to. [00:15:19.780] The reference line here, the reference line is centroidal x-axis, so let it be y naught. Now I want to find the moment of inertia of this. [00:15:30.884] Smaller element above the x axis. What is moment of inertia? Moment of inertia is second moment of an area, that is distance square into area. So I can write [00:15:44.164] inertia of the smaller element about x axis that is dx dx is equal to distance squared to area. Distance from ear to ear is y naught plus dx. So y naught plus dx [00:15:59.876] Whole square into area is dA. This is the moment of inertia of the smaller element about the x-axis. Now, once I integrate this, I am going to get the moment of inertia of this lamina about the x-axis. So just integrate this. [00:16:19.844] So d integration of d i x is i x inertia about the x axis will be equal to it is in the form of a plus b whole square. So just expand that. It's going to be a square. So y naught square into this d a multiply that d a plus again integral of [00:16:39.140] a a plus b whole square so b square dx square into d a then plus 2 times set of integral of y naught into dx into d. [00:16:54.180] Next, consider each term. First, I am going to consider the term integral of [00:17:00.228] Y not square into DA. What is this? It is dis it is in the form of distance square into area. What is distance square into area? Distance square into area is second moment of an area. Nothing but moment of inertia. [00:17:16.464] Since the distances with respect to centroidal x axis and area is here, it is moment of inertia bar centroidal x axis. So I can write it as I bar x. [00:17:30.628] Next consider the second term. Second term is integral of dx square into dm. Next. [00:17:42.116] This is constant, dx is constant. So pick that out, dx square integral of d a. What is d? Integral of d a integral of d a is total area. So that is going to be dx square into a ra into dx square. [00:17:58.080] Let's consider the last term. The last term is two into integral of y naug into dx into d. [00:18:08.932] This will be equal to here. Dx is constant. So take that out 2 into dx into integral of y0 into dx. Here I am going to divide it by total area and multiply with total area. It doesn't make any difference. [00:18:25.472] Now, if you look at this only this term, integral of y naught into dA divided by total area, it is the formula of y bar. Since the central is on the reference line, this will be equal to [00:18:40.480] 0. This entire term will be 0. So this is equals to 0. Now substitute these three terms in this Ix. So we are going to get IX as Ix is equal to first term integral of y naught square into d is i bar x. [00:18:57.796] I bar X plus this is dx square into A and this is 0. So I x is equal to I bar x plus area into distance square. Nothing

but moment of inertia about any axis will be always equal to. [00:19:15.524] Sum of the moment of inertia about parallel centroidal axis plus product of the area and square of the distance between the two parallel axes. Here you have to remember two points to apply the parallel axis theorem. The first condition is the axis. [00:19:33.348] And the centroidal axis both should be parallel to each other. And to use this formula, one of the axes should be the centroidal axis. Then only we can apply parallel axis theorem to find the moment of inertia above any axis. [00:19:50.852] Uh that's it for today's class. Thank you.

EV0000014 · EV0000015 · EV0000016 · EV0000017

Full transcript: 2998 normalized words, 84 timestamped segments, 1195.413333 seconds duration. Low-confidence segments flagged: 0.

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# 3 CLASS 38 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 38 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000018 · EV0000019

## Graphs and diagrams

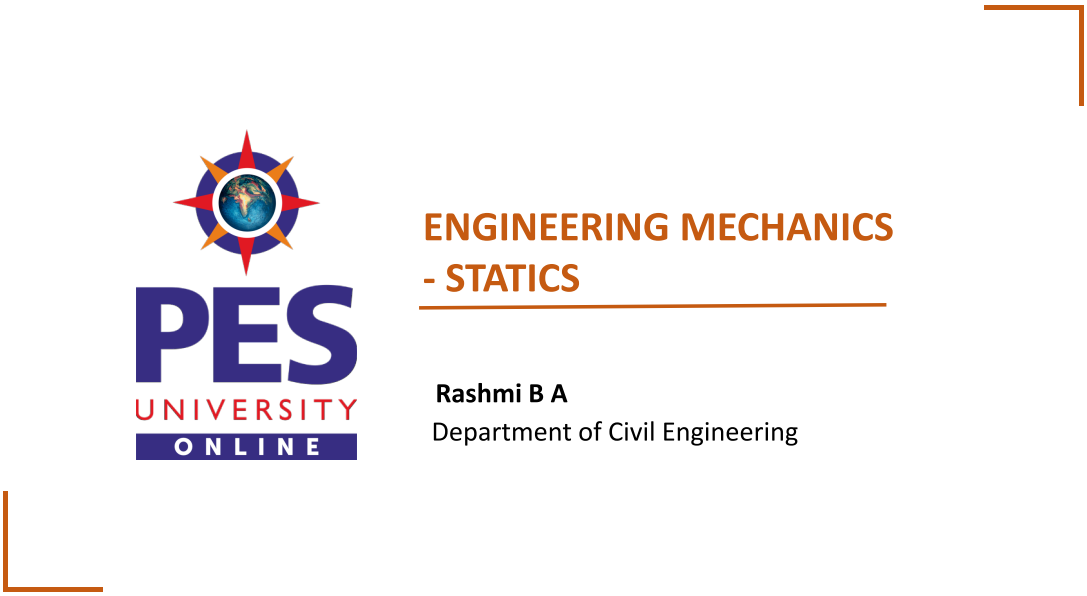


**Figure G000050.** UE20CV101 UNIT 4 SESSION 7 · source slide page 1.

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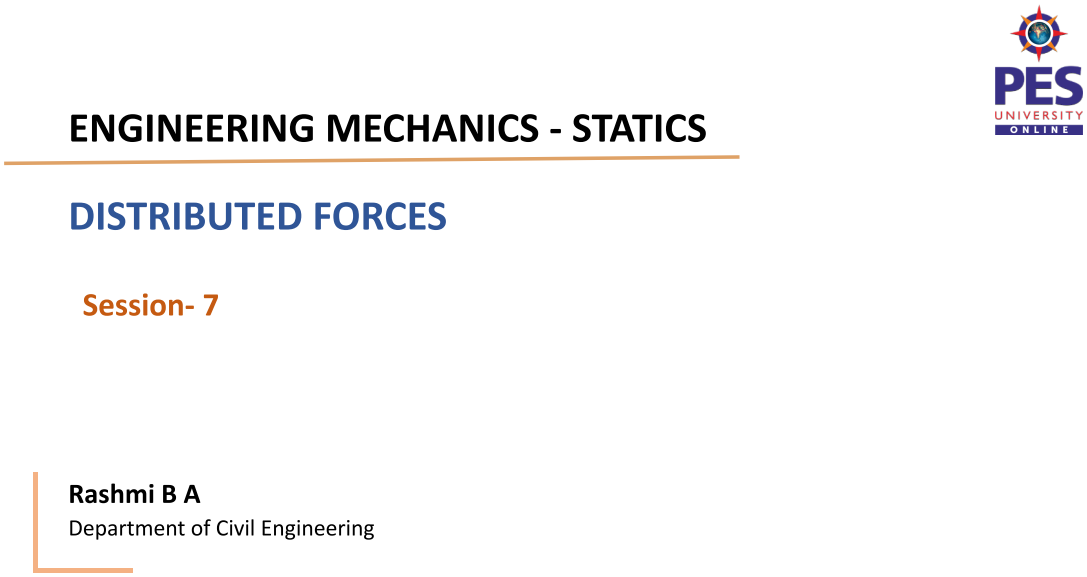


**Figure G000051.** UE20CV101 UNIT 4 SESSION 7 · source slide page 2.

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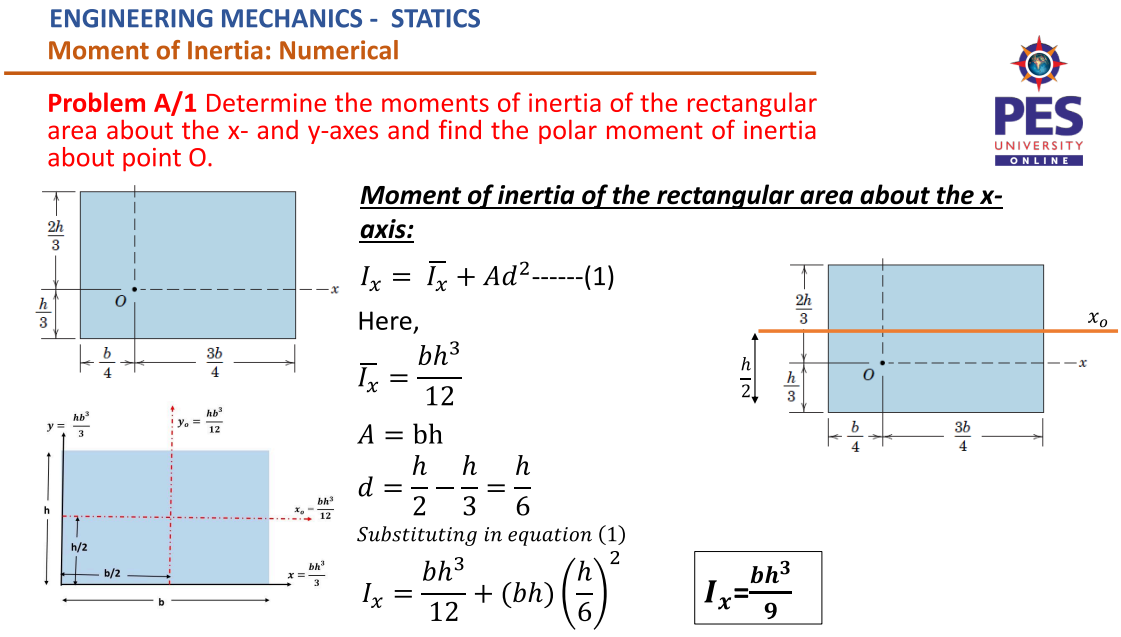


**Figure G000052.** UE20CV101 UNIT 4 SESSION 7 · source slide page 3.

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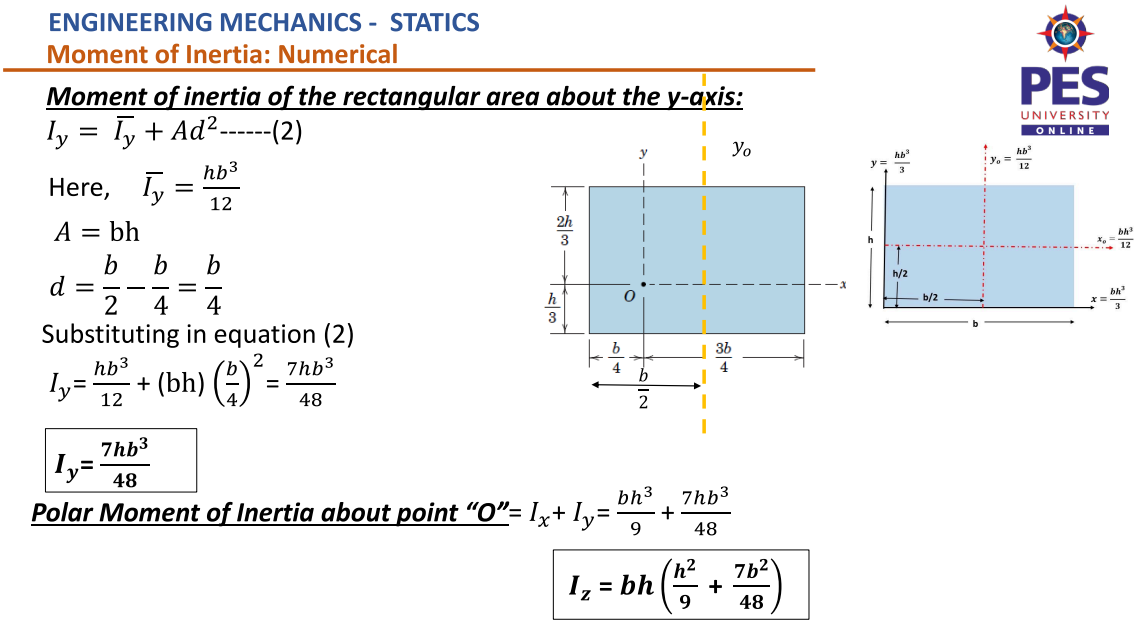


**Figure G000053.** UE20CV101 UNIT 4 SESSION 7 · source slide page 4.

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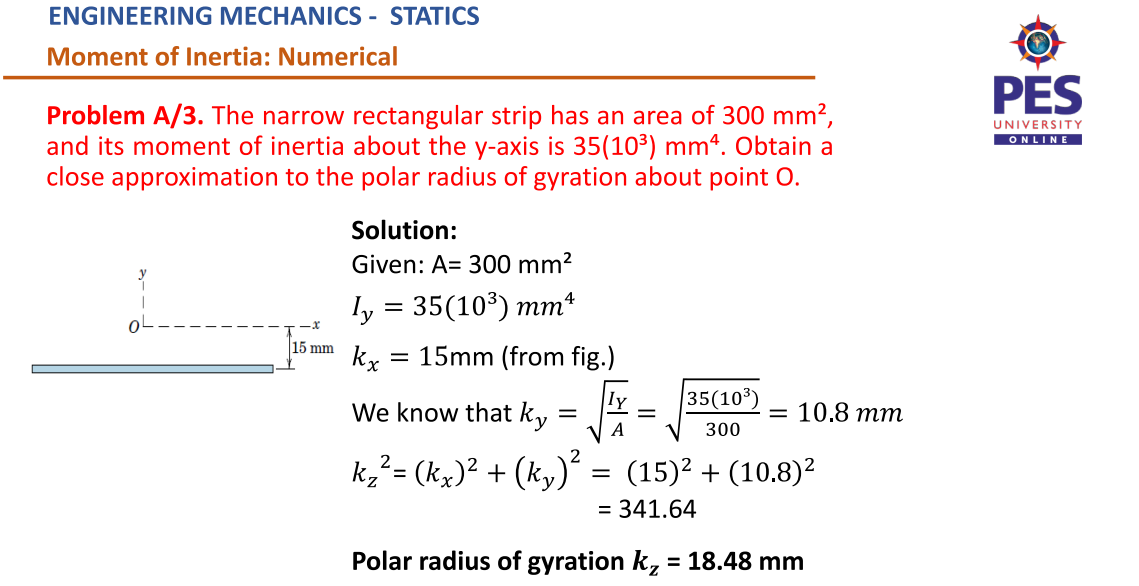


**Figure G000054.** UE20CV101 UNIT 4 SESSION 7 · source slide page 5.

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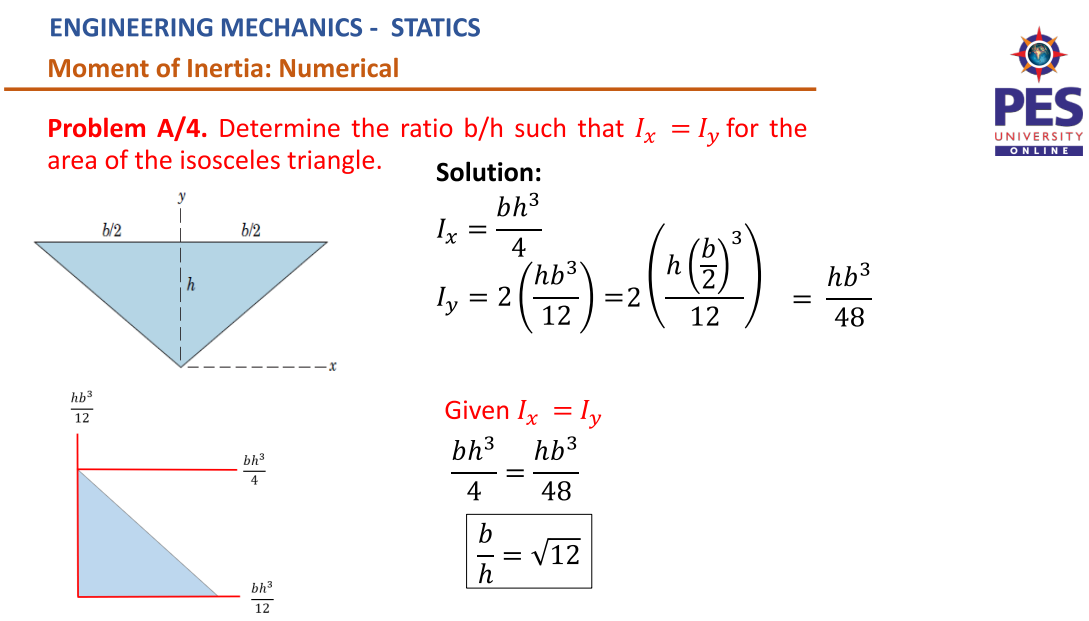


**Figure G000055.** UE20CV101 UNIT 4 SESSION 7 · source slide page 6.

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**Figure G000056.** UE20CV101 UNIT 4 SESSION 7 · source slide page 7.

EV0000352

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000333 · EV0000334 · EV0000335
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 7 Rashmi B A Department of Civil Engineering EV0000336 · EV0000337 · EV0000338
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/1 Determine the moments of inertia of the rectangular area about the x- and y-axes and find the polar moment of inertia about point O. Moment of inertia of the rectangular area about the x- axis: $Ix = \text{⊔}Ix + Ad2$------(1) Here, $xo\ bh3\ h\ \text{⊔}Ix = 12\ 2\ A = \text{bh}\ h\ h\ d = -h = 2\ 3\ 6\ Substitutinginequation1\ 2\ bh3\ h\ \mathbf{bh3}\ Ix = +\ (bh)\ \mathbf{Ix} = 12\ 6\ \mathbf{9}$ EV0000339 · EV0000340 · EV0000341
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Moment of inertia of the rectangular area about the y-axis: $Iy = \text{⊔}Iy + Ad2$------(2) $yo\ hb3$ Here, $\text{⊔}Iy = 12\ A = \text{bh}\ b\ b\ d = -b = 2\ 4\ 4$ Substituting in equation (2) $2\ hb3\ b\ 7hb3\ Iy = +\ (\text{bh}) = b\ 12\ 4\ 48\ 2\ \mathbf{7hb3}\ \mathbf{Iy} = \mathbf{48}\ bh3\ 7hb3$ Polar Moment of Inertia about point "O"$= Ix + Iy = +\ 9\ 48\ \mathbf{h2}\ \mathbf{7b2}\ \mathbf{Iz} = \mathbf{bh}\ \mathbf{9} + \mathbf{48}$ EV0000342 · EV0000343 · EV0000344
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/3. The narrow rectangular strip has an area of 300 mm², and its moment of inertia about the y-axis is 35(10³) mm⁴. Obtain a close approximation to the polar radius of gyration about point O. Solution: Given: A= 300 mm² $Iy = 35(10^3)\ mm^4\ kx = 15\text{mm}$ (from fig.) $IY\ 35(10^3)$ We know that $ky = A = 300 = 10.8\ mm\ 2\ 2\ 2\ kz\ 2 = kx2 + ky = 15 + 10.8 = 341.64$ Polar radius of gyration $\mathbf{kz} = 18.48$ mm EV0000345 · EV0000346 · EV0000347
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/4. Determine the ratio b/h such that $Ix = Iy$for the area of the isosceles triangle. Solution: $bh3\ Ix = 3\ 4\ hb\ hb3\ 2\ hb3\ Iy = 2 = 2 = 12\ 12\ 48$ Given $Ix = Iy\ bh3\ hb3 = 4\ 48\ b\ h = 12$ EV0000348 · EV0000349 · EV0000350
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000351 · EV0000352 · EV0000353

### Short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello, in today's session let's solve few problems and moment of inertia. Question A slash one. [00:00:11.268] Determine the moment of inertia of a rectangular area about the X and Y axis, and also find the polar moment of inertia about the point O. In this question, they have asked us to find inertia of this. [00:00:30.660] Inertia of this rectangular lamina about this x-axis, then about the y-axis. After that we have to find polar moment of inertia about the point O. [00:00:43.812] Polar moment of inertia about any point will be equal to some of the moment of inertia about the x and y axis passing through the same point, that is according to perpendicular axis. [00:00:57.264] theorem. So once we know the value of Ix and I, I z about O will be equals to Ix plus I y. [00:01:07.076] Let's start with finding Ix, that is moment of inertia of a rectangular area about the x axis. Then how to find inertia of this rectangular area about this axis. [00:01:21.028] When you think of rectangle, you may remember the formula about the bottommost member as BH cube by 12 and about the centroidal x axis as

bh cube by 3. But [00:01:34.704] If you look at this x-axis, it is not coinciding with the bottommost member. But what about is it a centroidal x-axis? Let's check that. Where will be the centroidal x-axis? [00:01:48.384] in a rectangle. We know the y bar formula of a rectangle that is h by 2. Here what is h? h is 2 h by 3 plus h by 3, it is h itself. So at a distance of y bar is equal to h by 2 [00:02:04.928] If we draw a horizontal line, this will be the centroidal x axis. That's why it is called as x naught. Now x axis is not the centroidal x axis. [00:02:19.524] We don't have a standard formula. Then how to find inertia about this axis, x axis. Whenever we don't have a standard formula, we have to apply the parallel axis theorem. Inertia about any axis is equal to inertia about [00:02:36.400] Parallel centroidal axis plus area into distance square. This is the statement. Now I am considered I bar x because. [00:02:47.364] This is horizontal, so centroidal axis also should be horizontal. Hence it will become I bar X. Next, how to find the value of I bar X then area and D. [00:03:02.308] If you remember the basic formulas, when we consider a rectangle, inertia of this rectangle about its centroidal x axis is BH cube by 2 y. So here also x i bar x is bh cube by [00:03:20.016] 12. This is from the formulas. Then A. A is nothing but area of a rectangle. Here what is area of a rectangle? Base into height. What is this width or a base? It's it is B and the height is H. [00:03:38.180] So A is equals to B into H. Then what is D? D stands for the distance between these two parallel lines. Nothing but X0 and X axis. [00:03:49.796] And what is this distance? From here it is h by 2 minus h by 3. H by 2 minus h by 3 if you simplify and get it as h by 6. [00:04:02.340] Now, substitute the value of I bar X area, then the distance in the equation 1. [00:04:11.780] After substituting in this, simplify this, we will get the value of ixs bh cube by 9. This is the moment of inertia of a rectangle about the x axis. [00:04:27.972] Next is we need to find moment of inertia of a rectangular area about the y axis. If you remember the formulas again, we know inertia about the leftmost member is Hbq by [00:04:43.216] 1. Inertia about the centroidal y-axis is HBQ by 3. Now this is not the leftmost member in a rectangle. At the same time, is it a centroidal y-axis? This width of the rectangle is B. Where will be the centroidal y-axis in a rectangle?

EV0000018 · EV0000019 · EV0000020 · EV0000021

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.168] At a distance of B by 2, if you draw a vertical line, this will be the central y axis. Since we don't have any formula about this y-axis, we have to apply parallel axis theorem to find inertia of this rectangular. [00:05:20.912] lamina about the y axis. Now start with the statement of the parallel axis theorem. Inertia about any line that is i y is equal to inertia about parallel centroidal axis that is i bar y plus area into distance square. [00:05:40.164] Always remember the statement of the parallel axis theorem. Next, what is this I bar Y inertia about centroidal y-axis? Remember the formulas. [00:05:51.652] From the formulas we know inertia about centroidal y axis, that is this value, is HBQ by 12. So I bar y is equal to HBQ by 12, then area is area of a rectangle, that is B into H. Then D. [00:06:08.976] D stands for the distance between the two parallel axes, that is y and y naught axis. What is the distance? That is b by 2 minus b by 4. [00:06:25.060] You will get it as b by 4 itself. Then substitute the value of i bar y, area and distance in the equation 2 and simplify we will get the value of i y. [00:06:40.240] Now the value of i is 7 hbq by 48. So we got the value of inertia about x axis and inertia about the y axis. [00:06:53.892] The last part of this question is polar moment of inertia about the point O. As I said before, IZ about O is equal to Ix plus Iy. So substitute the value of IX. [00:07:10.736] From the previous step Ix is BH Q by 9, substitute that here and Iy is 7 HBQ by 48. Substitute that. Once we simplify this, we will get the value of IZ as BH in.

[00:07:26.468] Ah into H square by 9 plus 7 B square by 48. This is the polar moment of inertia for a rectangular lamina about the point O. So we are done with this question. [00:07:44.228] Let's get into the next question. A slash 3. There is a narrow rectangular strip as an area of 300 mm square and its moment of inertia about the y axis is [00:07:59.280] 35 to 10 power 3 mm to the power of 4. Open a close approximation to the polar radius of guardation about the point O. How to find polar radius of guardation? [00:08:13.188] Polar radius of guaration is nothing but Kz. For Kz we have two formulas. Kz is equals to under root of IZ by A. That we can use when we know the polar moment of inertia about the [00:08:31.460] When we know the polar moment of inertia. This one more formula that is [00:08:37.264] Kz is equals to under root of kx square plus ky square. This formula we can use when we know the value of radius of guaration about x axis and radius of guaration about the y axis. Now for this caution which one to use? [00:08:54.960] Before that, let's tabulate what is given in the cursing. The area is given as 300 mm square, consider that. Then moment of initiative about the y-axis is given as 35 into 10 power 3 into mm to the power of 4, that is i y. [00:09:13.124] Next, look at the figure. If you look at the figure, the distance from the x axis to the narrow strip is given as 15 mm. That is nothing but radius of guarantee the x axis. It's kx is given. Now [00:09:32.804] Since we know iy and area, we can find the value of radius of guaration about the y axis, that is ky. Ky is equals to under root of iy by total area A. [00:09:46.144] Once we simplify this, we will get the value of Ky radius of variation about y-axis. [00:09:53.572] Now, look at this. We know radius of variation about x axis and radius of variation about y ax. Since we know kx and ky, let's use the second formula that is kz is equals to under root of kx square plus ky square.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.452] Simplify this, we will get the value of radius of garation about the point O as 18.48 mm. [00:10:23.652] This is the final answer. Let's start with the next question A slash 4. The question is determine the ratio B by H such that Ix is equal to Iy for the area of [00:10:40.656] the isoscerus triangles. [00:10:45.956] Now we we have been asked to find the ratio B by H. Yet B is [00:10:53.700] This base of this triangle and edge is the height. And what is given here is Ix is equal to Iy. What is Ix? Ix is moment of inertia about the x-axis, and what is Iy? [00:11:09.600] I is moment of inertia of this triangle about the y axis. Now, remember the formulas. When we have a triangle, assume a right angle triangle or any other triangle. [00:11:25.668] Moment of inertia about its base and centroidal x-axis and about the axis passing through its apex remains the same, irrespective of whether that is a right angle triangle or any other triangle like this. So that is [00:11:46.692] About this line is always about its base will be always BHQ by 12 about the [00:11:53.444] line horizontal line passing through the apex is BHQ by 4. Irrespective of the triangle these two will remain the same. So what I am going to do here is I am going to consider it as a one triangle [00:12:08.384] Moment of inertia this triangle about its ax, that is i will be equal to bh q by 4. I am going to use this formula. [00:12:18.896] Then in this what is B? B is nothing but B itself, that is B by 2 plus B by 2, nothing but B, and H, what is H? Height of the triangle here also it is H. So this formula remains the same. [00:12:32.496] Next, I need to find IY. When we think of initially about y-axis, we should have only the right angle triangles. Here, this is one right angle triangle and this is also one more right angle triangle. [00:12:47.396] Since these two are similar, iy will be equal to the formula is hbqed by 12. Consider that. Next, since these two are similar, I can take it as 2 times that of 1. [00:13:00.356] Because moment of inertia of this triangle about this line will be equal to moment of inertia of this

triangle about the same line or same axis y axis. Next, if we have to get into the substitution part. Here what is h? H is this. [00:13:16.960] That is nothing but H itself. Next, what is B? B stands for the base of the triangle. Since we have considered two triangles, the base should be taken as B by 2. So substitute that, it is going to be B by 2. [00:13:34.496] whole q. Next, simplify this. Once you simplify you will get it as h bqed by forty-eight. Now we got the value of Ix as B H Q by 4, I s h bq by forty eight. [00:13:51.328] Since Ix is equal to Iy, H B H cube by 4 will be equal to H B cube by 48. Solve it then. We need to find the ratio of B by H. The ratio will be equal to under root of 12. [00:14:10.404] So, this is about today's class. Try to solve the problems given in the assignment as well as in the question bank. If you come across any doubts, you can post your queries in the PESU forum. Thank you.

EV0000026 · EV0000027 · EV0000028 · EV0000029

Full transcript: 1961 normalized words, 59 timestamped segments, 866.474667 seconds duration. Low-confidence segments flagged: 0.

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# 4 CLASS 39 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 39 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000030 · EV0000031

## Graphs and diagrams

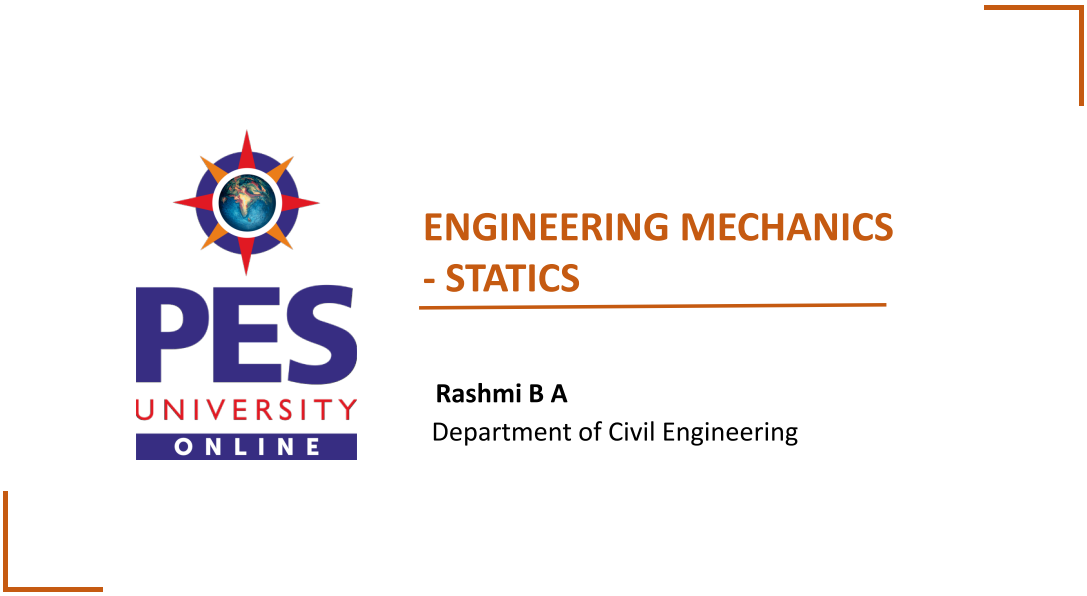


**Figure G000057.** UE20CV101 UNIT 4 SESSION 8 · source slide page 1.

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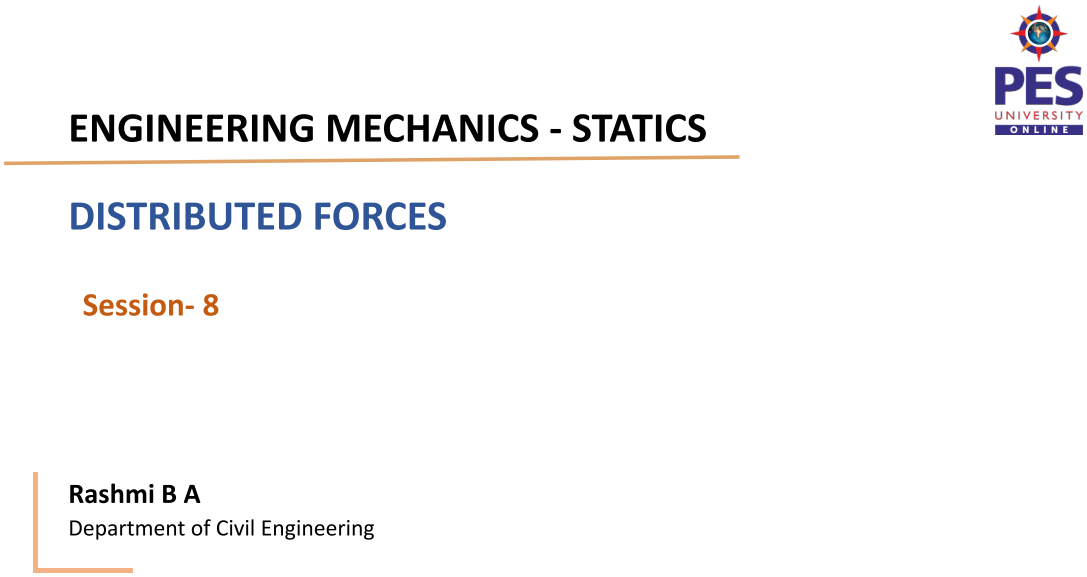


**Figure G000058.** UE20CV101 UNIT 4 SESSION 8 · source slide page 2.

EV0000358

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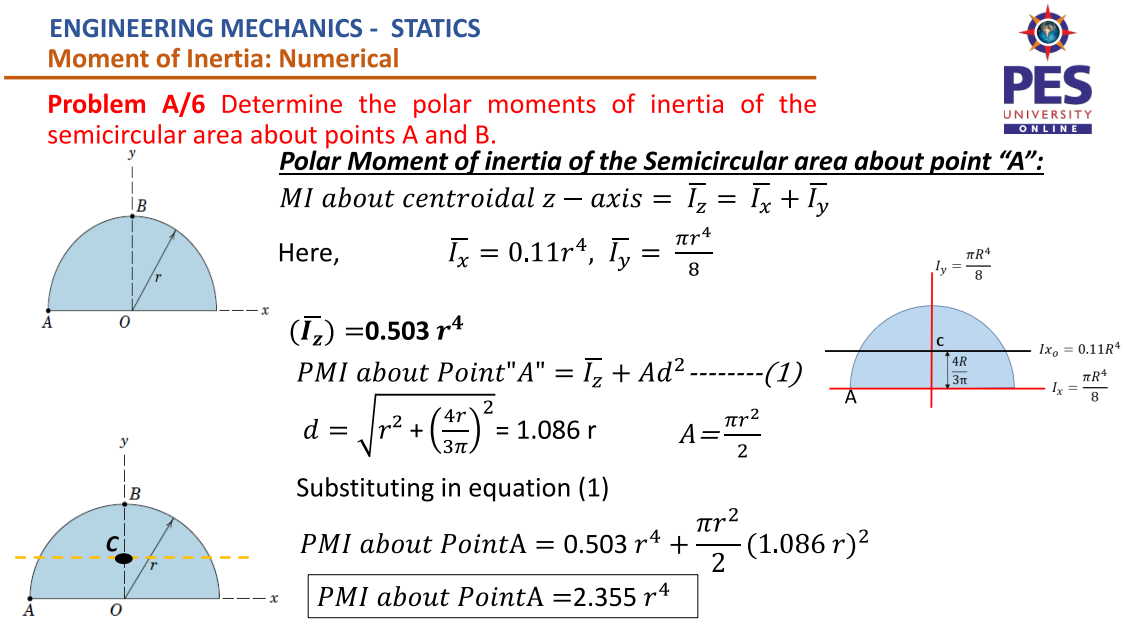


**Figure G000059.** UE20CV101 UNIT 4 SESSION 8 · source slide page 3.

EV0000361

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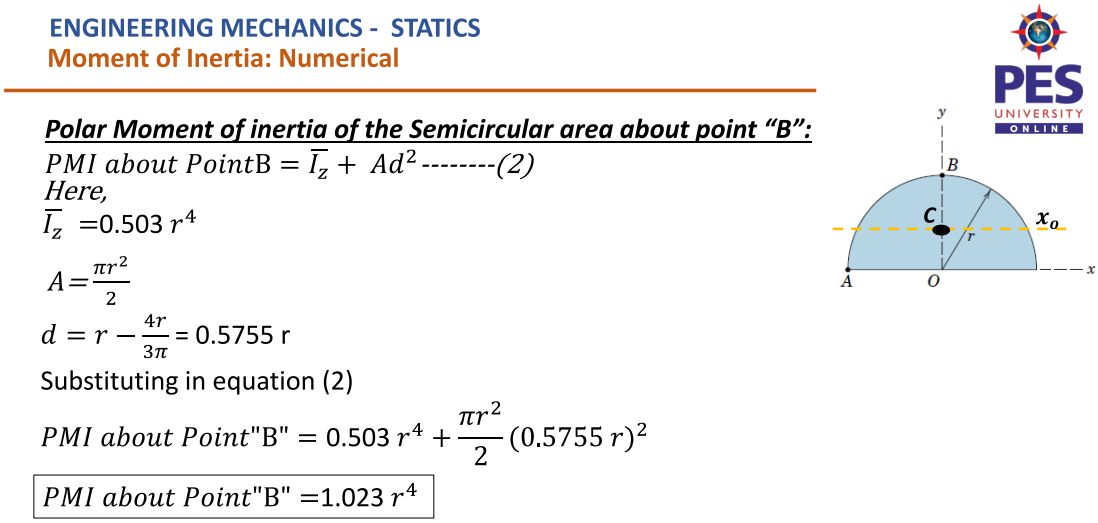


**Figure G000060.** UE20CV101 UNIT 4 SESSION 8 · source slide page 4.

EV0000364

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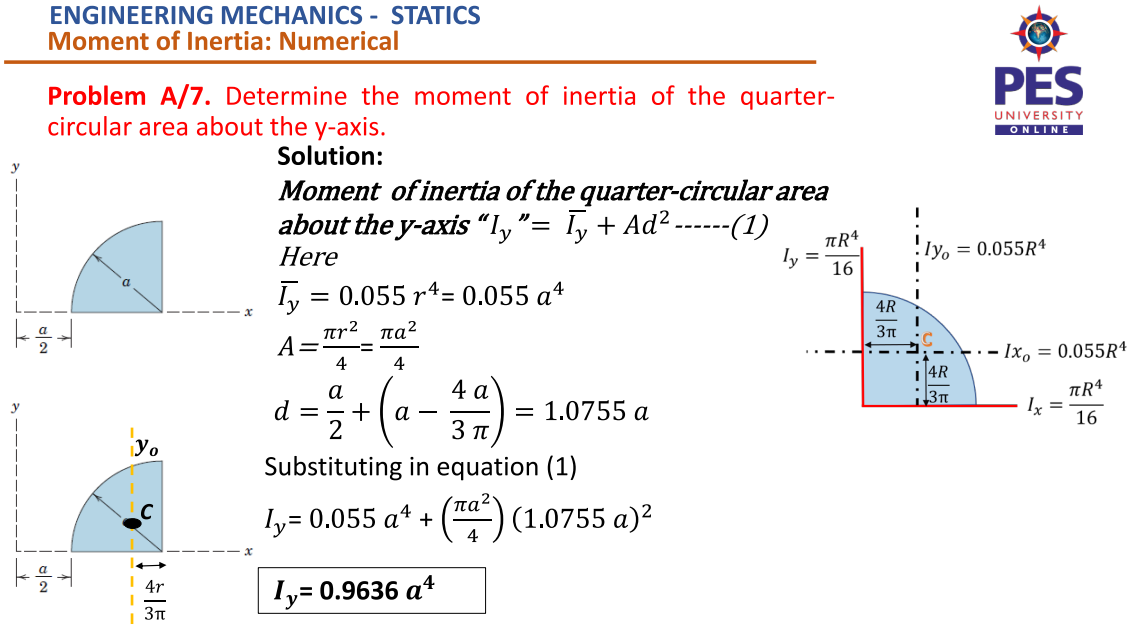


**Figure G000061.** UE20CV101 UNIT 4 SESSION 8 · source slide page 5.

EV0000367

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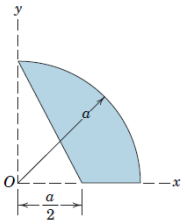
**ENGINEERING MECHANICS - STATICS**

**Moment of Inertia: Numerical**

**Problem A/19** Determine the moment of inertia of the shaded area about the x-axis.




***Moment of inertia of the shaded area about the x-axis:***

$$I_x = \begin{pmatrix} MI\ of\ a\ quarter\ circular \\ lamina\ about\ x-Axis \end{pmatrix} - \begin{pmatrix} MI\ of\ a\ triangle\ about \\ x-Axis \end{pmatrix}$$

$$I_x = \left(\frac{\pi r^4}{16}\right) - \left(\frac{bh^3}{12}\right)$$

Here,

r = a, b= a/2 & h = a

Substituting in the above equation

$$I_x = \left(\frac{\pi a^4}{16}\right) - \left(\frac{\left(\frac{a}{2}\right)(a)^3}{12}\right) = 0.154a^4$$

$$\boxed{I_x = 0.154a^4}$$

**Figure G000062.** UE20CV101 UNIT 4 SESSION 8 · source slide page 6.

EV0000370

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**THANK YOU**

**Rashmi B A**
Department of Civil Engineering
**rashmiba@pes.edu**

**Figure G000063.** UE20CV101 UNIT 4 SESSION 8 · source slide page 7.

EV0000373

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000354 · EV0000355 · EV0000356
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 8 Rashmi B A Department of Civil Engineering EV0000357 · EV0000358 · EV0000359
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/6 Determine the polar moments of inertia of the semicircular area about points A and B. Polar Moment of inertia of the Semicircular area about point “A”: $MIaboutcentroidalz-axis = \text{⧆}Iz = \text{⧆}Ix + \text{⧆}Iy\ \pi r4$ Here, $\text{⧆}Ix = 0.11r4$, $\text{⧆}Iy = 8$ ($\text{⧆}\boldsymbol{Iz}$) =0.503 $\boldsymbol{r4}$ $PMIaboutPoint"A" = \text{⧆}Iz + Ad2$--------(1) A 2 $4r\ \pi r2\ d = r^2 + = 1.086$ r $A = 3\pi$ 2 Substituting in equation (1) $\pi r2$ C $PMIaboutPoint$A = $0.503\ r4 + (1.086\ r)2$ 2 $PMIaboutPoint$A =2.355 $r4$ EV0000360 · EV0000361 · EV0000362
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Polar Moment of inertia of the Semicircular area about point “B”: $PMIaboutPoint$B = $\text{⧆}Iz + Ad2$--------(2) Here, $\text{⧆}Iz$=0.503 $r4$ C $\boldsymbol{xo}\ \pi r2\ A = 2\ 4r\ d = r - 3\pi$= 0.5755 r Substituting in equation (2) $\pi r2\ PMIaboutPoint"B" = 0.503\ r4 + (0.5755\ r)2$ 2 $PMIaboutPoint"B"$ =1.023 $r4$ EV0000363 · EV0000364 · EV0000365
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/7. Determine the moment of inertia of the quarter- circular area about the y-axis. Solution: Moment ofinertiaofthequarter-circulararea aboutthey-axis“ = $Ad2$------(1) $Iy$” $\text{⧆}Iy$+ Here 0.055 $r4$= 0.055 $a4$ $\text{⧆}Iy = \pi r2\ \pi a2\ A = = 4\ 4\ a\ a\ d = + a-4 = 1.0755\ a$ 2 3 $\pi\ \boldsymbol{yo}$ Substituting in equation (1) $\pi a2$ C $a4 + 1.0755\ a2\ Iy$= 0.055 4 $\boldsymbol{a4}$ $4r\ \boldsymbol{Iy}$= 0.9636 3π EV0000366 · EV0000367 · EV0000368
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/19 Determine the moment of inertia of the shaded area about the x-axis. Moment of inertia of the shaded area about the x-axis: $MIofaquartercircular\ MIofatriangleabout\ Ix = - laminaaboutx-Axis\ x-Axis\ \pi r4\ bh3\ Ix = -$ 16 12 Here, r = a, b= a/2 & h = a Substituting in the above equation $a\ a3\ \pi a4 - 2 = 0.154a4\ Ix$= 16 12 $Ix$=0.154$a4$ EV0000369 · EV0000370 · EV0000371
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000372 · EV0000373 · EV0000374

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.260] Hello in today's session let's solve few problems and moment of inertia. The first question is A slash six Determine polar moments of inertia of a semicircular area about points A and B [00:00:17.572] Now let's start with the first point, polar moment of inertia of this semicircular area, about a point A. [00:00:27.428] What is polar moment of inertia? It's a moment of inertia about Z axis passing through that particular point. Since we are talking about polar moment of inertia about A, it is the moment of inertia about Z-axis passing through point A. [00:00:43.968] And remember that z axis will be always normal to the plane of the paper. So that's why here we can't see the z axis. [00:00:52.512] Ah the point A itself, that's the point to which the Z axis is passing, and that is normal to this

plane of this semi-circular area. Now, how to find iZ about A? [00:01:06.724] When we don't have a standard formula, when we don't have a standard formula, we have to apply the parallel axis theorem. Inertia about any axis is equal to inertia about parallels intoidal axis plus C area into distance square. This is the statement which we know. [00:01:22.756] Since we are talking about the Z axis, here the centroidal axis also should be the centroidal Z axis. Next, where will be the centroidal? [00:01:35.024] Z axis. Centroidal Z axis will be passing through C and that will be normal to the plane of the paper. Again. [00:01:43.876] Now, how to find the C? Remember the basic formulas whenever we have a semicircle. First locate the center. This is the center of a semicircle. Then draw the axis of symmetry. Now this is the axis of symmetry. Then the distance. [00:02:01.924] From the centroid to the center on the axis of symmetry is 4 r by 3 pi. This is 4 r by 3 pi, we will get this point C. Now, this is the centroid of this semicircular area. [00:02:15.344] Now, how to find moment of inertia about the centroidal z-axis? That is nothing but z-axis passing through C. [00:02:23.972] For that, remember I z will be equals to Ix plus Iy, since uh here it is the Z axis passing to the centroid, it will be a centroidal z axis. That will be equal to I bar X plus I bar. [00:02:40.320] Y where I bar x is moment of inertia about centroidal x axis and i bar y is moment of inertia about centroidal y axis. So we got the value of [00:02:51.280] Moment of inertia about a centroidal z axis. Then what is i bar x? I bar x we know from the formulas it is 0.11 into r to the power of 4. Then what is i bar y? That is pi r to the power of 4 by 8. And these two. [00:03:07.440] We are going to get high bar Z. Remember, this is the moment of inertia about the centroidal Z axis. Once we know moment of inertia about centroidal z axis, we can find moment of inertia about z axis about any other point. [00:03:26.724] Now consider point A. Z axis is passing through point A, we have to find inertia about that Z axis. How to find that? That will be equal to inertia about centroidal Z axis, that is I bar Z plus [00:03:42.128] Area into distance square. This is the statement. Then what is I bar Z? We know moment of inertia about centroidal z axis is 0.503 R to the power of 4. [00:03:54.820] Plus, area is area of a semicircle, that is pi r square by 2. Then what is D? D is the distance between the two parallel axes. Here, one axis is passing through A and one more axis is passing through C. [00:04:11.632] Then how to find this distance AC? To find this distance A C, look at this. This is C which is at a distance of four r by three pi. [00:04:20.864] Then we need to find this distance A C. This A C will be equal to under root of AO square plus C O square because C O A is a right angle triangle. Then what is A C? A C is equal to under root of R square. [00:04:35.824] That is AO plus 4 R by 3 pi whole square. We will get the value of distance between the centroid and Z axis and the Z axis passing through A. And Area we know. Substitute the value of area, then the distance and I bar z in the [00:04:53.412] Equation 1 and simplify we are going to get the polar moment of inertia about the point A. That is 2.35 into R to the power of 4. This is nothing but moment of inertia about z axis passing through A.

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.644] Next, we need to find moment of inertia, polar moment of inertia about point B. [00:05:19.332] Let's consider that again we are going to apply the parallelaxis theorem. Inertia about polar moment of inertia about point B will be equals to moment of inertia about centroidal z axis, that is I bar Z plus area into distance square. [00:05:35.376] I bar z we know from the previous case. This is 0.503 R to the power of 4, substitute that. Then area is area of the semicircular lamina that is pi r square by 2. [00:05:49.296] Then D. D is the distance between the centroidal Z axis and the Z axis passing through B. Nothing but this C B distance. Then how to find this C B? [00:06:00.772] C B is equals to year to year OB

is R, R minus O C is 4 R by 3 pi. So, R minus 4 R by 3 pi will give us the distance. [00:06:13.636] Once we know the distance, area, then I bar said, substitute these three in the equation 2. And we have to simplify this. Then we will get the polar moment of inertia about the point B. [00:06:29.328] This nothing but moment of inertia of this semicircular area about the z-axis passing through B. That is 1.023 into r to the power of 4. This is the answer. [00:06:45.604] Next question A slash 7 determine the moment of inertia of a quarter circular area about the y axis. Y axis is given here, which is at a distance of A plus A by 2 with respect to this origin of this. [00:07:03.908] Quarter circle. Next. Now we don't have any standard formula about this axis y axis. Then we have to apply parallaxis theorem. Inertia about y axis is equals to initiate about centroidal y axis plus area into distance square. [00:07:20.448] This is the statement. Then what is I bar y? For a quarter circle, we know from the formulas I bar y is inertia about centroidal y axis that is 0.055 into r to the power of 4. [00:07:36.100] Remember the formula? Substitute that i bar y value. Since e r is a, it is going to be 0.055 a to the power of 4. [00:07:46.820] Okay, next we have A. A is area of this quarter circle that is pi r square by 4. Since r is a, it is it will be pi a square by 4. Next, D. D is the [00:08:01.904] Distance between the y axis and the centroidal y axis. For that, where will be the centroid with respect to the radial line? It will be at a distance of 4 r by 3 pi. So this will be 4 r by 3 pi. But what is the distance here? Distance is this distance we want. [00:08:22.596] That is the distance between y ax and y naught axis. Then how to find this distance? This distance will be equal to a by 2 plus a minus of 4r by 3 pi. [00:08:40.612] Here R is A, so substitute R S A in this we will get the value of D. It is 1.07 double phi phi phi into a. [00:08:53.060] Substitute the value of I bar y area then D in this equation one, then simplify this, we will get the value of I as 0.96368 to the power of 4. [00:09:10.660] This is the moment of inertia of this quarter circle about the about y axis. [00:09:20.548] This is the next question A slash 19. Determine moment of inertia of the shaded area about the X axis. Look at this shaded area. [00:09:33.616] Shaded area is only this. We have to find inertia of this shaded area above x axis. First thing, do we have any standard formula for the shaded area? We don't have. Then how to get this shaded area? If you observe this figure carefully. [00:09:52.196] This is a quarter circle of radius A in that this triangle has been cut or removed of base A by 2 and the height A.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.476] So, consider quite a circle in that direct triangle, we will get the shaded area. Next, how to find inertia of the shaded area? Moment of inertia of the shaded area about x axis will be equal to. [00:10:20.752] Moment of inertia of this entire quarter circle about x axis minus in that what do we need to detect? Inertia of a triangle about the same x axis. [00:10:34.980] That's what is given here. I x is equal to moment of inertia of a quarter circular lamina about x axis minus moment of inertia of a triangle about the same x axis. Next, what is moment of inertia of this quarter circular area about x axis? [00:10:53.060] This x-axis is coinciding with the radial line. So moment of inertia of quarter circle about this line is we know pi r to the power of 4 by 16. Here r is a, so it will be pi a to the power of 4 by 16. Next. [00:11:09.792] We are done with the quarter circle. Next come to the triangle. What is the moment of inertia of a triangle about its base? [00:11:18.532] The standard formula is BHQ by 12. So consider that. Here what is base A by 2, and what is H A. Substitute in this formula, we are going to get the value of Ix. [00:11:35.428] That's what is given here. Radius is a and base of the triangle is A by 2, and the height of the triangle is A. Substituting this the above equation, it's going to be pi A to the power of 4 by 16 minus basis. [00:11:51.728] A by 2, take that

as it is. H is J 8 cube by 12. Simplify this, we will get the value of Ix is 0.154 a to the power of 4. Remember, this is the moment of inertia of a shaded area about the x axis. [00:12:13.860] That's it for today's class. Try to solve the problems given in the assignment as well as in the question bank. If you come acros any doubts, you can post your queries in the PESU forum. Thank you.

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Full transcript: 1845 normalized words, 51 timestamped segments, 745.578667 seconds duration. Low-confidence segments flagged: 0.

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# 5 CLASS 40 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 40 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000042 · EV0000043

### Graphs and diagrams

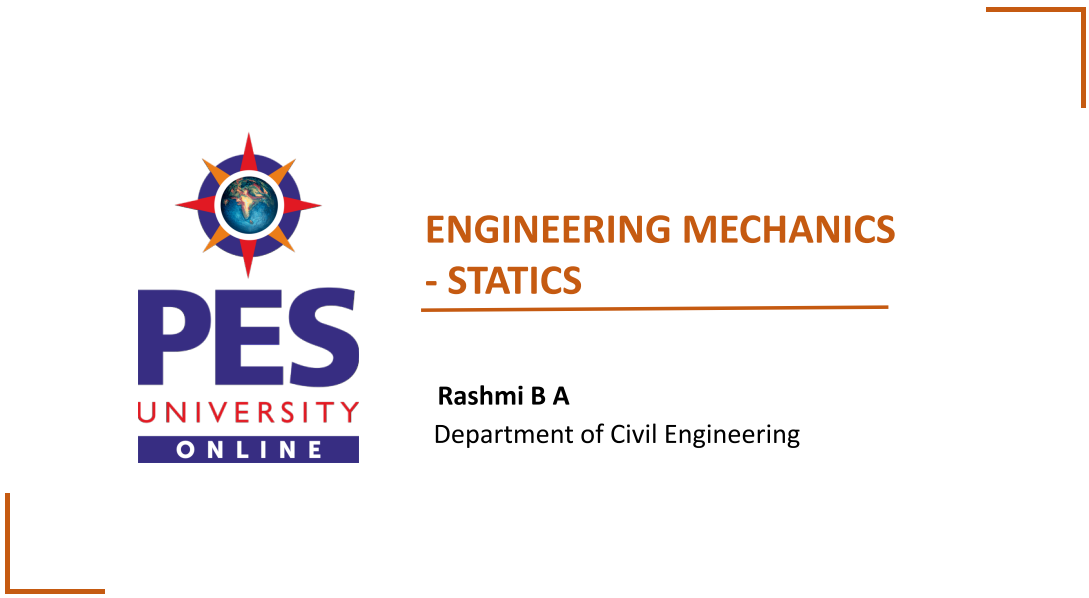


**Figure G000064.** UE20CV101 UNIT 4 SESSION 9 · source slide page 1.

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# ENGINEERING MECHANICS - STATICS

## DISTRIBUTED FORCES

Session- 9

**Rashmi B A**
Department of Civil Engineering

**Figure G000065.** UE20CV101 UNIT 4 SESSION 9 · source slide page 2.

EV0000379

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ENGINEERING MECHANICS - STATICS

Moment of Inertia: Numerical

**Problem A/35.** Determine the moment of inertia about the x-axis of the square area without and with the central circular hole.

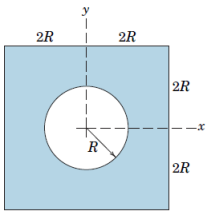


**Figure G000066.** UE20CV101 UNIT 4 SESSION 9 · source slide page 3.

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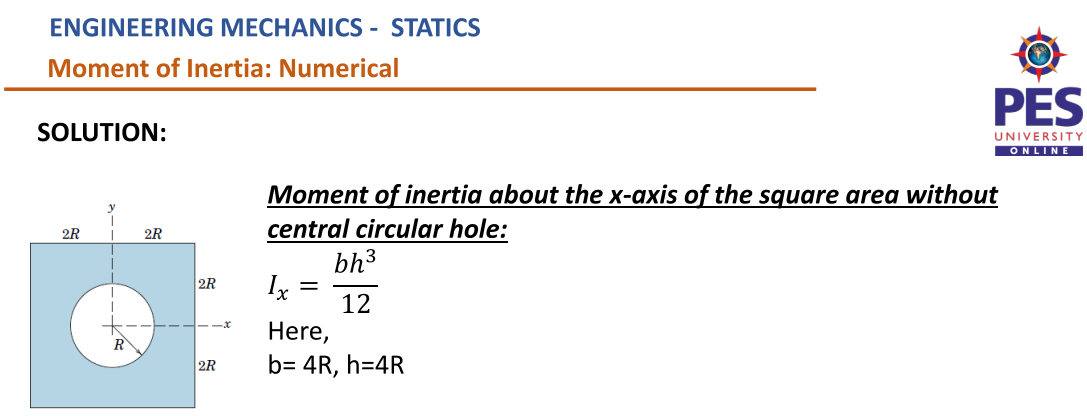


**Figure G000067.** UE20CV101 UNIT 4 SESSION 9 · source slide page 4.

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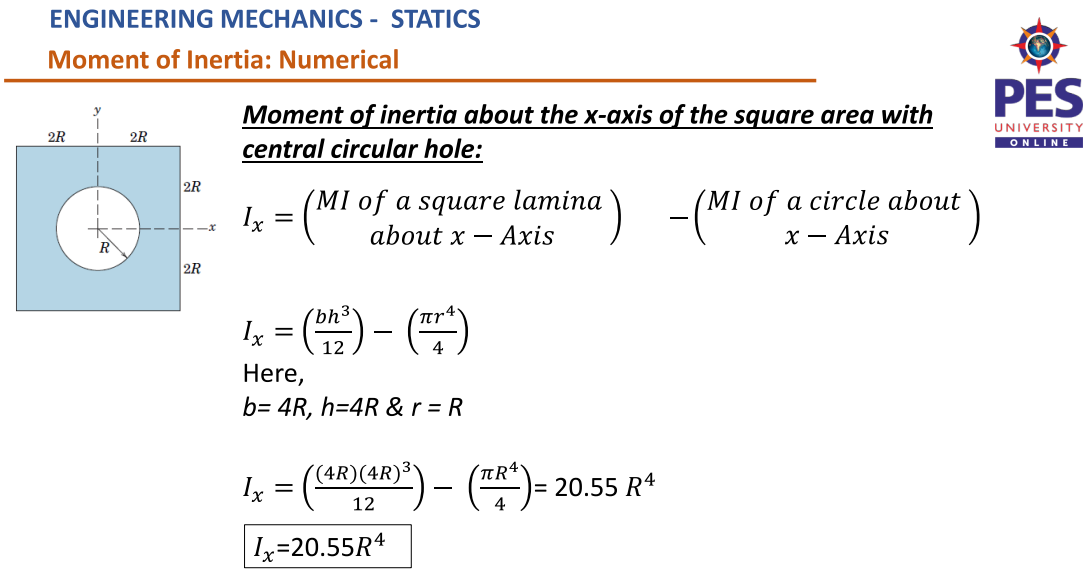


**Figure G000068.** UE20CV101 UNIT 4 SESSION 9 · source slide page 5.

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**ENGINEERING MECHANICS - STATICS**
**Moment of Inertia: Numerical**

**Problem A/39.** Calculate the polar radius of gyration of the shaded area about the center O of the larger circle.

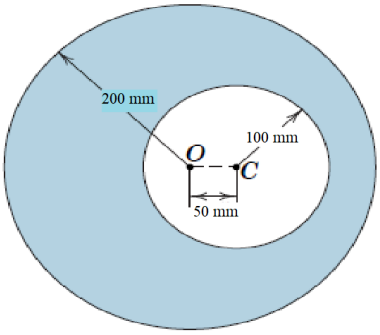


**Figure G000069.** UE20CV101 UNIT 4 SESSION 9 · source slide page 6.

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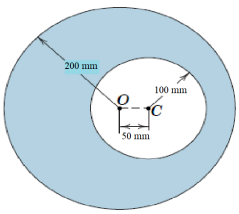
**ENGINEERING MECHANICS - STATICS**
**Moment of Inertia: Numerical**

**SOLUTION:**




***Polar Radius of Gyration of the shaded area about the Center "O":*** $K_z$

$K_o = \sqrt{\frac{I_z}{A}}$ *Here, $I_z$ is PMI of the shaded area about point "O" & A is shaded area.*

$I_z = I_x + I_y$ *Here $I_x, I_y$ are the MI of the shaded area about x and y – axis*

$I_x =$ (*MI of a larger circle about x – Axis*) - (*MI of a smaller circle about x – Axis*) ------(1)

*MI of a larger circle about x – axis* $= \frac{\pi R^4}{4}$ *here* $R = 200mm$

*MI of a smaller circle about x – axis* $= \frac{\pi R^4}{4}$ *here* $R = 100mm$

Substituting in equation (1)

$I_x = \left(\frac{\pi 200^4}{4}\right) - \left(\frac{\pi 100^4}{4}\right) = (12.566 \times 10^8) - (0.7854 \times 10^8) = \mathbf{11.78 \times 10^8 mm^4}$

**Figure G000070.** UE20CV101 UNIT 4 SESSION 9 · source slide page 7.

EV0000394

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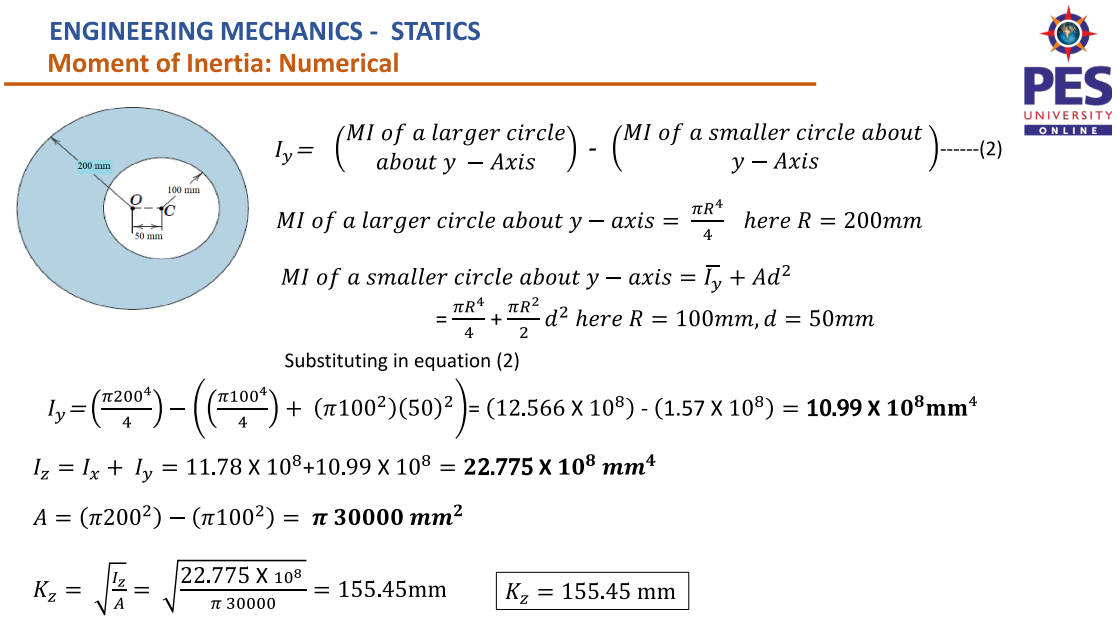


**Figure G000071.** UE20CV101 UNIT 4 SESSION 9 · source slide page 8.

EV0000397

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**Figure G000072.** UE20CV101 UNIT 4 SESSION 9 · source slide page 9.

EV0000400

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000375 · EV0000376 · EV0000377
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 9 Rashmi B A Department of Civil Engineering EV0000378 · EV0000379 · EV0000380
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/35. Determine the moment of inertia about the x-axis of the square area without and with the central circular hole. EV0000381 · EV0000382 · EV0000383
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Moment of inertia about the x-axis of the square area without central circular hole: $bh3\ Ix$= 12 Here, b= 4R, h=4R (4$R$)(4$R$)3 = 21.33$R$3 $Ix$= 12 21.33$R$3 $Ix$= EV0000384 · EV0000385 · EV0000386
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Moment of inertia about the x-axis of the square area with central circular hole: $MIofasquarelamina\ MIofacircleabout\ Ix$= − $aboutx−Axis\ x−Axis\ bh3\ \pi r4\ Ix$= − 12 4 Here, b= 4R, h=4R & r = R (4$R$)(4$R$)3 $\pi R4\ R4\ Ix$= − = 20.55 12 4 $Ix$=20.55$R$4 EV0000387 · EV0000388 · EV0000389
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/39. Calculate the polar radius of gyration of the shaded area about the center O of the larger circle. EV0000390 · EV0000391 · EV0000392
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Polar Radius of Gyration of the shaded area about the Center "O":$\boldsymbol{Kz}\ Iz\ Here, IzisPMIoftheshadedareaaboutpoint$"O" & Aisshaded $Ko$= $A$ area. Here $Ix$,$Iy$aretheMIoftheshadedareaaboutxandy−axis $Iz$= $Ix$+ $Iy\ MIofalargercircle\ MIofasmallercircleabout\ Ix$= - ------(1) $aboutx−Axis\ x−Axis\ \pi R4\ MIofalargercircleaboutx−axis$= $hereR$= 200$mm$ 4 $\pi R4\ MIofasmallercircleaboutx−axis$= $hereR$= 100$mm$ 4 Substituting in equation (1) $\pi$2004 $\pi$1004 $Ix$= − = 12.566 X 108 - 0.7854X 108 = 11.78 X $\mathbf{108mm^4}$ 4 4 EV0000393 · EV0000394 · EV0000395
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical $MIofalargercircle\ MIofasmallercircleabout$ ------(2) $Iy$= - $abouty−Axis\ y−Axis\ \pi R4\ MIofalargercircleabouty−axis$= $hereR$= 200$mm$ 4 $MIofasmallercircleabouty−axis$= $Iy$+ $Ad2\ \pi R4\ \pi R2$ = + $d2\ hereR$= 100$mm$, $d$= 50$mm$ 4 2 Substituting in equation (2) $\pi$2004 $\pi$1004 2 $Iy$= − + $\pi$1002 50 = 12.566 X 108 - 1.57 X 108 = 10.99 X $\mathbf{108mm^4}$ 4 4 $Iz$= $Ix$+ $Iy$= 11.78 X 108+10.99 X 108 = **2**2.775 X $\mathbf{108mm4}$ $A$= $\pi$2002 −$\pi$1002 = $\boldsymbol{\pi 30000mm2}$ $Iz$ 22.775 X 108 $Kz$= $A$= $\pi$30000 = 155.45mm $Kz$= 155.45 mm EV0000396 · EV0000397 · EV0000398
9. Source slide page 9 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000399 · EV0000400 · EV0000401

## Short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.260] Hello, in today's class, let's solve few problems and moment of inertia. This is the first problem in today's class A slash 35. Determine the moment of inertia about the x-axis of the square area without and with the central circular pole. Here [00:00:20.516] They have given a square and a side 4R. We have been asked to find moment of initiative this square about this x axis in the absence of this circular pole, and second [00:00:35.968] We have to consider the circular hole here, and we have to find the moment of inertia of only the shaded area. So the first part is in the absence of this, what is the moment of inertia of this square about the x-axis. If you look at this x axis, this x axis is [00:00:55.364] Nothing but centroidal x-axis because this square is symmetrical about this x-axis. So centroid will be somewhere on this x-axis. That's why this itself will be the centroidal x-axis. [00:01:10.160] Then how to find moment of inertia phase square about its centroidal x axis? [00:01:15.972] We know we can make use of the formula of a rectangle itself. The moment of inertia of a rectangle about its centroidal x-axis is BH cubed by 12. So we can make use of the same formula here, and we can substitute B and H's. [00:01:32.464] four R and we can find IX. Let's start with the solution. [00:01:41.476] So I'm going to consider the same [00:01:46.532] Square ear, x axis is here. This is without considering the central circular hole. Now [00:01:58.180] Since the sides are 4R and this is also 4R, we know inertia of this square about centroidal x axis, I bar x formula we know that is equal to bh cube by so consider that. [00:02:15.620] What is B here? 4R and H is also 4R. So next I bar X will be equal to in this case it will be 6. [00:02:31.716] 4 r to the power of 4 by 12. Simplify this, we are going to get it as 21.33 into r to the power of 4. This is the moment of inertia of this. [00:02:50.276] Square about the x axis. Next is we have to consider moment of inertia of shaded area. [00:03:07.620] About X axis that will be equal to moment of inertia of S square. [00:03:21.988] About x axis [00:03:27.364] minus moment of inertia of a circle above the same x axis. [00:03:43.044] Now moment of inertia of a square about x axis we know it is bh cube by twelve so bh cube by twelve then moment of inertia of a circle about x axis. [00:03:58.896] If you consider that the circle is like this, and this circle is also symmetrical about this x-axis. So moment of initial of the circle about this x-axis will be equal to [00:04:11.396] Pi r to the power of 4 by 4 because this itself is the centroidal x-axis even for circle. So it is minus pi r to the power of 4 by 4. [00:04:25.860] Now substitute the value of B as 4R and H also has 4R. Cube by 12 minus pi into here radius is R itself. So R cub by [00:04:42.788] Sorry, R to the power of 4 by 4. Now once we simplify this, once we solve it, we are going to get Ix as [00:04:52.260] 20.55 into r to the power of four. This is the moment of inertia of the square with the central circular hole.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.588] Then this is the next question A slash 39. Calculate the polar radius of gallation of the shaded area about the center O of the larger circle. Look at this. [00:05:21.412] There is a circle of radius 200 millimeter, and this is the center of the larger circle. Next, there is one more smaller circle. [00:05:34.080] And C is the center of this smaller circle of radius, 100 mm. And if you look at these two O and C, they are on a same horizontal line. [00:05:44.804] Here, they have asked us to find polar radius of gyration. Polar radius of gyration is nothing but Kz. Then how to find Kz? [00:05:56.384] To find Kz, we have two formulas. One is under root of IZ by A. Where iz is polar moment of inertia and capital A is area. Then there's one more formula that is Kz is equals to under root of [00:06:13.508] Kx squar plus Ky squared. This formula we are going to use when we know the radius of variation about x axis

and radius of variation about y-axis. Here, since we don't know the radius of variation about x-axis and y axis, we are going to use the first formula. [00:06:30.976] That is Kz is equals to under root of IZ by A. Next, as I said before, iz is the polar moment of inertia. Since here we are talking about point O, that polar moment of inertia should be about the point O. [00:06:48.452] Next, how to find polar moment of inertia of the shaded area about point about this point. [00:06:56.836] Here, we want only the shaded area. So, what we can do here is just find the polar moment of inertia of this bigger circle about 0.4, then detect [00:07:10.320] Polar moment of inertia of this smaller circle about the point O. Then we are going to get the polar moment of inertia of the shaded area. Then find only the shaded area, then use the formula iz by A. Let's get into the solution. [00:07:31.140] We need to find Kz, that is polar radius of garation of the shaded area about 0.0. So I'm going to write that as Kz. That is polar radius of garration about point. [00:07:50.500] That is equals to under root of I z that is polar moment of inertia about the point. [00:08:06.212] [ASR low confidence; check audio] Mm. [00:08:09.252] Divided by total area capital E. Then how to find this polar moment of inertia that is Iz about point four. Polar moment of inertia about 0.4 will be equals to [00:08:27.140] I said about [00:08:31.460] O will be equals to polar moment of inertia of a bigger circle that is 200 mm radius circle about point O minus [00:08:49.988] Polar moment of inertia of a smaller circle that is 100 mm circle about the same point. [00:09:04.068] Next, what is polar moment of inertia of this bigger circle? That is 200 mm circle, about 0. If you observe the figure carefully, that o itself is the center of the bigger circle. So [00:09:19.184] Polar moment of inertia about that point O will be equals to I bar X plus I bar Y. So I z our polar moment of inertia of 200 mm radius circle about O will be equals to [00:09:40.484] i bar x plus i bar y. i bar x is pi r to the power of 4 pi. [00:09:49.924] 4 plus I bar y is pi r to the power of 4 by 4. And what is radius? Radius is 200. So this will be pi into

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.120] 200 to the power of 4 by 4 plus pi into 200 to the power of 4 by 4. Once we simplify this, we are going to get the value as [00:10:12.004] 8 pi into 10 power 8 mm to the power of 4. This is the polar moment of inertia of a bigger circle about the point. So we are done with the first term. We need to find next. [00:10:29.124] Polar moment of inertia of a smaller circle above the point O, that is Iz. [00:10:37.764] I said of 100 mm circle about point O. This will be equals to here. Do we have a standard formula for polar moment of inertia about point O? [00:10:53.636] No, we don't have standard formula, so we have to apply the parallel axis theorem. Remember, one of the axis is passing through O, that is Z-axis passing through O, and one more Z-axis is passing through C. The Z-axis passing through C is the centroidal z-axis of the smaller circle. [00:11:10.976] So inertia about any line is equal to inertia about parallel centroidal axis that is i bar z plus area into distance square. Then what is i bar z? We know for a circle i bar z is pi r to the power of 4 by 4 plus pi r to the power of 4 by 4. [00:11:30.404] So this is the value of I bar z. Then plus area we have to consider here area is phi r square. [00:11:40.452] Then what is distance? Distance is the distance between O and C, that is 15. So 15 whole square. So if you can't see here, this is 15 whole square. Next. [00:11:57.700] Here, r is 100, so pi into 100 to the power of 4 by 4, plus pi into 100 to the power of 4 by 4, plus pi into 100 square into 50 square. [00:12:11.172] Just use the calculator and find the value. We are going to get this as these two terms will be equal to 15 pi into 10 power 6. [00:12:24.516] Then plus pi r square, that is pi into hundred square into fifty square, will get it as twenty-five pi. [00:12:32.868] into 10 power 6. Then add these two, it will be 75 pi into 10 power 6

mm to the power of 4. This is the moment of inertia of a smaller circle about the point O. [00:12:50.340] Next, I said of the shaded area. [00:13:02.308] About wo [00:13:10.180] This will be equal to moment of inertia of the bigger circle, that is 8 pi into 10 power 8 minus moment of inertia of the smaller circle, that is 57 pi into 10 po 6. [00:13:26.628] So just deduct this from this, we are going to get the value of I bar z. So that will be equals to [00:13:44.932] twenty two point [00:13:48.548] Seven, seven, five into ten per eight, mm to the power of four. [00:13:56.260] So we got the numerator in this KZ formula. Next, we should know the value of area. Here, area is area of the shaded region. That area will be equals to A is equals to [00:14:11.684] Pi r square minus pi r square. So pi, I'll take it as common. R square is 200 square minus 100 square. So once this we are going to get the value of a as a is equal to [00:14:32.292] 300 00 into 5 and it is mm square. Now we got both the numerator and denominator. So substitute here I z 22.775 into 10 power. [00:14:49.056] eight and a as thirty thousand into five and solve it we are going to get the value of K said as

EV0000050 · EV0000051 · EV0000052 · EV0000053

Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.060] One five five point four five millimeter. This is the radius of variation of the shaded area about the point O. [00:15:21.956] That's it for the today's class. Try to solve the problems given in the assignment as well as in the question bank. If you come across any queries, you can post your queries in PESO forum. Thank you.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Full transcript: 1845 normalized words, 67 timestamped segments, 934.421333 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:08:06.212

EV0000042 · EV0000043 · EV0000044 · EV0000045

# 6 CLASS 41 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 41 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

## Graphs and diagrams

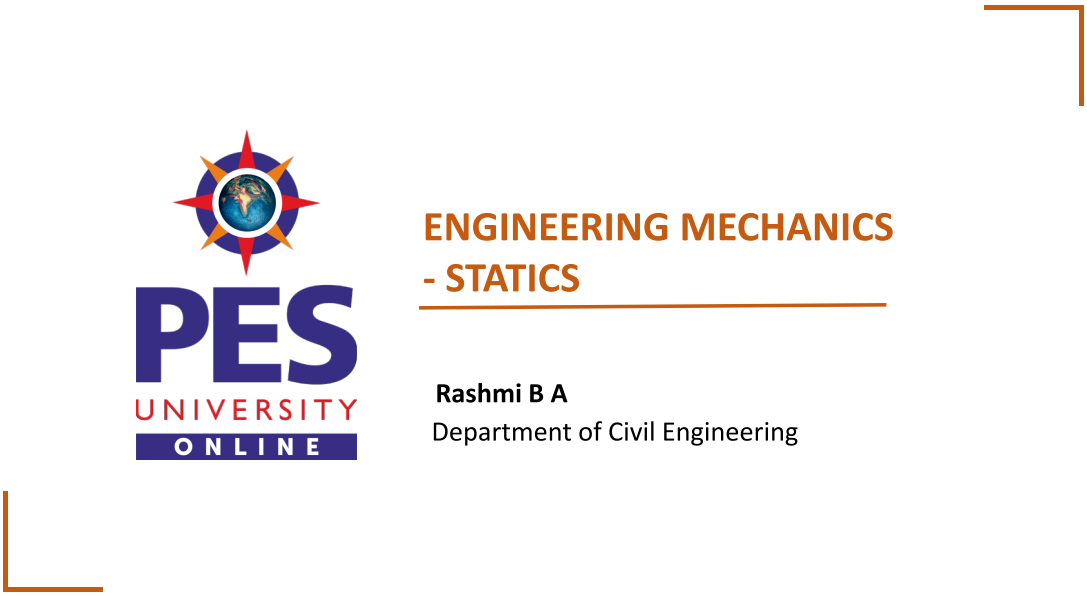


**Figure G000079.** UE20CV101 UNIT 4 SESSION 10 · source slide page 1.

EV0000421

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS - STATICS

DISTRIBUTED FORCES

Session- 10

**Rashmi B A**
Department of Civil Engineering

**Figure G000080.** UE20CV101 UNIT 4 SESSION 10 · source slide page 2.

EV0000424

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ENGINEERING MECHANICS - STATICS

Moment of Inertia: Numerical

**Problem A/42** Calculate the moment of inertia of the shaded area about the x-axis.

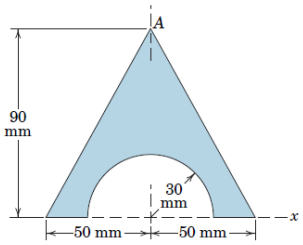


**Figure G000081.** UE20CV101 UNIT 4 SESSION 10 · source slide page 3.

EV0000427

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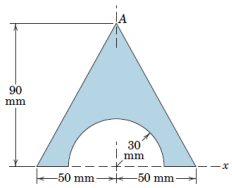
ENGINEERING MECHANICS - STATICS

Moment of Inertia: Numerical

**SOLUTION:**

***Moment of inertia of the shaded area about the x-axis:***




$$I_x = \begin{pmatrix} MI\ of\ a\ triangle\ about \\ x-Axis \end{pmatrix} - \begin{pmatrix} MI\ of\ a\ Semicircle \\ about\ x-Axis \end{pmatrix} ----- (1)$$

$$MI\ of\ a\ triangle\ about\ x-axis = \frac{bh^3}{12}$$

Here,

$$b = 10\ mm\ \&\ h = 90\ mm$$

$$MI\ of\ a\ semicircle\ about\ x-axis = \frac{\pi R^4}{8}$$

Here, $R = 30\ mm$

Substituting in equation (1)

$$I_x = \left(\frac{100\ (90)^3}{12}\right) - \frac{\pi(30)^4}{8} = 5.756\ X\ 10^6\ mm^4$$

**Figure G000082.** UE20CV101 UNIT 4 SESSION 10 · source slide page 4.

EV0000430

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ENGINEERING MECHANICS - STATICS

Moment of Inertia: Numerical

**Problem A/44.** The variable h designates the arbitrary vertical location of the center of the circular cutout within the semicircular area. Determine the area moment of inertia about the x-axis for (a) h =0 and (b) h = R/2.

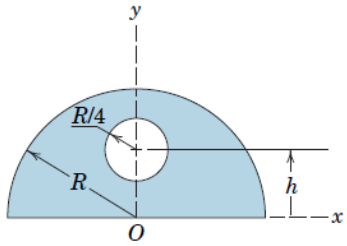


**Figure G000083.** UE20CV101 UNIT 4 SESSION 10 · source slide page 5.

EV0000433

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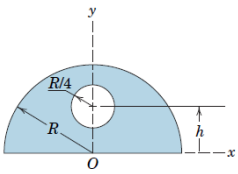
ENGINEERING MECHANICS - STATICS
Moment of Inertia: Numerical

**SOLUTION:**




Case (a) h=0
Moment of Inertia of shaded area about x-axis=
$I_x$ =(MI of a semicircle about x-axis) – (MI of a circle about x-axis)

$$I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi\left(\frac{R}{4}\right)^4}{8}\right)$$

$$I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi (R)^4}{(4^4)(8)}\right)$$

$$\boxed{I_x = 0.3911R^4}$$

**Figure G000084.** UE20CV101 UNIT 4 SESSION 10 · source slide page 6.

EV0000436

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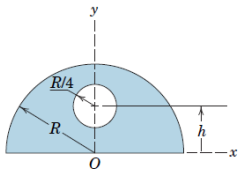
ENGINEERING MECHANICS - STATICS
Moment of Inertia: Numerical




Case (b) h= R/2
Moment of Inertia of shaded area about x-axis=
$I_x$ =(MI of a semicircle about x-axis) – (MI of a circle about x-axis)

$$I_x = \left(\frac{\pi R^4}{8}\right) - (\bar{I} + Ad^2)$$

$$I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi R^4}{4} + \left(\frac{\pi R^2}{2}\right)h^2\right)$$

$$I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi\left(\frac{R}{4}\right)^4}{4} + \left(\frac{\pi\left(\frac{R}{4}\right)^2}{2}\right)\left(\frac{R}{2}\right)^2\right)$$

$$I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi R^4}{4^5} + \left(\frac{\pi(R)^2}{(4^2)2}\right)\left(\frac{R}{2}\right)^2\right)$$

$$\boxed{I_x = \left(\frac{\pi R^4}{8}\right) - \left(\frac{\pi\ 17\ R^4}{1024}\right) = 0.341R^4}$$

**Figure G000085.** UE20CV101 UNIT 4 SESSION 10 · source slide page 7.

EV0000439

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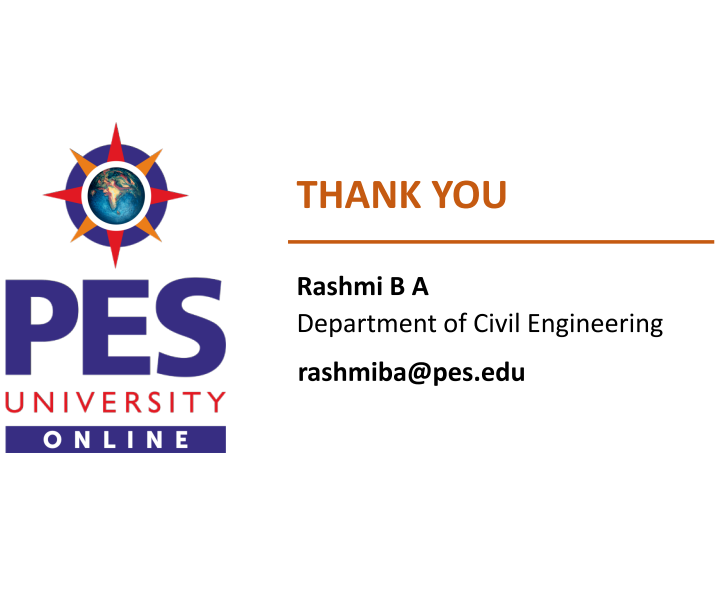


**Figure G000086.** UE20CV101 UNIT 4 SESSION 10 · source slide page 8.

EV0000442

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000420 · EV0000421 · EV0000422
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 10 Rashmi B A Department of Civil Engineering EV0000423 · EV0000424 · EV0000425
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/42 Calculate the moment of inertia of the shaded area about the x-axis. EV0000426 · EV0000427 · EV0000428
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Moment of inertia of the shaded area about the x-axis: $MIofatriangleabout\ MIofaSemicircle\ Ix$= - −−−−−(1) $x-Axis\ aboutx-Axis\ bh3\ MIofatriangleaboutx-axis$= 12 Here, $b$= 10 $mm$& $h$= 90 $mm\ \pi R4\ MIofasemicircleaboutx-axis$= 8 Here, R = 30 $mm$ Substituting in equation (1) 100 (90)3 $\pi$(30)4 $Ix$= - = 5.756 $X$106 $mm$4 12 8 EV0000429 · EV0000430 · EV0000431
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/44. The variable h designates the arbitrary vertical location of the center of the circular cutout within the semicircular area. Determine the area moment of inertia about the x-axis for (a) h =0 and (b) h = R/2. EV0000432 · EV0000433 · EV0000434

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Case (a) h=0 Moment of Inertia of shaded area about x-axis= $Ix$=(MI of a semicircle about x-axis) – (MI of a circle about x-axis) 4 $R$ $\pi$ $\pi R4$ 4 $Ix$= − 8 8 $\pi R4$ $\pi R4$ $Ix$= − 8 (44)(8) $Ix$= $0.3911R4$ EV0000435 · EV0000436 · EV0000437
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Case (b) h= R/2 Moment of Inertia of shaded area about x-axis= $Ix$=(MI of a semicircle about x-axis) – (MI of a circle about x-axis) $\pi R4$ $Ix$= − ıȷ$I$+ $Ad2$ 8 $\pi R4$ $-\pi R4$ + $\pi R2$ $h2$ $Ix$= 8 4 2 4 2 2 $\pi R$ $\pi R$ $\pi R4$ 4 4 $R$ $Ix$= − + 8 4 2 2 2 $\pi R4$ $\pi R4$ $\pi R2$ $R$ $Ix$= − + 8 45 (42)2 2 $\pi R4$ $-\pi 17$ $R4$ = $0.341R4$ $Ix$= 8 1024 EV0000438 · EV0000439 · EV0000440
8. Source slide page 8 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000441 · EV0000442 · EV0000443

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.996] Hello, in today's session let's solve few problems and moment of inertia. [00:00:07.492] This is the question A slash 42. Calculate the moment of inertia of the shaded area about the x axis. We have been asked to find moment of inertia of only this shaded area about the x-axis. [00:00:25.572] How to find inertia of this shaded area about the x-axis? What we can do here is we can find moment of inertia of this triangle about the x-axis, then in that we can detect the moment of inertia of this semicircle about the same x-axis. [00:00:45.008] Then we are going to get moment of inertia of this remaining shaded area. Let's get into the solution. [00:00:53.540] So Ix is equals to that is moment of inertia of the shaded area about x axis is equals to moment of inertia of a triangle. [00:01:11.172] About x axis [00:01:17.380] In this, we have to deduct minus moment of inertia of a semicircle. [00:01:28.836] Of a semi circle about the same x axis. Next, what is moment of inertia effect triangle about x axis? [00:01:45.476] Moment of inertia of a triangle about x axis, we know from basic formulas. That is equal to, I am going to call that as Ix1. That is moment of inertia of a triangle about x axis. That is equals to bh cube by [00:02:02.596] Twelve. This is from the basic formulas we know. Here, what is B? B is the base of the triangle that is hundred. [00:02:13.060] into H. What is H? H is height of a triangle that is 90. So 90 whole cube by 12. Use the calculator, then find this value [00:02:27.300] That is six point zero seven five into ten per six. [00:02:35.268] mm to the power of 4. This is the moment of inertia of a triangle about its base, that is about x-axis. Next we need to find the inertia of a semicircle about the same x-axis that I call it as Ix2. [00:02:52.100] This will be equal to Since the X axis is passing through the [00:02:56.644] Is nothing but the diameter axis. We have a direct formula. Moment of inertia of a semicircle about its diameter axis is pi r to the power of 4 by 8. H r is 30. So pi into 30. [00:03:13.440] to the power of four by eight. Just the calculator find this value. This is zero point three one eight into ten power six mm to the power of four. [00:03:31.044] Now, substitute these two values here. So Ix is equal to [00:03:38.692] 6.075 into 10 power 6. This is the moment of inertia of a triangle about x axis minus moment of inertia of this semicircle about the x axis is 0.318 into 10 po 6. [00:03:58.532] Once we solve this, we are going to get as 5.756 into 10 bar 6 mm to the power of [00:04:09.124] 4. This is the moment of inertia of the shaded area about the x axis. This is the final answer. X question A slash 44. [00:04:25.476] The variable h designates the arbitrary vertical location of the center of a circular cutout within the semicircular area. Determine the area moment of inertia about the x-axis for A H is equals to 0 and when H is equals to R by 2. [00:04:45.732] In the first case, before that,

look at this, there is a semicircle of radius capital R. [00:04:54.404] And there is this circular cutout. When the position of the center of this circular cutout is h is equal to 0. That is when the center of the circle is coinciding with the origin O, in that case, what is the moment of inertia of the

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.928] Shaded area, nothing but what is the moment of inertia of the remaining area we have to calculate. Then in the second case, when H that is the position of the center of this circular cutout is R by 2, what is the moment of inertia of this? [00:05:26.256] Remaining area about the x axis we have to calculate. Let's get into the solution. Let's start with the first case that is when h is equal to 0. So case A. [00:05:43.460] when h is equal to 0. When h is equals to 0, how will be the figure? Let's start with the semicircle. So this will be the radial line, nothing but this is x axis, then the semicircle will be like this. [00:06:02.372] Next, we should consider a circle. Since h is equal to 0, the half of the circle is going to be here. [00:06:10.916] And the remaining half it's going to be outside this semicircle. The part which is outside the semicircle that doesn't matter to us because we are considering moment of inertia of the remaining area within this semicircle. So this is [00:06:26.628] The radius of this semicircle is capital R. Then the radius of this circle is R by 4. So this is R by 4. Then we are considering only this shaded. [00:06:42.192] Area. You need to find moment of inertia of only this shaded area about the x-axis. That will be equal to moment of inertia of this bigger semicircle minus moment of inertia of this smaller semicircle about the x-axis. [00:06:57.184] So let's start with that. So Ix. Ix is nothing but moment of inertia of the shaded area about x-axis is equal to moment of inertia of a semicircle. [00:07:13.476] About X axis. [00:07:20.580] This is because the circle minus moment of inertia of a circle about the same X axis. [00:07:36.292] Next. What is this moment of inertia of a circle, semicircle about this x ax? We know moment of inertia about its diameter axis is pi r to the power of 4 by 8. So consider that. Then [00:07:53.508] N R is capital R minus. [00:07:57.760] Moment of initial circle, since as I said before, half of the circle is outside this semicircular area. So that doesn't matter. Now instead of circle, we have to consider it as a semicircle. So the formula is again remains the same pi r to the power of 4 by 8. [00:08:15.632] Because this x axis is his diameter axis even for the smaller semicircle. That's why. Now here for this r is r itself. So pi into r to the power of 4 by 8, then minus here for [00:08:34.436] R is radius of the smaller circle that is R by 4 to the power of 4 by 8. [00:08:46.468] Now use the calculator and find out this will be pi r to the power of 4 divided by 4 to the power of 4 into 8. Then deduct that from this, we are going to get the value of IS. So this is point. [00:09:03.664] 3911 into r to the power of 4. This is the moment of inertia of the shaded area about the x axis, and this is when h is equal to 0. [00:09:20.612] This is the answer when H is equals to 0. Now we have to find moment of inertia when H is equal to R by 2. So this is case B. [00:09:37.764] when h is equal to r by 2. Let's start with the figure. This is the semicircle of radius r [00:09:51.044] This is the center of the semicircle and this is the axis of symmetry y axis. Now the center of the circle is at a distance of h that is r by 2. So the circle is going to be here.

EV0000062 · EV0000063 · EV0000064 · EV0000065

Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.948] This is the center of the circle. Now we have to find moment of inertia of this remaining area. [00:10:15.652] about the x axis. Then how to find that? For that, we have to consider moment of inertia of this semicircle about x axis. In that we have to detect moment of inertia of this circle about the same

[00:10:32.528] x axis. Then only we are going to get moment of inertia of this shaded area about the x axis. Let's start with this solution. So Ix is equals to [00:10:46.596] Moment of inertia of a circ semicircle, moment of inertia of a semicircle. [00:10:57.220] about X axis minus [00:11:05.956] Moment of inertia I'll write here itself moment of inertia of a circle about the same x axis. I'll call it this one as I x1 and I'll call this as I x2. [00:11:25.156] Then how to find moment of inertia of a semicircle about the x axis that is i x1 that will be equal to. Since the x axis itself is the diameter axis for the semicircle, moment of inertia we have a direct formula that is I r to the power of 4 by [00:11:44.292] 8. Consider that. Yet what is R? R is R itself. So the formula is going to be pi R to the power of 4 by 8 itself. This is the moment of inertia of a semicircle above the [00:11:57.184] X-axis. Next, Ix2. This is the moment of inertia of a circle about this line. Since we don't have a standard formula, we have to apply the parallel axis theorem. [00:12:09.252] Where will be the centroid of this circle? Centroidal x-axis of this circle. This is the center of the circle. Draw a horizontal line passing through this. This will be the centroidal x-axis of a circle. Now. [00:12:23.392] We are talking about moment of inertia of this circle about this axis. So we have to apply parallel axis theorem. Inertia about any line is equals to inertia about parallel centroidal axis plus area into distance square. So here centroidal axis is I bar s plus area. [00:12:43.492] into distance square. What is I bar x per circle? I bar x per circle is pi r to the power of 4 by 4 plus area. That is area of a circle. That is pi r square. [00:13:00.932] into D. D stands for the distance between the two axes. Here one axis is X and one more axis is X0. So the distance between these two is H. That is R by 2. So R by 2 square. [00:13:19.524] Just substitute here pi into r r is nothing but the radius of the circle. What is the radius of the circle? It is given in the figure. Look at the figure, it is r by 4. So it is r by 4 to the power of 4 by 4. [00:13:39.172] Plus pi into r square is r is radius of the circle. That is r by 4. Square into distance is r itself. This is r. So r pi 2 whole square. Now just find out this value. [00:13:59.140] We are going to get this as five. [00:14:07.172] 17 pi r pi into 17 r to the power of 4 divided by 1024. This is the moment of inertia of the circle about the x axis. Now substitute here in this. [00:14:26.212] So moment of inertia of a circle about x axis is pi r to the power of 4 pi 8 minus moment of inertia of a circle about x axis is 17. [00:14:40.644] pi into r to the power of four divided by one zero to four. [00:14:48.068] Now just deduct this from this one. We are going to get this as 0.341 into R to the power of 4. This is the moment of inertia of the remaining.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.868] area about the centroidal, sorry, about the x axis which is passing through the diameter axis. This is the value ix is equal to 0.341 into r to the power of 4. That's it for today's class. Thank you

EV0000070 · EV0000071 · EV0000072 · EV0000073

Full transcript: 1931 normalized words, 64 timestamped segments, 924.672 seconds duration. Low-confidence segments flagged: 0.

EV0000058 · EV0000059 · EV0000060 · EV0000061

# 7 CLASS 51.1 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 51.1 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000074 · EV0000075

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello. In today's session, let's solve few problems and moment of inertia. [00:00:08.836] Cushion A slash forty-four. Determine the moments of inertia of the Z section about its centroidal x axis and centroidal y axis. Look at this Z section. It's given this way like [00:00:24.356] The width of this member is 100, and the depth is 40, the width of this web is 20, and the height will be from the bottom most member till here it is 140. Then there is one more rectangular section here. We have been asked to. [00:00:39.844] Find the moment of inertia this Z section about the centroidal X and centroidal y axis. Then where will be the centroidal x axis? Due to symmetry, we can say that. [00:00:53.540] The centroidal x axis will be at a distance of 140 plus 20 divided by 2. That will be equal to [00:01:03.876] 80 mm with respect to the bottommost member. So 80 will be the distance of this x naught axis with respect to this member. [00:01:13.824] Then where will be this y naught axis? Y naught axis will be exactly at the center, again due to symmetry, that will be equal to from ear to ear it will be 100. From year to ear it is 100. So it will be exactly at the center, nothing but it will be at a distance of 90. [00:01:30.608] R in one more way, hundred minus half of this width, that will be ten. Twenty by two will be ten. So hundred minus ten will give us [00:01:41.028] 90. So with respect to this member, y naught will be at a distance of 90. With respect to this member also, it will be at a distance of 90. Once we know the centroidal x axis and centroidal y axis. [00:01:55.440] Next, we can find moment of inertia of this Z section about the centroidal x axis and centroidal y axis. Let's get into the solution part. [00:02:06.980] As I said before, the coordinates of the centroid are if you consider this bottommost member as x-axis and the leftmost member as y-axis, then the coordinates of this centroid will be 90, uh, 80. [00:02:21.280] Now if you draw a horizontal line passing through this, this will be the central. [00:02:27.232] X axis. If you draw a vertical line passing through this, this will be the centroidal y-axis. Now we need to find moment of inertia of this z section about the centroidal x axis and centroidal y-axis. First. [00:02:43.632] Let's start with the moment of inertia about the centroidal x axis that is Ix naught. Then how to find moment of inertia of this satisfaction about this axis? We know the moment of inertia formula for rectangles. So here what we can do here is [00:03:00.768] We can just divide the given z section into three rectangles, then find the moment of inertia of the each rectangle about this axis and sum it up that will be the moment of inertia of the entire z section above the centroidal. [00:03:18.208] X-axis. So how to divide this? Either you can divide it either by drawing a horizontal line or by drawing a vertical line. So I'm going to draw a vertical line here. This is the first rectangle, and here also I'm going to draw a vertical line. So this is the second rectangle. I'll call it as 2. [00:03:38.084] Then this is the third rectangle. Now moment of inertia of this z section about centroidal x axis will be equal to moment of inertia of the first rectangle about centroidal x axis plus [00:03:53.456] Moment

of inertia of the second rectangle about the centroidal x-axis plus moment of inertia of the third rectangle about the centroidal axis. [00:04:02.768] X axis. Now I'm going to consider each rectangle and I am going to find moment of inertia of each rectangle above the centroid x axis. I am going to start with the first rectangle. [00:04:18.596] The first rectangle is like this. [00:04:23.524] Here, what is this width of the rectangle? It is not 100 because this part is included in the second rectangle. So the width is 100 minus 20. So this is 80. This is B. 80. [00:04:39.136] Into what is the depth of the rectangle? 20. Now, where is the centroidal x axis of the z section? The centroidal x axis of the z section is here, x0. And it is at a distance of [00:04:54.660] This is if you we have considered this is the x-axis. With respect to x axis, this is at a distance of

EV0000074 · EV0000075 · EV0000076 · EV0000077

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.228] We know y bar is 80, so 80 at a distance of 80, it is there. Then, how to find inertia of this rectangle about the centroidal x-axis of the z section? We don't have a standard formula, so we have to apply the parallel axis theorem. To apply the parallaxis theorem, we have to [00:05:23.876] Look at the centroid of this rectangle. [00:05:26.852] The center of the rectangle C is here, then what will be this distance? This will be 10. That is y bar is equal to h by 2, h is 20, 20 by 2, we will get as 10. Then from ear to ear distance we know that is 140. So consider that 10 [00:05:44.144] 40. Now if you draw a horizontal line passing through the centroid of this particular rectangle, this will be the centroidal x-axis of this particular rectangle. Whereas this is the centroidal axis of the z section. [00:05:58.864] With respect to this rectangle, this is any other axis. So moment of inertia about any other axis, that is ix1 is equal to moment of inertia about balance centroidal axis plus area into distance square. What is I bar x formula for rectangle? That is bh cube by [00:06:18.020] 12 plus area is B into H into distances, the distance between the centroidal axis of the rectangle and the centroidal x axis of the Z section. This distance we want that will be equal to 140 minus 80, we will get this distance. [00:06:37.412] 140 minus 80, then plus 10. So distance will be that. So here we have to consider the distance. Distance squared. Now substitute the value of that. So b is equals to [00:06:57.572] I'll write it here Bs eight into H s twenty B H cube by twelve. [00:07:09.572] Plus area we have to consider. Area is B into H, that is 80 into 20 into distance square. Distance as I said before, that is 140 minus 80. That is 60, 60 plus 10, 70 square. Now. [00:07:30.052] Once we solve this, we are going to get the value as 7, 8, 9, 3, 3, 3, 3 mm to the board of [00:07:41.444] 4. This is the moment of inertia of the first rectangle about the central x-axis of the z section. And one more, if you look at this. [00:07:54.660] The second rectangle uh forget about the second rectangle for time being. If you look at the third rectangle, the centroid of the third rectangle will be here, and the distance from here to here will be 10. [00:08:08.068] 10 from year to year it is 80. 80 minus 10 will give us 70. So the centroid is at the distance of 70 year, and these two are identical. So due to that I can write moment of inertia of the third rectangle about [00:08:25.764] The centroidal x-axis of the z section will be equal to Ix. One moment of inertia the first rectangle about the centroidal z centroidal axis of the z section. [00:08:39.428] Then we are left with only this rectangle. If you consider this rectangle, where will be the centroid of the rectangle? That will be equal to 140 plus 20. That is the height of this rectangle. That will be equals to [00:08:53.604] 160. So 160 by 2 we will get 80. So the centroid of the Z section itself is the centroid of the rectangle also. Both the centroid are coinciding with each other. So no need to apply the parallaxis theorem. [00:09:10.768] The centroidal x axis of the z section and the centroidal axis of this second rectangle are coinciding. So I can write Ix to us. We know the standard formula that is BH cube by 12. So B

here is 20. [00:09:31.204] Because this is 20 and h here is 140 plus 20. That is 160 cube divided by 12. [00:09:42.276] And we'll get the answer as six eight two six six six six mm to the power of four. [00:09:52.132] Now we got the moment of inertia of the first rectangle, second rectangle and third rectangle about the centroidal x axis of the z section. So some uh just sum it up like Ix naught is equal to first piece Ix1 plus Ix2.

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.236] Plus IX 3. If we add 1, then 2, plus again this, we are going to get the value of momentum inertia of the z section about the central x axis, that is 2261. [00:10:29.092] 333 2 into mm to the power of 4. This is the moment of inertia of the z section about the centroidal x axis. [00:10:44.356] Next, we need to find the moment of inertia of the same z section about its central y-axis. Here also we are going to apply the same concept. [00:10:53.648] Moment of inertia of the Z section about its centroidal y axis will be equal to sum of the moment of inertia of these three rectangles about the same centroidal y axis. So first I'm going to start with the first so I [00:11:12.196] I y0 will be equal to I1. This is moment of inertia of the first rectangle plus I2, moment of inertia of the second rectangle plus I 3 moment of inertia of the third rectangle. Let's start with the first rectangle. [00:11:29.088] So this is the rectangle, first rectangle, it's like this. Then here the width of the rectangle is 100. Then the height of the rectangle is 20. So consider that 20. Next, where is the centroidal y-axis? The centroidal y axis is here. [00:11:49.732] This is the y axis we have assumed in the figure. This is y axis, and this is the central y axis. Then the distance between these two is [00:12:03.620] Sorry, this is not hundred because this part is included in the second rectangle. So this is eighteen. And this is ninety. [00:12:16.260] Because from year to year it is 80. 80 plus 10 is 90. That is what X bar value. So 90. Next, how to find inertia? [00:12:27.460] This rectangle about this axis. Since we don't have a standard formula, we have to apply the parallel axis theorem. Since the axis is vertical here, we should consider the centroidal y-axis of this particular. [00:12:40.496] rectangle. Where will be the centroidal y-axis of this particular rectangle? At a distance of x bar. So what is the value of x bar? X bar will be equal to b by 2 that is 80 by 2 will be 40. [00:12:54.436] Now, this is the centroidal y-axis of the rectangle, whereas this is any other axis with respect to this rectangle. So we have to apply the parallel axis theorem. [00:13:07.296] Let's start with the statement of parallel theorem. Inertia about any line will be equal to inertia about parallel centroidal axis plus area into distance square. This is the statement. [00:13:20.324] Next, what is I bar Y? I bar Y formula for rectangle we know HB cube by twelve. [00:13:27.780] Plus area is B into H. Into distance. Distance is the distance between this centroidal y-axis of the rectangle and the y-nut axis. That will be equal to total 90, 90 minus 40. [00:13:43.840] That will give us the distance between these two axes. So 90 minus 40 will be distance. So I'm going to write it here once again. So h is equal to 20, then p is equal to 80 cube by 12. [00:14:03.524] Plus B is 20 and H is 80 into distances 90 minus 40, that is 50. [00:14:19.684] Square. Now just find out the value. We'll get this as four, eight, five. [00:14:33.124] 3333 point three three mm to the power of four. This is the moment of inertia of the first rectangle about the centroid and y-axis of the z section. [00:14:48.324] Okay, now again here also we can apply the concept of symmetry because this centroidal y-axis is exactly at the same distance with respect to rectangle 1 and 3, and these two are identical, 1 and 3. So I can write IY3 will be equal to IY.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.420] Next, we are left with the second rectangle. [00:15:13.540] Again, it's the same case now also. The centroid of this Z section itself is the centroid of this rectangle. That's the reason the centroidal y-axis of the Z section itself is the centroidal y-axis of the rectangle. [00:15:30.720] So i y2 will be equal to we know the formula that is h b cubed by 12. Consider that formula. What is h here? H is height of the rectangle that is 140 plus 20, that is 160 into [00:15:47.920] B's width of the rectangle 20 whole cube by 12. So 160 into 20 cube by 12. If we simplify we are going to get it as 10. [00:16:03.780] 66 66.67 mm to the power of 4. This is the moment of initiative the second rectangle about the centroid and y axis. Now we need to find the moment of inertia of the entire z section. [00:16:19.968] That is equals to moment of inertia of the first rectangle plus moment of inertia of the second rectangle plus moment of inertia of the third rectangle about the same central y axis. That is equals to it is 981. [00:16:37.028] 333 3 into mm to the power of 4. This is the moment of inertia of the z section about the central y axis. [00:16:59.364] Uh try to solve the problems given in the assignment. If you come across any doubts, you can post your queries in the PESO forum. Thank you.

EV0000086 · EV0000087 · EV0000088 · EV0000089

Full transcript: 2357 normalized words, 71 timestamped segments, 1029.184 seconds duration. Low-confidence segments flagged: 0.

EV0000074 · EV0000075 · EV0000076 · EV0000077

# 8 CLASS 51.2 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 51.2 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.388] Hello. In today's session, let's solve few more problems and momentum inertia. This is question A slash 60. Calculate the polar radius of garage of the shaded area about its centroid C. If you look at this figure, [00:00:17.328] They have asked us to find polar radius of gradation, that is case about its centroid C. So this is the centroid C, but do we know the coordinates of the centroid? [00:00:29.552] We don't know the coordinates of the centra, so the first step is we need to locate the coordinates of the centra here in this case. [00:00:39.492] Here first, I am going to divide this into elemental areas. I am going to consider this bottommost member as the x-axis. This is just for my reference, and the leftmost member is [00:00:52.836] Y axis. Now the entire figure is in the first coordinate. First, I need to find the coordinates of the centroid C. So we know the coordinates of the centroid C are always X bar, comma, Y bar. Since this is Y axis, this is going to be X bar. [00:01:11.812] And this is will be y bar. So I should know the value of X bar and Y bar. First, do we have the axis of symmetry here? [00:01:23.524] Yes, vertical line. If I draw a vertical line here passing through this,

this right of the section is the mirror image of left of the section. So where should be that line at a distance of two fifty. Total is [00:01:35.876] 500. So just draw a line at a distance of 250 vertical line. So that will be the axis of symmetry. So no need to find the value of x bar. So x bar can be directly written as x bar is equals to [00:01:51.268] Bar is equal to 500 by 2, that will be 250 mm. Next, we need to know the value of y bar. To find y bar, we have a formula. Y bar is equals to summation of a i yi by total area. [00:02:08.144] To apply this formula, we have to divide this into elemental areas. Then how to divide this into elemental areas? I am going to draw a horizontal line here and here also. So instead of C section, we have three rectangles. Let it be first area. [00:02:24.592] Let it be second area and let it be third area. Now [00:02:29.924] We know here the formula is going to be a1 by 1 plus a2 by 2 plus a3 y 3 divided by total area. To find y bar, let's start with the first one. A1 is equals to A1 stands for area of the first rectangle. So A1 is equals to [00:02:50.436] Base into height, since it is a rectangle, B into H. B we know 100 into H. What is H? H is from ear to ear distance. Since total is 600, total is 600. 600 minus this is 100, even in the figure. [00:03:10.212] So 600 minus 100 will give us 500. So from here to here it is 500. So A1 is equal to B is 100 into H is 500. [00:03:24.836] This will be fine to 10 power 4 mm to the power of mm square. Next. After this we should do the value of y 1. Y1 is [00:03:37.188] This cent this rectangle will have it, this rectangle will have a centroid somewhere here that is C1. The y coordinate of this C1 will be equal to 100 plus H by 2 because from here to the base of the rectangle is [00:03:52.288] H by two. So y one is equal to 100 plus h by two. What is h here? H is five hundred. So five hundred by two. So this will be two fifty, two fifty plus hundred will be three fifty. Next is a two. [00:04:09.488] A2 is area of the second rectangle. This is the second rectangle. So A2 is B into H, B is 500 into H is 100. Again it is 5 into 10 power 4 mm square. Next, Y2. [00:04:26.816] Y2 is this rectangle will have a centroid C2 here, then this y coordinate of this is y2. So that will be equal to H by 2. So H is 100. So 100 by 2 is 15 millimeter. Next, due to [00:04:44.580] These two are identical, like the same dimensions, they are at the same distance with respect to x. A3 will be equal to a1 and y3 will be equal to y1. If you want to calculate, you'll get the same value. Then substitute ER in y bar. So y bar will be equals to

EV0000090 · EV0000091 · EV0000092 · EV0000093

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.420] A one is phi into ten power four into y one is three fifteen plus a two is phi into ten pour into y two is fifteen plus [00:05:19.908] A3 is A1 and Y3 is Y1. So I can write it as this itself. 2 times set of this divided by total area is A1 plus A2 again plus A1. So once we do this, we will get the value of Y bar. So Y bar, we will get it as [00:05:39.908] 350. Sorry, y we will get it as two fifty mm. So two fifty mm. This is the y coordinate of the centroid. Next we need to find moment of inertia of this. [00:05:58.916] given figure about its centroidal x-axis. This is the centroidal x-axis. It is at a distance of y bar that is 250 with respect to bottommost member. So with respect to this, it is at a distance of 250. Now [00:06:14.640] How to find inertia of this C section about this centroidal x axis, which is at a distance of two fifty with respect to bottom most member. What we can do here is since we don't have standard formula, moment of inertia of this lamina about its centroidal x axis will be equal to [00:06:32.368] So, sum of the moment of inertia of the first rectangle, second rectangle, and the third rectangle about the same centroidal x-axis of this lamina. So I can write it as inertia about I bar X or I X naught one and the same. [00:06:49.504] Will be equals to inertia of the first one plus second one plus third one about the same x0 axis. Now let's start with the moment of inertia of the rectangle one about x0 axis one. [00:07:09.604] About snap axis

[00:07:16.708] We know moment of initial about any line will be equals to moment of initial parallel central dial axis plus area into distance square. For time being forget about this second rectangle and the third rectangle. Just consider this rectangle. It's like this. Then [00:07:36.228] The x-axis is here. Now the centroidal x axis of the lamina is here at a distance of this is x naught axis at a distance of 250 mm. Next [00:07:54.020] Where is the centroidal x-axis of this rectangle? We know that is at a distance of y1, that is 350, because from here to 100 plus h by two will be 250. So that will be equal to 350 means above this line somewhere here. [00:08:11.492] This is the centroidal x axis of this rectangle. [00:08:17.508] Please remember when I'm talking about this rectangle, this is any other axis, whereas this will be the centroidal x axis. How to find initial any axis? Inertia about any axis will be equal to inertia about paral centroidal axis plus area into distance square and this is three fifty. [00:08:39.428] Now, I bar X formula for a rectangle is BH cube by 12. So BH cube by 12 plus area is B into H into distance square. Distance is 250 minus, sorry, 350 minus [00:08:56.448] 250 is the distance between these two palms that will be equal to 100 one square. And substitute your B's width of the rectangle is 100. H is 500. [00:09:14.308] cube byte 1 plus area is area of the first rectangle already we know 5 into 10 power 4 into distances 100 square. Once we solve this we are going to get the value of this and this is equal to [00:09:33.732] 15.42 into 10 power 8 mm to the power of 4. This is the moment of inertia of only the first rectangle about the centroidal x axis of this entire lamina. Next we need to consider [00:09:51.504] Moment of inertia rectangle two about the centridal x axis. Now let's consider the rectangle two. This is the centroidal x axis of the lamina x naught, which is at a distance of

EV0000094 · EV0000095 · EV0000096 · EV0000097

Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.788] 250 with respect to the bottom most member and the rectangle is here, rectangle 2 is here, and the dimensions of this are this is 500 from the figure, and this is 100. [00:10:30.980] When we are considering this rectangle, this is any other axis with respect to this rectangle, and the central axis of this rectangle will be here, which is at a distance of 50 mm, because 100 by 2 will be 50. Now moment of initial of this rectangle, about [00:10:47.776] The central x axis will be equals to I bar x same formula I bar x plus area into distance squad. So I bar x formula is bh cube by 12 for rectangle plus area is b into h. [00:11:03.808] And distance is distance between the two parallelines, this line and the centroid and x-axis of the lamina. That will be equal to two fifty minus fifty. So two fifty minus fifty will be equals to [00:11:15.748] Two hundred whole square. Then here B is five hundred and h is hundred cube by twelve plus b into h b is five hundred and h is hundred into [00:11:34.212] 200 squared. So once we simplify this, we are going to get the values 20 point. [00:11:42.180] 42 into 10 power 8 mm to the power of 4. This is the moment of inertia of this rectangle, second one about centroidal x-axis. Again, next moment of inertia of the third one will be equal to moment of inertia of the [00:11:56.816] First one because these two are identical and they are at the same distance with respect to its axis. So I can write this will be equals to Ix naught will be equal to first one is Ix naught is equal to better I'll write it here I x naught is equals to [00:12:16.100] Moment of initial of the first rectangle is 15.42 into 10 power 8 and take it as common. Plus moment of initial of the second rectangle is 20 point four two. [00:12:30.692] Moment definition of the third rectangle is again fifteen point four two into ten power eight. So I x naught is equals to [00:12:45.028] Moment of finish of the entire lamina about its centroidal x axis is fifth fifty one point two six into ten per eight mm to the power of four. [00:13:03.364] This is the moment of inertia of the C section about the centroid and X-axis. Next, we need to find moment of inertia of the same C section about your

centroidal y-axis. [00:13:17.348] Now, moment of inertia of the C section about its centroidal y-axis, we have to find. [00:13:24.580] Again, same concept, moment of inertia of this lamina about its centroidal y axis will be equal to sum of the moment of inertia of the first rectangle, second rectangle, and the third rectangle about the same y-axis. So let's start with first one. [00:13:41.040] Moment of initial the rectangle one about its centroidal y axis of the lamina rectangle one I am going to consider. The rectangle one is like this. [00:13:57.444] Next, the centroidal y-axis is here. The distance? The centroid of the first rectangle is here, which is at a distance of 100 by 2 will be 15. From here to here it will be 15. [00:14:13.956] Next, from this line to the central y axis of the lamina is 250. So this we know. [00:14:27.556] And this is the centroidal y-axis of this particular rectangle. And this will be any other axis with respect to this rectangle. Then inertia about any line will be equal to inertia about parallel centroidal axis, that is I bar y plus area into distance square. [00:14:44.388] Now, I bar y formula for rectangle is HBQ by 12 plus area is B into H, and the distance here is the distance between these two vertical axis. That will be equal to this we know 250. So 250 minus 50 will be equal to 200.

EV0000098 · EV0000099 · EV0000100 · EV0000101

Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.076] With square. So here h is the height of the first rectangle that is 500. So 500 into B is 100 cube by 2 plus 500 into 100 is the area into distance is 200 square. [00:15:22.372] Once we simplify this, we are going to get inertia of the first rectangle about its central y-axis of the lamina, that is equal to 20.42 into 10 power 8 mm to the power of 4. [00:15:39.812] This is the moment of inertia of this section about the centroid. Sorry, moment of inertia of the first rectangle about the centroidal y-axis of the lamina. Next we should consider moment of inertia of the second rectangle about the centroidal y-axis. [00:15:54.148] Now, for the time being, forget about rectangle 1 and rectangle 2. Just consider the rectangle 3. Now, this rectangle 3 is actually symmetrical about the centridal y-axis. So centroid C2 will be on the same axis. [00:16:10.880] Since the centrate of this rectangle and the axis are coinciding with each other, no need to apply parallel theorem. Inertia of the second rectangle about this will be equals to this is directly it will be equal to HPQ by [00:16:27.716] 12. No need to apply parallaxis theorem since we have a standard formula. Here H is 100 and B here is 500 Q by 12. So HP cube by 12. Once we solve it, we are going to get the value as [00:16:46.244] 10 point four two into ten power eight. So this is equal to ten point four two into ten power eight mm to the power of [00:16:58.020] 4. This is inertia of a second rectangle about the centroidal y-axis. Next, again, these two are identical. So inertia of the third one will be equal to inertia of the first one. So inertia of the entire lamina about centroidal y-axis will be equal to [00:17:14.384] Moment of initiative the first rectangle is 20 point four two into ten per eight and that has common plus second one is ten point four two plus third one is again the first one twenty point four two [00:17:33.700] into n power eight mm to the power of four. So i y naught equals to [00:17:43.748] 51.26 into 10 power 8 mm to the power of 4. This is moment of inertia about the centroidal y axis. But the question is K Z radius of guiration about the centroidal y. [00:18:03.716] About the point C, C is the centroid here. So Kz is equals to under root of IZ by A. So I z we have to calculate I Z is equal to IX0 plus I naught. [00:18:18.724] Plus I naught. That will be equals to. We know from the previous thing, I x naught is equal to uh that is I x naught is 51 point two six plus here. I y naught is fifty-one point two six. [00:18:49.252] So this is fifty one point two six into ten power eight plus fifty one point. [00:18:56.228] 26 into 10 by 8 is I z that will be equal to 102 point five two into ten bar eight mm to the power of four. This is I z. Next area we know area is a one plus a2 plus a three that is one

five [00:19:16.004] 000. So Kz can be written as Kz is equal to conduct of IZ by area. So I z is equal to one zero two point. [00:19:29.572] Five two into ten part eight divided by one point five into ten part five. [00:19:42.116] Once we solve this, we are going to get the values 261.43 millimeter. This is the polar radius of gradation passing to the center C [00:19:56.068] That's it for today's class. Try to solve the problems given in the assignment and question bank. If you come across any queries, you can post your queries in the PESO forum. Thank you.

EV0000102 · EV0000103 · EV0000104 · EV0000105

Full transcript: 2729 normalized words, 76 timestamped segments, 1205.674667 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

# 9 CLASS 51.2 Moment of Inertia_ Numerical

This lesson is indexed from the source video titled CLASS 51.2 Moment of Inertia_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000106 · EV0000107

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello, in today's session, I am going to explain how to find moment of inertia of a rectangular lamina about its centroidal axis. Similarly, moment of inertia of a circle about its diameter axis. [00:00:16.612] So the first is moment of inner ship a rectangle about its centroid and x-axis. For this I am going to consider a rectangle. Let us consider this rectangle of width B. Let the height of this rectangle be h. Next. [00:00:35.620] Like the total area of this rectangle that is capital A will be equal to B into H. Next, we want to find inertia about the centroidal x-axis. First, we should mark the centroidal x-axis here. Where will be the centroidal x-axis? If you consider this is just an x-axis. [00:00:55.524] Then where will be the centroidal x axis? We know why by formula for rectangle is h by 2. So at a distance of h by 2, if you draw a horizontal line, this will be the centroidal x axis. So this distance is h by [00:01:11.184] Two. Now this is my reference line, so this will be 0 and this is this will be minus h by two and this will be plus h by two. Next to find inertia of this rectangle about the centridal x axis, I need to consider one smaller element here. [00:01:28.708] I'm going to consider an element like this. The shaded area is the element. [00:01:34.116] Since this is a small area, let it be dA. Now, this is at a distance of y with respect to reference line. The distance is y. Now, the moment of inertia of this shaded area about the centroidal x-axis, that is d. [00:01:52.644] I x0 will be equal to we know moment of inertia's distance square into area. Here the distance is y, so y square into area is d. [00:02:04.516] Next, this is the inertia of the smaller element about the centroidal x-axis. Next, I need to find the inertia of the rectangle about the centroidal x-axis. So just integrate this. [00:02:17.156] Once I integrate this, integral of d i x naught will be equal to i x naught. This is the moment of inertia of a rectangle above the centroid x axis. This will be equal to integral of y square into d. Integral of y square. [00:02:34.608] into d a. What is d? D is the area of

the small element. The area of the element will be equal to since it is in the form of rectangle, it will be equal to b into h. Here what is b? Since these two lines are parallel, the width of the element will also be equal to [00:02:52.480] B itself, so d a is equals to b into h. What is this h? The width of the element is a variable along y direction. So let it be dy. Now d a is equals to b into dy. Consider that here. [00:03:09.412] Next, i x naught will be equal to integral of b's constant. So b into integral of y square into dy will be equal to y cube by three. [00:03:22.116] Limits will be from C minus H by 22 plus H by 2. So here also minus H by 22 plus H by 2. I x0 will be equal to B by 3. [00:03:41.092] It will be equal to h by 2 4 cube minus minus of h by to cube. So once we solve this, I x naught will be equal to b by three into h cube by [00:04:01.476] 8 minus into minus it is going to be plus h cube by 8. I x naught will be equal to b h cube by [00:04:15.556] So this is going to be 4, 4, 3 sa 12. This is the moment of inertia of a rectangular lamina above the centroid and x axis. So it will be written as i bar x. That will be equal to bh cube by 12. You have to remember this. [00:04:32.576] Moment of inertia of a rectangle about its centroidal x axis is b by 12. Now look at this. There is a rectangle like this, and here is the x-axis, and centroidal x axis is at a distance of [00:04:50.756] H by 2 from here. Moment of inertia about this axis, we got it as BH U by 12. Now, how to find inertia about X axis?

EV0000106 · EV0000107 · EV0000108 · EV0000109

Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.484] We can apply parallel axis theorem. According to parallel axis theorem, inertia about any axis that is ix will be equals to inertia about parallel centroidal axis that is i bar x plus area into distance squared. This is the statement of parallel axis theorem. And here [00:05:23.876] I bar X already we know B H cube by 12. Area is area of a rectangle that is B into H. Then D stands for the distance between the two parallel lines that is H by [00:05:36.548] 2. So substitute the values here, Ix will be equal to BH Q by 12. [00:05:44.676] Plus area is b into h, then the distance is h by 2 square. Just simplify this, we are going to get ix as this will be bh cube by 4, and this is bh cube by 12. So this will be equal to bh cube by [00:06:04.548] [ASR low confidence; check audio] Three. [00:06:08.548] [ASR low confidence; check audio] B.H.Q. B.H.Q.B. [00:06:12.516] Three. This is the moment of inertia of a rectangle above the centroidal, sorry, above the line passing through its base. So once we come to rectangle, you have to remember inertia about centroidal x axis is bh by twelve, then inertia above [00:06:29.776] Its bottom most member will be BHQ by three. Next is moment of inertia of a circle about its diameter axis. For this I am going to consider a circle. [00:06:46.052] [ASR low confidence; check audio] Of radius R. [00:06:50.532] Let O be the center of the circle. Let the radius of the circle let it be R. Now if I draw a horizontal line passing through this, this itself is the centroidal axis as well as the diameter axis. Because for circle, this is where the centroid. [00:07:06.112] will be. Now I come to find the initial of this circle about its diameter axis. So I'm going to consider one smaller element here. [00:07:17.540] So let us consider there's a element like this, which is at a distance uh [00:07:24.996] R0 with respect to center. Then the radial width of this element let it be DR0. [00:07:32.996] Dr0. Next, let the inclination of this element with respect to this centroidal x axis adjust the x-axis. Let it be theta, then the inscribed angle of this element let it be d theta. So the element is like this. [00:07:53.476] This is R0 and this is D R naught and this is D theta. Now we need to know this. [00:08:07.012] width. Since it is like an arc and the arc this arc length will be equal to r naught into inscribed angle d theta. Now this is a smaller element [00:08:20.032] The area of this element it's on when D theta is small it means almost like a rectangle. So DA will be equals to area of rectangle B into H. So it is going to be DR naught into

R0 into D theta. This is the area of the smaller element. [00:08:36.944] Now, moment of inertia of this smaller element about x axis we have to calculate. So I'm going to consider it is at a distance of y here with respect to x axis. [00:08:49.412] So moment of inertia of the smaller element about central x axis d i x naught will be equal to distance square into area. So distance square is y square into area is d e. This is the inertia of the smaller element. Now [00:09:06.628] Once we integrate this, we are going to get it as i x naught will be equal to integral of y square into d e. Next what is 1? Before that [00:09:20.580] D is R0 into DR0 into D theta. There are two variables. One, this element can be anywhere from 0 to R within this radius. At the same time, theta, this element can be anywhere from 0 to [00:09:36.068] 2 pi. So it is a double integration 0 to 2 pi, then 0 to capital R, then we should find the value of y. Y is nothing but this is the x-axis, it's like this, and this is y. [00:09:55.588] When this d theta is very very small, the element will be just like a point, then y will be the opposite side of this right angle triangle. This is theta, we know. So y will be equal to r0 into sine theta.

EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 3 (source interval 000600-000899 seconds): [00:10:13.412] So y is equals to r naught into sin theta into this is full square into d is dr0 into r naught into d theta. Now just simplify this. [00:10:30.884] Zero two pine zero two or [00:10:36.388] Then R0 square into R0 will be equal to R0 Q into D R naught. This is the term related to R0, then sin square theta into D theta. Next. [00:10:55.044] Integration of R0 cube will be equal to R0 to the power of 4 by 4, and if you apply the limits from 0 to R, that term is going to be. [00:11:06.148] R to the power of 4 by 4, one more term is anyhow 0. Then we need to integrate 0 to 2 pi sine square theta into d theta. Sin square theta will be equal to 1 minus cos 2 theta by 2. [00:11:27.652] And for this the limits will be from 0 to 2 pi. Now this is IX0 is equal to R0 to the power r to the power of 4 by 4, this is first term. Then integration of 1 is [00:11:45.988] Uh I'll take this 1 by 2 outside, so it's going to be 1 by 2. Then integration of 1 is theta because it is with respect to into d theta here and down d theta is there. So integration of 1 with respect to d theta is theta minus of [00:12:05.188] Cos two theta will be equals to sine two theta by two, limits will be from zero to two by. Next, this will be equal to R to the power of four by eight. Next. [00:12:23.332] Uh substitute it as two pi, two pi, then minus of sine of two times it of two pi divided by two minus of zero and zero. [00:12:36.388] So this will be r to the power of four pi into two pi minus of sine of four pi is zero. [00:12:46.820] This will be equal to 5 R to the power of 4 by 8. This is the moment of inertia fit circle about the diameter axis. So this due to symmetry. [00:13:00.768] I x naught will be equal to i y naught. I can write it as i bar x is equals to i bar y is equals to pi r to the power of 4 by [00:13:12.452] See this is the moment of inertia of a circle about the diameter x-axis as well as the y axis. This is about circles. Similarly, to find the moment of inertia of a semicircle like this. [00:13:31.524] Only difference is the limits will be varying from zero to pi instead of two pi. So you can try for this. Uh here you can see the formulas for the semicircle. [00:13:47.684] If you just derive it just like a circle by changing the limits from 0 to pi, you are going to get the values. Here, moment of inertia of a circle, sorry, semicircle, which is symmetrical about y-axis here. [00:14:03.460] Uh try to learn the formulas this way. Inertia about an axis of symmetry and the diameter axis is always equals, that is pi r to the power of 4 by 8 and pi r to the power of 4 by 8. Now centroidal axis which is parallel to the diameter axis. [00:14:20.448] Here, this is the centroid which is at a distance of 4 r by 3 pi. If you draw a horizontal line, this is the centroidal axis which is parallel to diametral axis. Moment of inertia about this line is 0.11 into R to the paraph. [00:14:35.408] 4. Similarly, the formulas will

change for semicircle which is symmetrical about x-axis. In that case, inertia about diameter and axis of symmetry is pi r to the power of 4 by 8, and the centroidal axis parallel to the diameter line is 0.11 into r to the power of 4. [00:14:55.364] That's about semicircle. Similarly, if we change the limits for quarter circle from 0 to 5 by 2.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.712] In the derivation of circle, we are going to get ix and iy value. One is pi r to the power of 4 by 16, and this is also pi r to the power of 4 by 16. This is moment of inertia of a quarter circle about the radial lines. Next moment of inertia of a quarter circle. [00:15:20.672] About the centroidal x axis and centroidal y axis is point five fi into r to the power of four. So these are the formulas you have to remember. That's it for today's class. Thank you.

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Full transcript: 2207 normalized words, 63 timestamped segments, 935.68 seconds duration. Low-confidence segments flagged: 3. Review locators: 00:06:04.548, 00:06:08.548, 00:06:46.052

EV0000106 · EV0000107 · EV0000108 · EV0000109

# 10 CLASS 47 Friction_ Numerical

This lesson is indexed from the source video titled CLASS 47 Friction_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000122 · EV0000123

## Graphs and diagrams

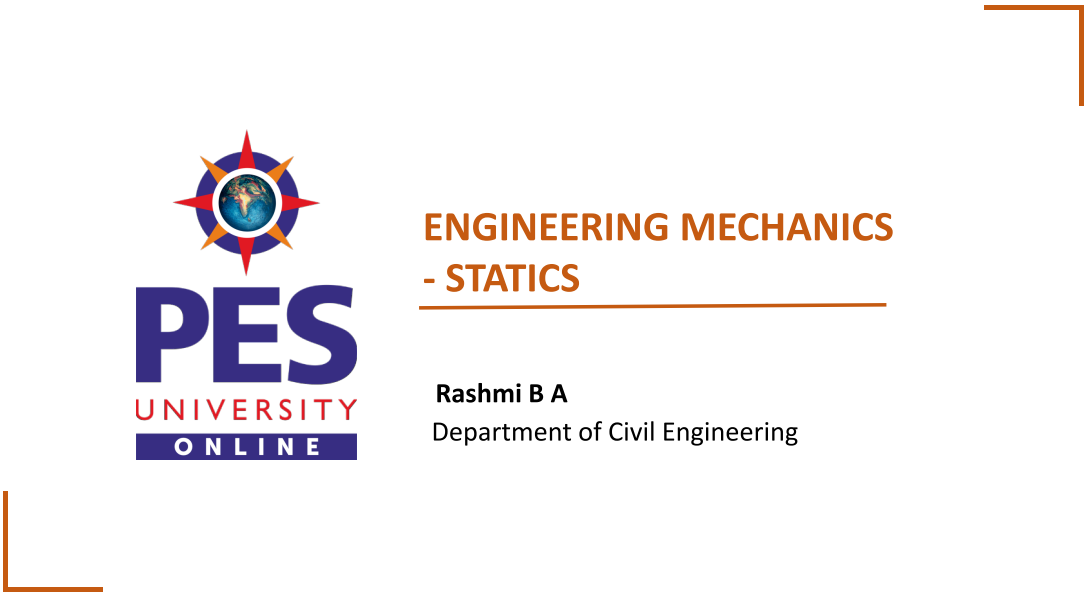


**Figure G000120.** UE20CV101 UNIT 5 SESSION 4 · source slide page 1.

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# ENGINEERING MECHANICS - STATICS

## FRICTION

Session- 4

**Rashmi B A**
Department of Civil Engineering

**Figure G000121.** UE20CV101 UNIT 5 SESSION 4 · source slide page 2.

EV0000543

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## ENGINEERING MECHANICS - STATICS
## Friction- Numerical

- Problem 6/8: Determine the coefficient $\mu_k$ of kinetic friction which allows the homogeneous body to move down the 30° incline at constant speed.

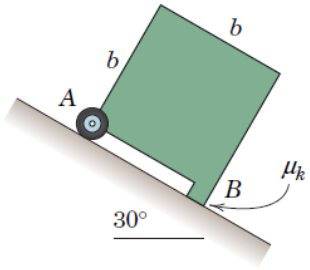


**Figure G000122.** UE20CV101 UNIT 5 SESSION 4 · source slide page 3.

EV0000546

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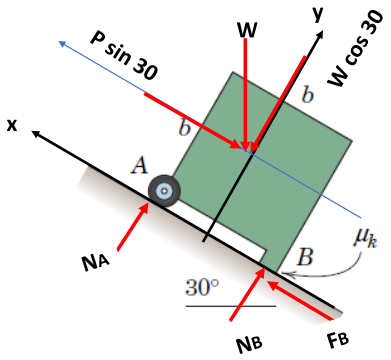
ENGINEERING MECHANICS - STATICS

Friction- Numerical

Problem 6/8 :Solution




***Applying the conditions of equilibrium:***

$\sum F_x = 0 \quad +F_B - W\sin(30) = 0$

$F_B = 0.5\,mg$---(1)

$\sum M_o = 0$

$+N_B(b) - W\cos(30)\left(\frac{b}{2}\right) - W\sin(30)\left(\frac{b}{2}\right) = 0$

$N_B = 0.683\,mg$ ——(2)

from *equation* 1 & 2

$$\mu_s = \frac{F}{N} = 0.732$$

**Figure G000123.** UE20CV101 UNIT 5 SESSION 4 · source slide page 4.

EV0000549

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ENGINEERING MECHANICS - STATICS

Friction- Numerical

- **Problem 6/13:** The 50-kg wheel rolls on its hub up the circular incline under the action of the 12-kg cylinder attached to a cord around the rim. Determine the angle at which the wheel comes to rest, assuming that friction is sufficient to prevent slippage. What is the minimum coefficient of static friction which will permit this position to be reached with no slipping?

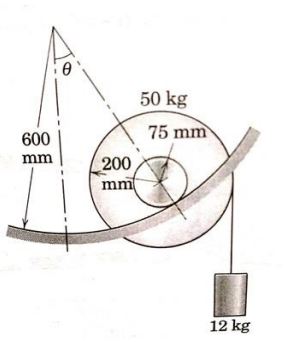


**Figure G000124.** UE20CV101 UNIT 5 SESSION 4 · source slide page 5.

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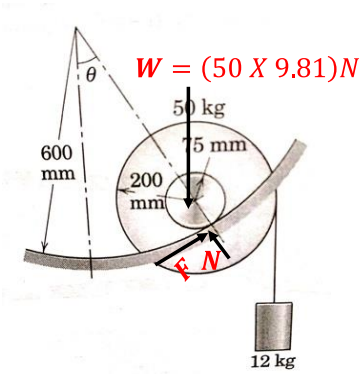
ENGINEERING MECHANICS - STATICS

Friction- Numerical

**Problem 6/13: Solution**




*Applying the conditions of equilibrium*

$\sum M_o = 0 \quad -(12 \text{ X } 9.81)(200 - x_1) + Wx_1 = 0 - -(1)$

*To find* $x_1$ *consider a triangle*

$\sin\theta = \frac{x_1}{75} \Rightarrow \quad x_1 = 75 \sin\theta$

substituting in equation (1)

$\sin\theta = 0.516$

$\theta = 31.07$

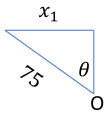



**Figure G000125.** UE20CV101 UNIT 5 SESSION 4 · source slide page 6.

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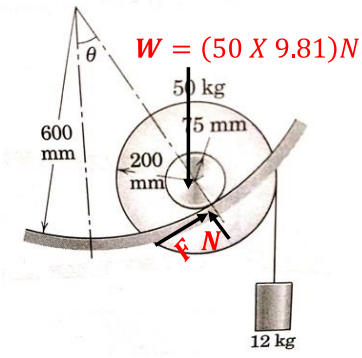
ENGINEERING MECHANICS - STATICS

Friction- Numerical

**Problem 6/13: Solution**




$\sum F_x = 0 \quad +F - W\sin\theta - (12\,X9.81)\sin\theta = 0$

$F = 313.89\ N$

$\sum F_y = 0 \quad +N - W\cos\theta - (12\,X9.81)\cos\theta = 0$

$N = 520.963\ N$

$\mu_s = \frac{F}{N} = 0.602$

**Figure G000126.** UE20CV101 UNIT 5 SESSION 4 · source slide page 7.

EV0000558

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**Figure G000127.** UE20CV101 UNIT 5 SESSION 4 · source slide page 8.

EV0000561

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000539 · EV0000540 · EV0000541
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS FRICTION Session- 4 Rashmi B A Department of Civil Engineering EV0000542 · EV0000543 · EV0000544
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • Problem 6/8: Determine the coefficient $\mu k$of kinetic friction which allows the homogeneous body to move down the 30° incline at constant speed. EV0000545 · EV0000546 · EV0000547
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/8 :Solution $\boldsymbol{Applyingtheconditionsofequilibrium}$: = 0 σ $\boldsymbol{Fx}$= $\mathbf{0}$ $+FB-Wsin30$ y W $FB$= 0.5 $mg---(1)$ σ $\boldsymbol{Mo}$= $\mathbf{0}$ x $b$ $b$ $+NBb-Wcos30$ $-Wsin(30)$ = 0 2 2 $NB$= 0.683 $mg---(2)$ from $equation1$ & 2 $\boldsymbol{F}$ $\boldsymbol{\mu s}$= $\boldsymbol{N}$= $\mathbf{0.732}$ EV0000548 · EV0000549 · EV0000550
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • Problem 6/13: The 50-kg wheel rolls on its hub up the circular incline under the action of the 12-kg cylinder attached to a cord around the rim. Determine the angle at which the wheel comes to rest, assuming that friction is sufficient to prevent slippage. What is the minimum coefficient of static friction which will permit this position to be reached with no slipping? EV0000551 · EV0000552 · EV0000553

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/13: Solution $Applyingtheconditionsofequilibrium$ σ $Mo$= 0 −(12 X 9.81)(200 $-x1$) + $Wx1$ = 0 −−(1) $\boldsymbol{W}$= 50 $X$9.81 $N$ $Tofindx1$ $consideratriangle$ $x1$ $x1$ sin ϑ = => $x1$= 75 sin $\theta$ 75 $\theta$ substituting in equation (1) O sin ϑ = 0.516 $\boldsymbol{N}$ ϑ= 31.07 EV0000554 · EV0000555 · EV0000556
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/13: Solution σ $\boldsymbol{Fx}$= $\mathbf{0}$ + $F-W$sin $\theta-12$ $X$9.81 sin ϑ = 0 $F$= 313.89 $N$ $\boldsymbol{W}$= 50 $X$9.81 $N$ σ $\boldsymbol{Fy}$= $\mathbf{0}$ +$N-W$cos $\theta-12$ $X$9.81 cos ϑ = 0 $N$= 520.963 $N$ $\boldsymbol{F}$ $\boldsymbol{\mu s}$= $\boldsymbol{N}$= $\mathbf{0.602}$ $\boldsymbol{N}$ EV0000557 · EV0000558 · EV0000559
8. Source slide page 8 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000560 · EV0000561 · EV0000562

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.284] Good morning. In today's session let's solve few problems on fiction. The first question for today's session, question number six slash eight. [00:00:13.412] determine the coefficient of kinetic friction that is which will allow the homogeneous body to move down the thirty degree incline at constant speed. [00:00:24.900] And the figure is given like this. There is an inclined plane. Inclination of the plane with the horizontal is thirty degree. And this is a homogeneous body and we have to assume here it is. [00:00:37.328] A two dimensional body. One end of this body is resting on a roller and the other end is resting directly on this inclined plane. So for what uh we need to find coefficient of kinetic friction which will allow the body to move at a constant speed. [00:00:56.100] Now let's start with the free body diagram of this body. [00:01:03.172] So first thing is we have to consider here X and Y axis. Since this homogeneous body is resting on an inclined plane, here we are going to assume X axis along the inclined plane. [00:01:17.984] plane. So x axis is going to be like this. Next we need to consider y axis. We know y axis will be normal to x axis. So this is the y axis. Next, since we are considering this homogeneous body [00:01:35.072] It will be having a weight and we know weight will be always vertically downward. So the magnitude of that is going to be W. [00:01:44.580] This W is is an inclined force with respect to XY plane. Now we need to resolve it along X Y so we should know the angle made by this W [00:01:56.064] Either with respect to x axis or with respect to y axis. Since this is thirty degree here, the angle between w and y axis will also be thirty degree. So the x component of this w will be equal to down the plane and the magnitude. [00:02:13.712] will be W sin 30 degree and the y component will be downward and magnitude will be W cos 30 degree. Next. As I said before, this is a roller. Since the roller is [00:02:29.536] On an inclined plane we are going to neglect the frictional force. We are going to consider only the normal reaction. So the normal reaction between the roller and this inclined will be like this normal that is along y direction or [00:02:46.948] Normal to x axis and I'm going to call it as NA. That is the normal reaction at A. Next comes this part. It's like a foot here which is in contact with the incline. It will be having both frictional force as well as the normal reaction. So the normal reaction will be [00:03:06.704] along the y direction and it will be upward. I'm going to call it as N B. That is normal reaction at B. And because of this incline this [00:03:17.648] Homogeneous body will have a tendency to slide down the plane. To avoid that the frictional force will be up the incline. This is the frictional force at B. Now this is the complete free body diagram of this homogeneous body. So we have to find [00:03:35.424] mu k to find mu k we know the formula mu k it will be equals to

since e at e frictional force is equals to zero fr from b here we can find mu as mu is equals to f by n we know the formula. So once we know the value of f b and b [00:03:53.008] We can use the formula mu k is equals to F B by N B. So we need to find first what is frictional force at P and what is normal reaction at B. To find these two we have dot conditions of equilibrium. [00:04:11.492] Now, let's start with the first condition of equilibrium that is sigma f of x is equal to 0. [00:04:19.556] Once we consider sigma of x is equals to 0, we should consider the forces which are parallel or along the x direction. So first one is this f b is towards the positive side of x axis, so plus f b. Next is [00:04:35.344] This W sine thirty since it is in the opposite direction, it is minus W sine thirty. These are the only two forces which are along the X direction. [00:04:46.660] C we will get the value of F B. So F B here is F B will be equals to W sin 30 and W here is weight of this homogeneous body that we can write it as

EV0000122 · EV0000123 · EV0000124 · EV0000125

Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.688] Point five times uh that we can write it as mg and we know sign that is point five so FP will be equals to point five times set of mg. Call it as equation one. Next [00:05:14.404] Ah, which is the next condition of equilibrium we should consider. No need to consider sigma f y is equals to zero because we have two unknowns along y direction, one is n a and one more is n b. So I'm going to start with the moment equation. Since we need to find mu k we need to [00:05:33.604] Know the value of F B and N B. It's not required to know the value of N A. So to avoid N A I am going to consider moment about A as 0. Sum of the moments of all the forces about A as 0. That is the next condition of [00:05:51.152] equilibrium. Now once we consider moment about A is equals to zero, the equation is going to be moment due to this N A will be zero. Moment due to F B will also be zero because it is a [00:06:05.296] Uh the point is in the same line of action. So it will be zero. And N B is acting in this direction. It is counterclockwise. So plus N B into distance. Distance here it is given as B. So N B into B will be the [00:06:21.796] Moment due to this normal reaction at B. That's what is written here. Next comes this W cos thirty. It is acting in the clockwise direction, so it should be taken as minus. [00:06:35.680] Minus W cos thirty into perpendicular distance will be due to symmetry. This distance will be B by 2. [00:06:44.228] Next comes this force. It is also clockwise. So minus W sine 30. And the distance. We are talking about this distance. Again, due to symmetry, this distance will be B by 2. That should be equal to 0. [00:07:01.632] Now again here W is M into G. And the equation is going to be N B will be equals to [00:07:16.644] Point six eight three times the dup M G and call it as equation two. Now we got both the equations one and two in the form of MG. We know next mu case. [00:07:35.588] Yes, B by N B. So we can write it as now once we simplify this we are going to get mug as point seven three two. [00:07:46.688] This is the coefficient of kinetic friction between the homogeneous body and this incline for which this body will be under motion with the uniform speed. Next question. Question number six slash thirteen. [00:08:05.764] The fifty kg wheels on its hub up the circular incline under the action of the twelve kg cylinder attached to a card around the rim. [00:08:18.084] Determine the angle at which wheel comes to rest, assuming that friction is sufficient to prevent slippage. What is the minimum coefficient of static friction which will prevent this position to be reached with no slipping? [00:08:35.524] Uh here you can look at this figure. This is the fifty kg field, which is attached to a twelve kg cylinder. [00:08:47.428] And due to this twelve kg cylinder, this fifty kg wheel rolls on this circular incline. [00:08:59.236] Now assume this is the angle theta for which the this circular wheel is going to be in a state of rest. Because here the friction between this circular wheel and this

incline is sufficient [00:09:16.400] for this to be in a state of rest. So now we need to find for what value of this theta this fifty kg [00:09:27.424] wheel is going to be in a state of rest. At the same time we need to find the coefficient of friction here between this 50 kg wheel and this circular incline. That's what we need to find. [00:09:43.588] Now let's get into the solution part. I'm going to consider the free body diagram of this 50 kg wheel. Now I need to start with the self weight. Since the weight of this circular wheel is 50 kg.

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.496] Oh sorry, wheel is fifty KC that will be acting vertically downward. Fifteen to nine point eight one that is the weight of this wheel vertically downward. Next this wheel is in contact with here, so there will be a friction as well as the normal reaction. [00:10:20.144] So the normal reaction will be along this line itself upward and the friction force. Since this will have a tendency to move this direction, the frictional force will be in the opposite direction, F will be up the incline. [00:10:39.364] So next the same thing is considered here. Uh there is a 12 kg cylinder attached to this wheel and the weight since it is freely suspended here the weight is going to be downward vertically downward to one into 9.81 N. [00:10:57.392] Now we need to find what is the value of F, N and Teta. So to find these three we need to find we need to apply the conditions of equilibrium. Let's start with the first condition of equilibrium. [00:11:13.636] I'm going to consider moment about this contacting point as zero. I'll call this point as O and I'm going to apply the first condition as sum of the moment of all the forces about O as zero. If you consider that the first one is [00:11:29.968] This I'm going to start from here. It is acting in the clockwise direction with respect to this point of contact. So clockwise means negative, force is twelve into nine point eight one into distance. What is the distance between horizontal distance between this point of contact and [00:11:47.600] the line of action of this force. That should be the horizontal distance, this distance. How to find this distance, if you look at this figure, the distance from the center of this wheel [00:12:03.392] Till here it is a radius that is two hundred millimeter. So in that two hundred mm, if I deduct this [00:12:12.384] A B. I am going to get that horizontal distance between the force and the moment center. So I'll call this A B distance is X1. So it's going to be 200 minus X1. This is the moment due to the 12 kg cylinder. Next moment due to F [00:12:30.064] And n will be zero because they are acting at the same point. [00:12:35.328] self weight of the wheel since it is acting in the counterclockwise direction it should be taken as plus. This is W from here we know this is W into distance will be horizontal distance because the line of action is vertical the distance should be horizontal. I am going to call that distances. [00:12:53.008] X1. That should be equals to zero. Once I know the value of X1 here, then we will get the value of theta. [00:13:05.188] Ah so how to find this X1? Is it possible to express X1 in the form of theta? Yes we can do that. For that I am going to consider you can see there is a small triangle here, right angle triangle, AB V. Same AB V is taken here. [00:13:22.960] If this angle is theta, then this angle will also be [00:13:26.912] Theta. Next A B as I said before that is X1. Consider that X1. Then A U distance. A u is nothing but the radius of the smaller circle. The radius of the smaller circle we know is 75. So it will be 75. Now X1 will be we can use. [00:13:46.832] sine theta, sine theta is equals to opposite by hypotenuse. So sine theta is equals to opposite is X1 by hypotenuse is 75. X1 will be 75 sine theta. So we got the value of X1. Substitute the value of X1 in the equation [00:14:04.464] Once we substitute the value of x1 in the equation one, we will get it as sin theta is equal to 0.516. Now [00:14:15.712] Once we simplify this we are going to get the value of theta. Theta is thirty one point zero seven. Nothing but when the

value of theta is thirty one point zero seven, the wheel which is rolling up this incline will be nest of [00:14:33.984] Rest because at this angle the frictional force here will be sufficient to prevent this from slipping under the motion. Next we need to find the value of F, coefficient of friction. For that we need to know the value of F and N. Once we know F [00:14:51.632] And n then mu will be is equal to f by n. So first, let's start with the first condition sigma f of x is equals to zero.

EV0000130 · EV0000131 · EV0000132 · EV0000133

Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.172] Sigma f of x is equals to 0. Here we are going to consider incline as like x axis will be along this incline like a tangent here. Since this f is along this x axis itself, it should be taken as plus. [00:15:20.640] F, so plus F, then W will be inclined force with respect to X and Y. And the inclination here will be since this angle is theta, angle with respect to Y axis will be [00:15:34.288] theta since we are talking about x component of that that's going to be w sine theta and the component is going to be left side so minus w sine theta. [00:15:43.808] Then minus of we need to consider this, the x component of this will be again left side. So minus 12 into 9.81 and the angle will be theta. That should be equal to zero. Already we know the value of theta from the previous uh [00:16:00.932] So substitute that here we'll get the value of F as 313.89 N. [00:16:08.740] Next, we should know the value of normal reaction. So we need to apply the second condition of equilibrium, sigma f of y is equal to 0. When we consider sigma f y 0, n is upward, so plus n [00:16:23.492] The y component of this W will be downward like this. So minus W into angle here is theta. Minus W into cos theta will be the component. Then Y component of this will be again downward. So minus [00:16:38.416] twelve into nine point eight twelve into cos theta should be equal to zero. We know the value of theta, we need to find only the value of n. So n we will get it as five twenty point nine six three Newton. Once we know [00:16:54.244] Frictional force in the normal reaction, then coefficient of friction will be equal to F. We will get it as point six zero two. [00:17:06.244] So when this wheel which is on this inclined plane is in a state of rest, then the coefficient of friction at that point of contact will be 0.602. This is the answer. [00:17:23.684] Thank you.

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Full transcript: 2571 normalized words, 72 timestamped segments, 1045.098667 seconds duration. Low-confidence segments flagged: 0.

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# 11 CLASS 48 Friction_ Numerical

This lesson is indexed from the source video titled CLASS 48 Friction_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000138 · EV0000139

## Graphs and diagrams

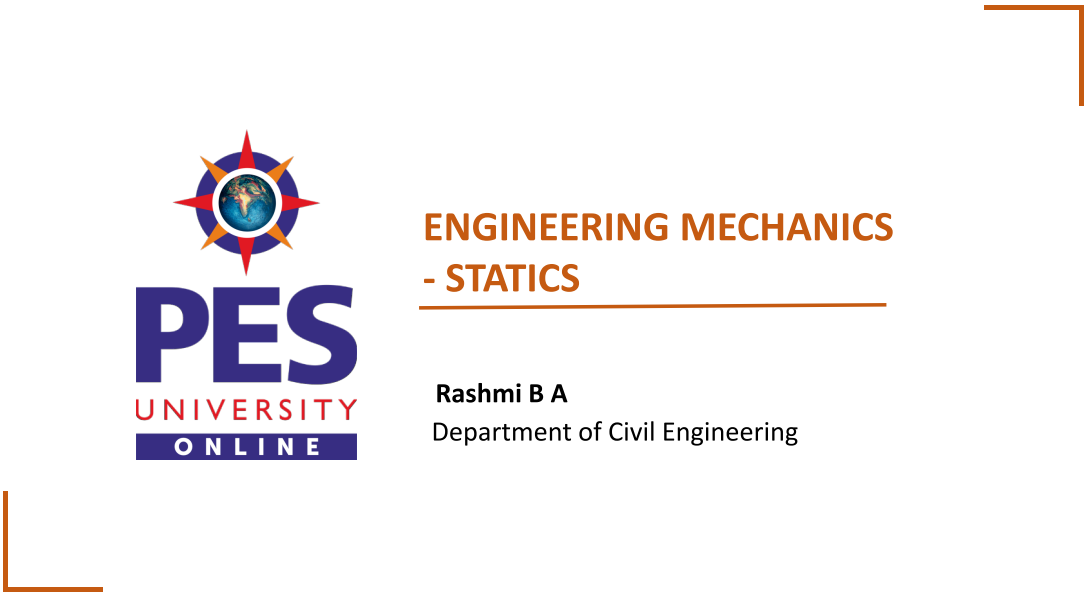


**Figure G000138.** UE20CV101 UNIT 5 SESSION 5 · source slide page 1.

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# ENGINEERING MECHANICS - STATICS

## FRICTION- Numerical

### Session- 5

**Rashmi B A**
Department of Civil Engineering

**Figure G000139.** UE20CV101 UNIT 5 SESSION 5 · source slide page 2.

EV0000597

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ENGINEERING MECHANICS - STATICS
Friction- Numerical

- **Problem 6/18:** The uniform slender bar has an ideal roller at its upper end A. Determine the minimum value of the angle θ for which equilibrium is possible for $\mu_s = 0.25$ and for $\mu_s = 0.50$.

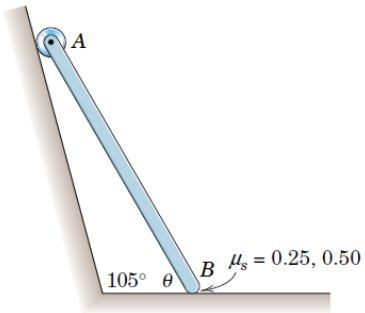


**Figure G000140.** UE20CV101 UNIT 5 SESSION 5 · source slide page 3.

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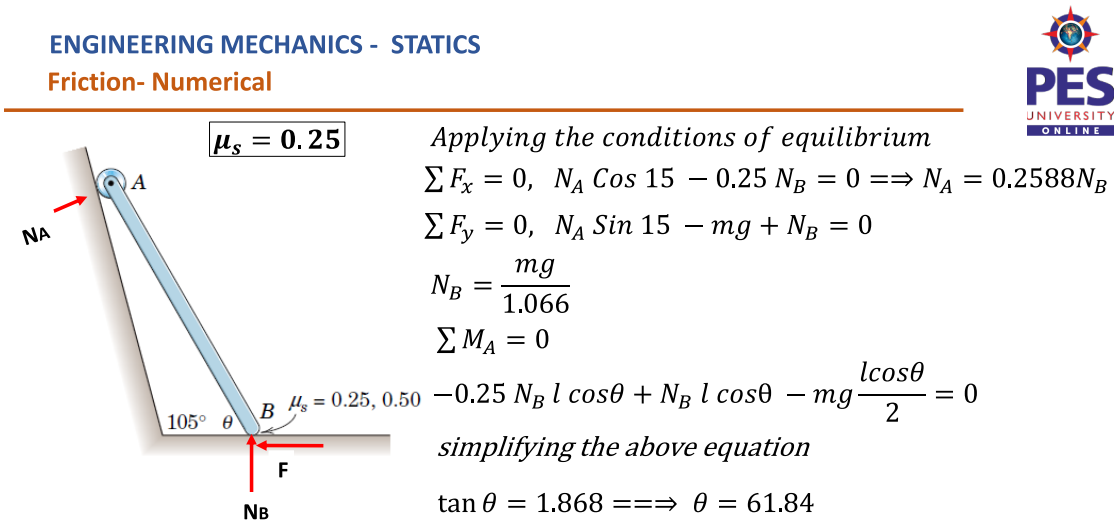


**Figure G000141.** UE20CV101 UNIT 5 SESSION 5 · source slide page 4.

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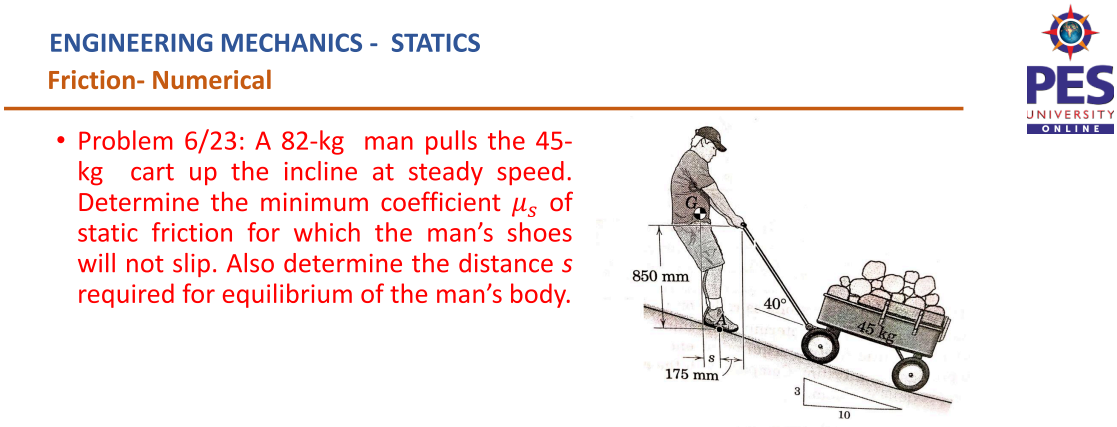


**Figure G000142.** UE20CV101 UNIT 5 SESSION 5 · source slide page 5.

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Friction- Numerical

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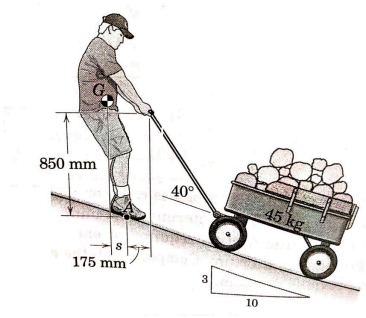
**Problem 6/23: Solution**




$$\theta = \tan^{-1}\left(\frac{3}{10}\right) = 16.7$$

***Considering the FBD of Cart & Applying the conditions of equilibrium***

$$\sum F_x = 0$$

$$F\cos 40 - (45\ X\ 9.81)\sin\theta = 0$$

$$F = 165.6\ N$$

**Figure G000143.** UE20CV101 UNIT 5 SESSION 5 · source slide page 6.

EV0000609

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Friction- Numerical

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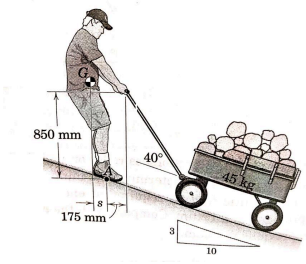


***Considering the FBD of Man & Applying the conditions of equilibrium***

$$\sum \boldsymbol{M_A} = \mathbf{0}$$

$$(82\ X\ 9.81)(s) - F\cos(40+\theta)(850) - F\sin(40+\theta)(175) = 0$$

$$s = 126.2\ mm$$

$$\sum \boldsymbol{F_x} = \mathbf{0}$$

$$F_A - (82\ X\ 9.81)\sin(16.7) - F\cos(40) = 0$$

$$F_A = 358.91\ N$$

$$\sum \boldsymbol{F_y} = \mathbf{0}$$

$$N_A - (82\ X\ 9.81)\cos\theta + F\sin(40) = 0$$

$$N_A = 876.93$$

$$\mu_A = \frac{F_A}{N_A} = 0.409$$

**Figure G000144.** UE20CV101 UNIT 5 SESSION 5 · source slide page 7.

EV0000612

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**Figure G000145.** UE20CV101 UNIT 5 SESSION 5 · source slide page 8.

EV0000615

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### Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000593 · EV0000594 · EV0000595
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS FRICTION- Numerical Session- 5 Rashmi B A Department of Civil Engineering EV0000596 · EV0000597 · EV0000598
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • Problem 6/18: The uniform slender bar has an ideal roller at its upper end A. Determine the minimum value of the angle ϑ for which equilibrium is possible for $\mu s$= 0.25 and for $\mu s$= 0.50. EV0000599 · EV0000600 · EV0000601
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical $\boldsymbol{\mu s}$= **0**. **25** $Applyingtheconditionsofequilibrium$ σ $Fx$= 0, $NACos15$ −0.25 $NB$= 0 =⇒$NA$= 0.2588$NB$ σ $Fy$= 0, $NASin15$ −$mg$+ $NB$= 0 $mg$ $NB$= 1.066 σ $MA$= 0 −0.25 −$mglcos\theta$ = 0 $NBlcos\theta$+ $NBlcos\vartheta$ 2 simplifyingtheaboveequation F tan $\theta$= 1.868 ==⇒$\theta$= 61.84 NB EV0000602 · EV0000603 · EV0000604
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • Problem 6/23: A 82-kg man pulls the 45- kg cart up the incline at steady speed. Determine the minimum coefficient $\mu s$of static friction for which the man's shoes will not slip. Also determine the distance s required for equilibrium of the man's body. EV0000605 · EV0000606 · EV0000607

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/23: Solution 3 $\theta$= tan−1 = 16.7 10 $\boldsymbol{ConsideringtheFBDofCart}$& $\boldsymbol{Applyingtheconditionsof\ equilibrium}$ $\sigma Fx$= 0 $F$cos 40 −45 $X$9.81 sin $\theta$= 0 $F$= 165.6 $N$ EV0000608 · EV0000609 · EV0000610
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical $\boldsymbol{ConsideringtheFBDofMan}$& $\boldsymbol{Applyingtheconditionsofequilibrium}$ $\sigma\boldsymbol{MA}$= **0** 82 $X$9.81 $s$ −$F$cos 40 + $\theta$ 850 −$F$sin 40 + $\theta$ 175 = 0 $s$= 126.2 $mm$ $\sigma\boldsymbol{Fx}$= **0** $FA$−82 $X$9.81 sin 16.7 −$F$cos 40 = 0 $FA$= 358.91 $N$ $\sigma\boldsymbol{Fy}$= **0** $NA$−82 $X$9.81 cos $\theta$+ $F$sin 40 = 0 $NA$= 876.93 $FA$ $\mu A$= = 0.409 $NA$ EV0000611 · EV0000612 · EV0000613
8. Source slide page 8 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000614 · EV0000615 · EV0000616

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] Hello. In today's session, let's solve few more problems of friction. [00:00:06.576] Problem six slash eighteen The uniform slender bar as an ideal roller at its upper LDA. Determine the minimum value of the angle theta for which equilibrium is possible for mu s point two five and for mu s point five. This is the figure given. [00:00:23.684] Now if you look at this, there is a slender bar A B. We don't know the length of this bar, we can assume it as L. Let capital L be the length of this slender bar. [00:00:35.172] Now the inclination of this lender bar with respect to horizontal let it be theta. And when muus is 0 to 5, then in that case this entire system will be in a state of equilibrium. [00:00:48.676] So this for what value of this theta it will be in a state of equilibrium we have to calculate. Similarly when the value of mues is 0.25, for what value of this theta this entire system is in a state of equilibrium we have to calculate. Now let's start with the free body diagram. [00:01:07.172] Since it's a roller, there will be only one reaction from the surface onto this roller that will be normal to the surface. Next year the slender ball is resting on the surface, so I am going to assume it as a [00:01:23.780] Tough surface there will be two reaction normal reaction N B and one more friction. The direction of friction depends upon the tendency of motion. Now [00:01:35.200] Because of this inclination the slender bar will have a tendency to slide this way. So the frictional force will be opposite to that. So the friction will be left side here B. And we know frictional force will be as equals to mu s into n. Right now mu s is 0.25 and n here is n b [00:01:54.852] After this, let us consider the length of the slender bar. Let it be L. [00:02:06.404] Next, the weight of this center bar will be acting to the geometric center G that will be at a distance of L by 2 with respect to A, L by 2 with respect to B. Going to consider like this. Next, this is W vertically downward. [00:02:25.988] W is vertical, F B is horizontal, N B is vertical. So we now need to resolve this. Whereas N A is inclined. So I have to resolve this NA along X and Y. For that I should know this angle here. [00:02:39.524] Then how to find this angle? To find the angle I am going to drop it perpendicular to this point, then extend this horizontal surface here like this. Then you can see there is a right angle triangle A [00:02:52.196] C D this right angle travel this is 90 degree this is theta then if you look at this here this angle will be 90 so the remaining angle will be 15 degree if this is 15 degree this angle will be 15 degree [00:03:09.476] This is 15 degree. This will also be 15 degree and the component of this na will be right side na cos 15 and the component of NA along y direction will be upward and that will be na sine 15 degree. [00:03:29.700] So this is the complete free body diagram. Now we can start with the conditions of equilibrium. Let's start with the first condition of equilibrium. Before

that, remember here right now here is point. [00:03:46.884] Now the first condition of equilibrium is sigma f of x is equals to 0, considering right side component as positive and left side as negative. Start with this NE, NA is towards right side, so plus N A is N A cost 15, F B is left side, so minus F B is equals to 0. [00:04:10.244] yes b can be written as 0.25 n b in that case n a will be equals to 0.25 n b1 cos 50 that will be equal to 0.2588 times that of n b I'll call it as equation one [00:04:28.580] Next, second condition of equilibrium sigma f of y is equals to 0 taking upward component positive, downward, negative. N component is upward, so plus n mu. [00:04:40.644] Sine fifteen degree, then W is downwards so minus W, then N B is upwards so plus N B is equals to C. Next we know [00:04:52.868] N is 0.258 times N V from the equation one substitute that here into sine fifteen minus W plus N B is equal to zero. Then N B can be written as

EV0000138 · EV0000139 · EV0000140 · EV0000141

Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.788] N V will be equal to W divided by [00:05:16.388] One plus point two five eight eight into sine fifteen that will be equal to one point zero six seven zero. This will be equation. [00:05:28.644] Next to the last condition of equilibrium is moment. Moment about which part we have to consider here. Here I am going to apply moment about yes 0. [00:05:39.204] Once we consider moment about A as 0, considering counterclockwise positive and clockwise negative, moment because of these two will be zero. Next comes the force W. W is acting in the clockwise direction, so it is minus W. This is force, W is force. [00:05:56.720] Into perpendicular distance since the linear fraction of W is vertical, the distance should be horizontal. This distance if this is theta, this angle will also be theta, since this is L by two, A G distance is L by two, this will be equal to L by two cos. [00:06:14.244] theta. So the perpendicular distance between L by 2 cos theta. We are done with the moment of this W. Next comes F B, F B is acting in the clockwise direction. So minus F B is 0.25 times the N B. [00:06:31.812] into distance since the line of function of f b is horizontal the distance will be vertical distance from this rectangle triangle A C B this side will become L sin 15 degree that will be the perpendicular distance for this force F. [00:06:54.468] N sin 15. Next we are left with one more force NB. N B is acting in the counterclockwise direction. So plus N B. Since the line of action of N B is vertical, the perpendicular distance will be horizontal, that will be this distance. This will be equals to N. [00:07:14.308] Cos theta. Well cos theta is equals to zero. Next. [00:07:22.212] We have N B in the second term, we have N B in the third term also. So W we can write it as from the equation to W can be written as 1.067 times the N B. Then [00:07:37.412] Write the letters as it is L by two cos theta, then minus 12 psi n b sine fifteen degree, then plus of n b cos theta is equal to zero. [00:07:56.100] In all the three terms we have N V and L. Let's cancel that. So our final equation will be like this. [00:08:11.940] sine theta will be equal to 0.4665 cos theta sine theta by cos theta will be equal to tan theta that will be equal to 1.866 then theta will be equal to 61 point [00:08:28.976] So when the R theta is 61.81 and V S is 0.25, then that system will be in a state of equilibrium. Similarly, we can solve when V S is 0.25. [00:08:48.260] Next question problem six slash twenty three eighty two kg man pulls the forty five kg card up the incline at steady speed. Determine the minimum coefficient of static friction for which the man's shoes will not slip. [00:09:04.996] Also determine the distance S required for equilibrium of the man's body. If you observe this figure, there is an inclined plane. [00:09:16.388] The inclination of the plane with the horizontal we can find it out, assume this is theta, then theta will be equals to tan inverse of 3 by 10. Now there is a cart which is resting on this inclined plane, the mass of the cart is given as 45 kg. [00:09:33.412] Now there is a handle which is connected to this card. The inclination of this

handle with respect to a line parallel to the incline plane in 40 degree. Now there is a man who is pulling this cart up the incline at a steady speed. [00:09:50.656] Now for what value of this mu s and when s is what is for what value of this yes this card will be moving in a steady speed we have to calculate mu s coefficient of static friction mu s will be always equal to f max by

EV0000142 · EV0000143 · EV0000144 · EV0000145

Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.712] N I F by N in general. Then what is F? F is frictional force and N is normal reaction. So at a point A if we know the value of friction and if we know the value of N we can find the value of [00:10:23.888] mu s. So there are two unknowns here friction and normal reaction. Then after that we have to find this yes. S is the horizontal distance between the A. That is the [00:10:36.752] Position of the man's shoe with respect to the center of gravity of man. So this also we have to find. So there are three re uh three unknown forces E and E that is normal reaction at E, frictional force at E, then yes. [00:10:53.508] After that, once you consider a man, we don't know the force acting in this handle of the card. This is also unknown. So when we talk about man, there are four unknowns, but we have three conditions. [00:11:08.080] So we can't find four unknowns using three conditions. So better let's start with the [00:11:14.500] cart, then find the force in this handle. Once we know the force in this handle from the free body diagram of cart, the same force will be acting throughout from this point to the other point. So we can the number of unknowns will be [00:11:31.216] three when we consider the free body diagram of man. That's why let's start with the free body diagram of the cart. Once we consider a card let's start with x axis that is along the inclined plane and y axis. [00:11:46.532] normal to the incline plane. Next is weight of the cart. The self weight of the cart will be always acting vertically downward I'll call it as WC. WC stands for weight of the card that is equals to mass of the card 45 into [00:12:02.628] G9.81. This will be m. Next, this is inclined. As I said before, this will be theta, so this will also be theta. Then theta is equals to tan inverse from the triangle we know 3 by 10. [00:12:18.832] Next, after this there will be a force in this hang. That force is unknown right now, I will call it as U. [00:12:27.172] Then the components of this will be along the positive direction of x axis that will be f cos 40 and one more will be in this way f sin 40 next uh by considering the free body diagram of cart [00:12:46.212] We want to find only the value of F we only need to find the reaction at these two wheels or rollers. So I'm going to consider the first condition, the only condition which is required here is sigma fx is equal to 0. [00:13:01.476] This f is in the positive direction of x axis, so it will be plus f cos. [00:13:08.932] 40 degree. Then this W Wc the component of this WC is down the incline, so minus WC, since the angle is with respect to y axis, it will become sine. So sin theta is equal to 0. As I said before. [00:13:26.628] I need to find only the value of f so after simplifying this f will be equal to 165.6 newton. Once we know the value of f let's move to the free body diagram of the map. Next. [00:13:41.988] Consider man in the free body diagram. [00:13:49.092] It is surface here, it will there will be having two reactions. One is normal reaction, one more is frictional force, one is along and one more is normal to the nuclear. Then at G the weight of the man will be acting that is vertically downward weight of the man w m will be equals to [00:14:07.984] 82 that is mass of the man into acceleration due to gravity that is 9.81 newton. Next there will be a force through this member through the handle that will be equal and opposite to this F. So this is also F. [00:14:25.636] Now these two are one, this is along y direction, this is along x direction, we no need to find resolve them. But once we come to Wn, this is inclined with respect to y and x axis. So we should resolve it. To resolve this, since this angle is theta, this angle will also be [00:14:43.940] theta this angle will be theta the

components are going to be like this this will be wm cos theta this is the component along y direction and one more will be w m sine theta

EV0000146 · EV0000147 · EV0000148 · EV0000149

Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.540] This will be W M sine theta. [00:15:06.084] In the same vein, if this is 40 degrees, it's given here if I draw a line parallel to incline this will be 40 degree. The component is going to be f cos 40 and one more component will be log y direction f sine 40 d [00:15:27.300] Now let's start with the conditions of equilibrium. Sigma f of x is equal to zero. This n is towards the positive side, so [00:15:40.260] F a is towards the positive side, so plus then f is in the opposite direction. So minus f cos 40. [00:15:51.652] then w is w component is also in the opposite direction so minus w sine theta is equals to zero so f then minus f is 165.6 into cos 40 [00:16:08.484] then minus w m is 82 into 9.81 into sine theta is equal to 0. Then we know everything except f a so we can calculate f a f a is equal to 358 point nine Newton [00:16:27.556] So this is the frictional force acting a dip point A. Next we have to find [00:16:33.636] The one more force that is ne, apply the second condition of equilibrium, sigma f y is equals to 0. Now n e is in the positive direction, so plus n me, then w m is in the negative direction, so minus w m cos theta, then minus [00:16:50.400] F into sin 40 is equal to 0. Here except NA we know all the other terms, so N A will be equal to 870 6.93 Newton. Once we know F A then N A we know mu [00:17:06.628] Coefficient of friction muis will be equal to all this, F A by N A. [00:17:13.188] This will be equals to 0.409. We got the coefficient of static friction. Next we should know the value of yes. S we will get only when the moment equation comes because moment is forced into perpendicular. [00:17:29.168] distance. So apply the next condition of equilibrium considering moment about any point zero, here we are going to consider moment about this point A as 0. So by considering moment about A is equal to 0, considering counterclockwise positive and clockwise negative. Now [00:17:47.140] Here when we talk about momentum [00:17:49.668] Uh we know need to consider these components along X and Y. Let's consider the force vertical and horizontal because we know the coordinates of the points along x axis and along y axis, that's why it is acting in the counterclockwise direction. So plus w [00:18:07.376] M into perpendicular distances yes. Next comes f so here we have to resolve it along x and y because we know the coordinates of this point with respect to a along x axis that is 175 and along y axis that is [00:18:26.336] eight fifty so remember we have to resolve this cf along normal x and y now to resolve this cf along normal x and y like this and like this we should know the angle here how to find the angle that is the cushion [00:18:45.988] Uh here what do we know here is this angle with respect angle here will be theta. If this is theta and this is theta [00:18:57.636] 40 this angle will also become theta plus 40 degree. So the x component of f will be f cos theta plus 40 and y component will be f sin theta plus 40. [00:19:16.068] Now apply the condition moment about D is equals to zero. Since this is in the clockwise direction, it will be minus force is E of sine of [00:19:31.556] Theta plus 14 into perpendicular distances 175. Then again it is clockwise so minus of cos. [00:19:41.252] theta plus 40 into perpendicular distance will be equal to 850 that is equal to 0. Here we know the value of f we know the value of wm we know the value of theta so we just need to find the value of s so s will be equals to [00:19:59.684] 126.2 meters. So when the value of S is 126.2 meters and muus is 0.409, this man will be pulling the card with the steady speed up the incline plane.

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 5 (source interval 001200-001499 seconds): [00:20:18.564] That's it for today's class. Try to solve the problems given in the cushion bank as well as the assignment. Thank you.
EV0000154 · EV0000155 · EV0000156 · EV0000157
Full transcript: 2930 normalized words, 82 timestamped segments, 1226.133333 seconds duration. Low-confidence segments flagged: 0.
EV0000138 · EV0000139 · EV0000140 · EV0000141

# 12 CLASS 56.1 Friction_ Numerical

This lesson is indexed from the source video titled CLASS 56.1 Friction_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000158 · EV0000159

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.220] Good morning. In today's session, let's solve few more problems and friction. Question 6/33 Determine the distance S to which the 90 kg painter can climb without causing the 4 meter ladder to slip at its lower end A. The top of the 15 kg ladder has a small small roller. [00:00:20.304] And at the ground, the coefficient of static friction is 0.25, the mass center of the painter is directly above our feet. This is the figure we were supposed to follow for this particular question. [00:00:32.036] Here what is given is there is a ladder AB, the length of the ladder is 4 meter, then at A the ladder is directly resting on the ground, that's why they have given the coefficient of static friction between the ladder and the ground as point. [00:00:48.704] 25 that is mu s is equal to 0.25 at the ground. Then at the B, this roller is resting on the vertical wall and we are going to neglect the frictional force at this roller because we will assume it is an ideal condition. Next. [00:01:05.956] There is a painter, her mass is given as 90 kg. She is trying to climb the ladder. When she is at a distance of s with respect to A, then she will be in a state of equilibrium. If she further climb up the ladder, then the ladder is going to slip. [00:01:24.932] In this direction. So now for what value of this yes she will be in a state of rest, or she will be in a state of equilibrium, we have to calculate. [00:01:37.412] So let's start with the free body diagram of the ladder. I am going to consider the ladder for the free body diagram. We know the length of the ladder is 4 meter. [00:01:47.428] So weight of the ladder will be acting at a geometric center that will be 2 meter with respect to A. Similarly, 2 meter with respect to B. I call this point as G. G is the center of gravity. [00:02:06.052] of the ladder through which the weight of the ladder will be acting vertically downwards. So W L. W stands for weight of the ladder. That will be equal to mass of the ladder into G. Mass of the ladder is given as 15, so weight of the ladder is 15 into 9.81 N. [00:02:25.668] Okay. Next uh we have to consider [00:02:30.532] The reactions here at A, since the ladder is directly resting on the ground, there will be normal reaction NA as well as there will be a frictional force. Since the ladder is having a tendency to slip in the left side direction, frictional force here will be right side here, A. [00:02:47.632] Next, we know the coefficient of friction at A, so F A can be written as mu into N A.

Minus is known to us, it's given in the coefficient, so that is 0.25. So F A will be equal to 0.25 times the N A. [00:03:04.688] Next and B there there will be a normal reaction from the wall onto the roller. So NB will be horizontal because the wall is vertical. Next, the last thing is we have to consider the weight of the painter that will be acting. [00:03:20.432] Through our feet, that's what is given in the question W P weight of the painter. Weight of the painter will be equal to mass of the painter, that is 19 to 9.81 newton. So this is the complete free body diagram of the ladder. Now we have to find yes. [00:03:40.164] Before that, let's find what is the value of FE, NE, and NB. Then after that we can find the value of S. So let's start with the conditions of equilibrium. The first condition of equilibrium here is [00:03:52.656] sigma f of x is equals to 0 considering the right side component of the forces positive left side negative. F is towards right side so plus e of a then n b is towards left so minus n b is equal to 0. [00:04:08.624] These two are the only forces along X direction. So I can write FA is equals to NB. What is FA here is 0.25 times the of NA. That will be equals to N B. R in other ways N B is equals to 0.25 times the of [00:04:25.280] And I'll call this as equation one. Now, let's start with the second condition of equilibrium sigma f of y is equal to 0, considering upward component as positive, downward negative. [00:04:36.708] Na is upwards so plus Na, then weight of the painter is acting downwards, so minus WP, weight of the ladder is acting downwards, so minus W L is equal to 0. [00:04:47.204] Here we know WB that is 90 into 9.81. We know WL that is 15 into 9.81. After substituting WB and W L here we will get the value of NE. That is 1030.5 newton. Now

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.988] We can substitute this value of NA in the equation 1 and we can find the value of N B. So from equation 1, N B will be equal to 0.25 times that of Na that is 1030.5. So NB will be equal to [00:05:24.900] 257.512 Newton This is the value of N B. Next, the last condition of E helium is moment. Now [00:05:39.428] Since at A we have two forces, I am going to apply moment at A as 0 so that I can apply those two forces in this equation. Now, let's start with the moment of this force WP. Moment of this W is [00:05:55.524] W P into distance. This distance I will call it as X1. Then how to find this distance X1? To find this distance X1 I should know this angle theta. Then how to find this angle theta? To find the theta, I am going to consider this big triangle A, this is B. [00:06:12.080] I'll correspond to C. So from the triangle A B Camo [00:06:21.188] Theta. Theta will be equals to I know the hypotenuse, I know the adjacent side, so it will be equal to cos inverse of adjacent is 1.5 by hypotenuse is 4. So theta we will get it as 67.97 [00:06:39.108] Nine seven. This is the value of theta. Now once we know the value of theta, come back to this moment equation. Moment because of this WP is clockwise, so minus WP. This is the force into perpendicular distances x1. [00:06:59.140] And what is X1? X1 will be equal to S cos theta. [00:07:10.052] From the small triangle here. Then come to this WL. W n is also acting in the clockwise direction. So minus W L force into perpendicular distance will be this distance. So this will be equal to since it is [00:07:29.508] This distance is 2 and this angle is known to as theta, this will be equal to 2 cos theta. W n into 2 cos theta. [00:07:41.092] This is the moment because of the weight of the ladder. Then it comes to this NB, it is acting in the counterclockwise direction. So plus NB into perpendicular distance. Since the line of action is horizontal, the distance will be vertical, that is this C D. [00:07:57.680] Side C B. Then how to find the C B? We know C B will be equals to [00:08:05.796] under root of 4 square minus 1 point five square because a a C B is the right angle triangle. So now this will be equal to 0. So minus WP [00:08:22.948] weight of the painter weight of

the painter we know as 882 point [00:08:28.516] 9 newton into X1. X1 we know S cos theta. Then minus weight of the ladder. Weight of the ladder is 15 into 9.81. That is equal to 1 for 14. [00:08:43.364] one forty seven point one five into two cos theta is sixty seven point nine seven then plus N B N B we know N B is two fifty seven point [00:09:01.796] 512 into C B. C B is under root of 4 square minus 1.5 square that is 3.708 will be equal to 0. In this except yes, we know all the other terms. So s will be equals to [00:09:21.764] 2.5 meter. So when the painter, when she is at a distance of 2.55 meter, she will be in a state of equilibrium. If she further climbs up the ladder, then the ladder is going to slip down the plane. That is what the meaning. [00:09:40.676] Next question 6 slash 36. The homogeneous square body is positioned as shown in the figure. [00:09:47.652] If the coefficient of static friction at base power four determine the critical value of angle theta which will which slipping will below which slipping will occur. Then friction at A. So this is the given figure.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.444] Here what is given here is there is the homogeneous square body and that body is having a roller at A and the body is resting directly on this horizontal surface at B. Now [00:10:16.356] The side A B of this body is making an angle of theta with respect to this horizontal surface. It is that value of theta for which the body will be in a state of equilibrium. If there is a if the value of this theta decreases, then the body is definitely going to slide or slip. [00:10:35.812] Along this surface, it's going to move in this direction. So for what value of this theta, maximum value of theta, it will be in a state of equilibrium, we have to calculate. Let's start with the free body diagram of this homogeneous body. [00:10:57.892] Since the body is in direct contact with the surface, there will be a normal reaction as well as there will be a friction force. So normal reaction will be like this N B. Similarly [00:11:09.700] This body will have a tendency to slip in the right side direction, the frictional force here will be towards left side. I will call that as FB. FB will be equal to mu into normal reaction.4 N this year normal reaction is NB. Next, there will be [00:11:28.964] Reaction from this surface onto this roller that will be always making an angle of 90 degrees with the surface. Normal reaction NE. [00:11:43.236] Normal reaction at A. Next we have to consider the weight of this body, homogeneous body. For that I am going to first find out the geometric center of this. So since a square, this distance will be S by 2, and this is also S by 2. Then I am going to [00:12:02.116] Consider draw one line the distance of s by 2 then s by 2. Now this is the geometric center of this rectangular body. So this is the point through which the weight of this body will be acting vertically downward. [00:12:19.812] Since it we don't know exactly what is this mass or weight, I'm going to call it this is just W. Next. [00:12:30.980] This is along its direction, so we no need to find the angle. This is along y direction, no need to find the angle. But here it is inclined. What is given here is this angle is 60 degrees. This is 60 degree. I am going to draw a horizontal line here. [00:12:47.524] So this angle will become 60. If this is 60, this angle will become 30 because the angle between this norm reaction and the surface is 90 degree. That's why this angle will become 30 degree. Now let's start with the conditions of equilibrium. [00:13:07.524] First, let's find what is Na, what is NB. Then we will find the value of theta. Let's start with the first condition of equilibrium, sigma f of x is equal to 0, considering the right side component as positive. [00:13:23.908] X component of this Na will be towards right side. So plus Na into angle will be called 30 degree. Then F B is towards left side. So minus 0.4 N B will be equals to 0. [00:13:44.420] I can write here N B is equals to 2.165 times of NE. This is the relationship between N and we will

call it as equation one. [00:13:57.828] Now let's start with the second condition of equilibrium that is sigma f of y is equal to 0, taking upward as positive. [00:14:05.924] The upward component of N A will be Na sin thirty. Since it is upwards it's plus plus Na sin thirty and W is acting downwards so minus W then N B is acting upwards so plus N B is equals to 0. [00:14:23.940] Next. So N A will be equals to [00:14:29.860] once N A will be equals to W divided by 2.664 because we know N B is 2.165 times zero in A. So after substituting that value here [00:14:47.524] We will get this expression and call this is equation 2. Next, the last condition of equilibrium, moment about any point 0, I'm going to apply this at B because at B I have two forces. So moment about B is equal to 0, taking counterclockwise is positive. Now this is having

EV0000166 · EV0000167 · EV0000168 · EV0000169

Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.332] Two components, one is horizontal and one is vertical. This is NA cos thirty and this is NA sin 30. [00:15:20.708] Let's start with the horizontal component, it is acting in the clockwise direction. So minus Ne cos 30 into [00:15:30.852] This is force. Into perpendicular distance we need to find. Then how to find the perpendicular distance. For that I am going to consider this right angle triangle. You can see there is a rectangle triangle here, it's like this. This side is this is A and this is B. [00:15:48.708] A B we know it's the one of the side of a square, it will be S and this angle we know theta. Then if I call this point as C then this will be equal to S cos theta and this will be equals to S sine theta. [00:16:09.092] That's why here for this particular force the perpendicular distance will be the height of this triangle that will be equal to s sin theta. S sine theta. Then come to this force it is acting in the clockwise direction again. So minus N A [00:16:28.772] sine thirty degree into perpendicular distance. Perpendicular distance will be the base of this channel that will be S cos theta. Next come to this W for W [00:16:44.804] You have to remember one thing. If you consider W as it is, then we don't know this perpendicular distance and we can't find it so easily. So what we can do here is since we know the distance of this B with respect to this point, it is S by 2. Similarly, we know [00:17:04.100] This distance that is also S by 2. So it's better to resolve this W1 along this line and one more along this line. Since we know the coordinates of this particular point G, that is the point to which [00:17:20.352] The mass of this body is acting vertically downward. Since this is S by 2 and S by 2, we know the coordinates of this G with respect to B. That's why resolve it along this line and normal to this line. For this, we should know the angle. The angle here will be connect. [00:17:39.684] We have to look into this figure. This angle is theta. So if I draw a horizontal line here, this angle will be theta. If this is theta, this angle will be also theta because mutually perpendicular. So this component will be equals to W. [00:17:56.036] Cos theta this will be W sine theta. Let's start with this component which is parallel to the side AB. It is clockwise so minus W sine theta. [00:18:12.996] perpendicular distance will be S by two. Then come to the component. [00:18:19.908] This component it is counterclockwise, so plus W into cos theta into perpendicular distance will be s by 2. This will be equal to 0. Now we have to simplify this expression. This will be um [00:18:40.388] We have sine theta here, we have sine theta here also. [00:18:48.964] And we have cos theta here and we have cos theta here. Once we simplify this, we are going to get the expression like this. [00:19:01.572] Sine t sine theta into point A to five will be equals to cos teta into point three one [00:19:13.604] 23 then sin by cos will be equals to tan theta that will be equals to 0.3785 then theta will be equals to tan inverse of 0.3785 that is equals to twenty point seven three. [00:19:30.052] This is the value of theta for which this homogeneous body

will be in a state of test. If there is a decrease in the value of theta, then this is going to slide down the plane. That is what the meaning. [00:19:47.876] That's it for today's class. Thank you.

EV0000170 · EV0000171 · EV0000172 · EV0000173

Full transcript: 2850 normalized words, 78 timestamped segments, 1191.488 seconds duration. Low-confidence segments flagged: 0.

EV0000158 · EV0000159 · EV0000160 · EV0000161

# 13 CLASS 56.2 Friction_ Numerical

This lesson is indexed from the source video titled CLASS 56.2 Friction_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000174 · EV0000175

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Good morning. In today's session, let's solve few problems and fiction. Problem six slash three. [00:00:08.356] The force peace applied to the 50 kg block when it is at rest. Determine the magnitude and direction of the friction force exerted by the surface on the block if the [00:00:22.116] P is equals to zero, second P is equals to two hundred newton, third case when P is to fifty newton, fourth, when what value of P is required to [00:00:32.832] Initiate motion up the incline. The coefficient of static and the kinetic friction between the black and the inclined mu is equals to 0.25 and mu case equals to 0.2 respectively. [00:00:46.756] Here what is given is there is an inclined plane. This inclination of the plane with the horizontal is given as fifteen degree. Then there is a fifty kg crate resting on this inclined plane. Then [00:01:04.900] It is in a state of rest. Then we are going to apply this force P. Due to this force, this block will have a tendency to move up the [00:01:15.312] Plane. So when P is equals to zero and two hundred and when P is two fifty newton, they've asked us to find the friction force between the contacting surface, that is between the block and the inclined plane. [00:01:31.012] Then in the last they have asked us for what value of this P this block will start to move up the inclined plane. So these four things we have to find in this question. [00:01:44.944] And they have given the coefficient of static friction as 0.25 and kinetic friction as point two. Now let's start with the free body diagram of the block. That's when p is equals to zero. Now [00:02:00.644] Since we are considering a block on an inclined plane, we should consider the x ax along the inclined plane and y axis normal to the inclined plane. Next, the weight of this block will be vertically downward. Since the mass is given as 50, weight will be 19 to [00:02:20.004] 50 sorry fifty into nine point eight one so w will be downward like this. Then the components of this W parallel x axis will be up the incline sored plane and the component of W [00:02:36.240] Parallel to y axis will be downward like this. Then P, the components of P will be right and upward. Since here it is P is 0, you can neglect. [00:02:49.124] This for time being, then there will be a normal reaction from the surface onto the plane that will be upward. Since the value of P is 0 this block will have a tendency to slide down the plane. So the frictional force will be up the plane.

[00:03:07.172] This is the complete free body diagram of the block. Now we can apply the conditions of equilibrium. Let's start with the first one sigma f is equals to zero. [00:03:18.372] Since n is upward, it will it it will be plus n and the y component of w is downwards, so minus w into sine so cos 15 degree because the angle 15 is with respect to [00:03:32.736] Why the axis so cost 15 degree. No need to consider P because P is 0. So we will get the value of N as 473.78 newton. Now apply the second condition of equilibrium sigma if x is equal to 0. [00:03:50.020] F is right side, so plus f the p component is right side, since the value of p is zero, no need to consider this. Then the x component of this w is down the plane, so that will be minus w into sine fifteen degree. [00:04:07.172] From this we can find the value of F. F is 126.95 newton. Now we should know whether the block is in a state of rest or is it moving downward. That we will get to know. [00:04:21.792] Only when we compare this frictional force with the maximum frictional value. [00:04:28.032] We know f max is equals to mu s into n, here mu s is 0.25 and n we know from here it is 475.78. Simplify we are going to get the value of f maxes under an 18.44 newton. [00:04:45.488] Since F is greater than F max, this condition is not at all possible. It is impossible. So the block is actually moving down the plane. If the block is moving down the plane, then the friction between the block and the surface will be.

EV0000174 · EV0000175 · EV0000176 · EV0000177

Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.248] Kinetic friction that will be equals to mu c and two n. Here mu k is point two and n we know four seventy three point seven eight. So actual friction between the block and the [00:05:17.504] plane when p is equals to zero is seventy four point seven five newton and that is up the plane because the block is moving down the plane. [00:05:28.996] So remember this is the value of F when P is zero. Next, the second case when the value of P is two hundred newton. It is the same free body diagram normal to the consider x axis along the plane, but [00:05:48.164] Like this, x axis and the y axis in this direction, normal to the x axis, then the cell fate vertically downward, then the components of W will be like this, then components of P going to be P cos 20 and P sin 20. Normal reaction up the plane and the frictional force will be. [00:06:07.488] Since the value of P is 200 newton we are going to assume because of this P this block will have a tendency to move up the plane, so the frictional force will be down the plane. [00:06:18.500] That's why I have considered the friction in this direction. Then again the same conditions of equilibrium, sigma f of y and sigma f of x is equal to 0. Let's start with sigma f y 0. N is upwards to plus n. Then [00:06:35.812] This P sin 20 is also upwards, so plus P sin 20, W is downward, so minus W into cos 15 degree will be equal to 0. Here only one unknown that is N. [00:06:50.788] Substitute the value of P s 200, we'll get the value of NS 542.2 newton. Next sigma Fx is equal to 0. [00:07:01.380] Once we consider sigma fx is equal to 0, f is left side, so minus f, x component of p is towards right, so plus p cos twenty. [00:07:12.224] X component of W is towards left, so minus W into sin twelve fifteen will be equals to zero. Here also only one unknown that is F. So F will be equals to 60.99 newton. [00:07:27.168] Now we should know whether this block is in a state of rest or is it moving up the plane. Further what we can do here is we have to compare this [00:07:36.800] Value of F with respect to F max. We know that F max is equal to mu S into N. N muus is point two five and n is five forty two point two. Find the value of F max. We'll get F max as 135.55 newton. Now compare F. [00:07:53.024] With respect to F max, since F is smaller than F max, this indicates the body is in a state of rest. Once the body is under rest, the frictional force will be this itself, that is [00:08:05.296] Sixty point nine nine N and since the answer is positive that indicates our assumption is correct. We assumed the block is having a

tendency to move up the plane, so frictional forces down the plane. This is correct. The third condition [00:08:24.100] When P is two fifty N, the same wavy body diagram. [00:08:31.300] Just consider X axis along the plane, Y axis [00:08:34.916] normal to the x axis then w vertically downward components of w then uh resolve p along x and y then normal reaction. Since the value of p is two fifty newton we will assume the block is moving up the plane so fictional force will be [00:08:52.336] down the plane. Apply the conditions of equilibrium sigma f y 0. N is upward, so plus n, then p sin twenty is also upward, so plus p sin 20, then the y component of w is downwards, so minus w [00:09:07.696] Cause fifteen will be equal to zero. Here also only un one unknown that is n. So n we will get as five fifty nine point three Newton. Next second condition of equilibrium, sigma f of x is equal to zero. Uh f is towards left, so minus f [00:09:26.980] P component is towards right. So plus P cos 20 and W is left side, so minus W sin 15 will be equals to 0. Here also only one unknown that is CF. We'll get the value of F as 107.97 N. [00:09:43.760] Now we should know whether the block is in state of rest or it is under motion. That we will get to know only when we compare the value of this F with respect to F max. So F max is equal to mu s into n. N is uh [00:09:59.504] double five five nine point three and s is point two five we'll get the value of f max is one thirty five point five five newton. Since f is smaller than f max this indicates the body is under rest. So therefore f will be equals to

EV0000178 · EV0000179 · EV0000180 · EV0000181

Transcript block 3 (source interval 000600-000899 seconds): [00:10:20.932] 107.95 newton and we assume the block is having a tendency to move up the plane so fictional forces down the plane. And we got the answer also as positive. [00:10:35.236] This is the last case for what value of P this fifty kg block will start to move up the incline. That's what we have to calculate. The solution remains the same. [00:10:49.252] Again we have to start with a free body diagram. Consider X along the plane and the Y axis normally to the plane and the weight vertically downward. Components of W are going to be like this. Then the components of P are going to be right side and upward, then up. [00:11:04.420] Yeah. It's going to be P cos twenty and this is going to be P sin twenty. Normal reaction will be upward. Since the block is moving up the plane, the frictional force will be down the plane. [00:11:17.508] Next, start with the conditions of equilibrium. Sigma F of Y is equals to 0. N is upwards, so plus N. P component of P is upwards, so plus P sin 20, component of W is downwards, so minus W [00:11:32.260] Cos 15 will be equals to 0. Here we have two unknowns, one is P and one more is N. So we are going to get the equation N plus 0.342 P minus 473.78 will be equal to 0. Call it as equation one. [00:11:51.940] Now apply the second condition of equilibrium that is sigma fx is equal to 0. [00:11:57.860] Since the component of F is left side, so minus F, it is right side, so plus P e c cost twenty, it is left side, so minus W sine fifteen will be equal to zero. Here also we have two unknowns F and P. [00:12:13.328] Since the block is under motion, we can replace F as mu K into N because here the frictional force will be kinetic friction that will be always equal to mu K into N. Here mu K is given as [00:12:28.336] Point two. So it is 0.2 N plus 0.94 times that of P minus 126.95 will be equal to zero. Once we solve equation one and two, we are going to get the value of P s [00:12:43.264] two thirty nine point three newton. So for this value of this P, this block will start to move up the incline. That is for the meaning. [00:12:57.188] Next question, problem 6 slash 6. So here we have to calculate coefficient of static friction between the drum and between the surface, and at the same time, what is the value of P and F we have to calculate. Now let's start with the free body diagram of the drum. [00:13:16.576] Uh here as I said before always x axis should be taken along the inclined

plane. So this is going to be x axis and the y-axis will be always normal to the inclined plane and y axis is going to be this way. [00:13:30.564] And uh the weight since the mass is given as m, we have to consider the weight w vertically downward. So w will be equals to m into g. [00:13:42.052] The components of W are going to be X component is going to be down the plane and the Y component is going to be downward and the angle here, since this angle is 15 degree, the angle between W and the Y-axis will also be 15 degree. [00:13:57.152] So it's going to be W sine fifteen and one more component is going to be W cos fifteen degree. Next uh there will be a normal reaction from the surface unto the drum that will be upward and before that [00:14:12.720] P is that so we have to resolve this P. To resolve the P we should know the angle. If I draw a line passing through this, which is parallel to X axis, the angle between the P and the line parallel to X axis will be fifteen degree because that is going to be thirty minus fifteen, that will be fifteen degree. So [00:14:30.448] The component is going to be P P cos fifteen and one more will be P sine fifteen degree. The next is normal reaction of the plane and it is going to move in this direction, the drum is moving in this direction, so the frictional force will be up the incline plane. [00:14:50.404] Next, we have to apply the conditions of equilibrium. The radius of the drum is given as R and the radius of the inner up is R by 2. If you think of sigma Fx and FY, we have two unknowns. P is also your unknown and n is also unknown.

EV0000182 · EV0000183 · EV0000184 · EV0000185

Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.992] So along y direction we'll have two unknowns. If you think of sigma fx zero, f is a unknown, but we can express f as mu s into n, that is correct. [00:15:19.024] But N is also an unknown. So one unknown then again we have P along X direction also. So we have two unknowns along X and Y direction. So let's start with the moment condition. [00:15:32.964] We can consider moment about the center of the drum as 0. So moment about O let's consider this point as O. So moment about O as 0. Once we consider moment about O as 0, the moment due to W sine 15, then [00:15:49.712] W cos fifteen, then P sin 15 will be equal to 0 because they're acting at the same point. So the moment will be 0. So we are left with only three two forces P cos 15 and F. [00:16:07.364] Since P is clockwise, it is going to be negative. Minus P cos 15 into perpendicular distance will be R by 2. Then moment of F will be equal to this counterclockwise direction. So plus F into R will be equal to 0. [00:16:29.124] So we'll get the value of f in terms of p that is equals to 0.483 times set of p. Now we'll apply sigma f x 0. Once we apply sigma fx 0 plus f minus w sine 15 plus p cos 15 will be equals to 0. [00:16:47.428] As we know the value of f substitute here f is point four eight three times that of p minus w is m into g into sin fifteen plus p cos fifteen. So once we [00:17:02.564] Solve this we are going to get the value of P as 0.78.1786 times it of mg. Next apply the next condition of equilibrium sigma f y 0 n is upwards so plus n then p sin 15 is also upward so plus [00:17:20.708] Then W cos 15 is downwards or minus w will be equal to 0. We already know the value of P and P in terms of Mg, so substitute here W is also equal to Mg. We will get the value of NS 0.919 times of Mg. [00:17:40.196] Next we should know the value of coefficient of static friction mu s it will be equal to f by n. F is 0.483 times the of P and N is 0.919 times the of EMG. We'll get the value of mu s as 0.094. [00:17:57.040] This is the coefficient of static friction between the drum and the surface for which this will have a tendency to roll up the plane. That's it for today's class. Try to solve the problems given in the forum. Thank you.

EV0000186 · EV0000187 · EV0000188 · EV0000189

Full transcript: 2841 normalized words, 73 timestamped segments, 1094.314667 seconds duration.
Low-confidence segments flagged: 0.
EV0000174 · EV0000175 · EV0000176 · EV0000177

# 14 Source slide deck: Moment of Inertia

This entry preserves the complete slide deck titled Moment of Inertia as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_04, to avoid an unsupported match. EV0000210 · EV0000211

### Graphs and diagrams

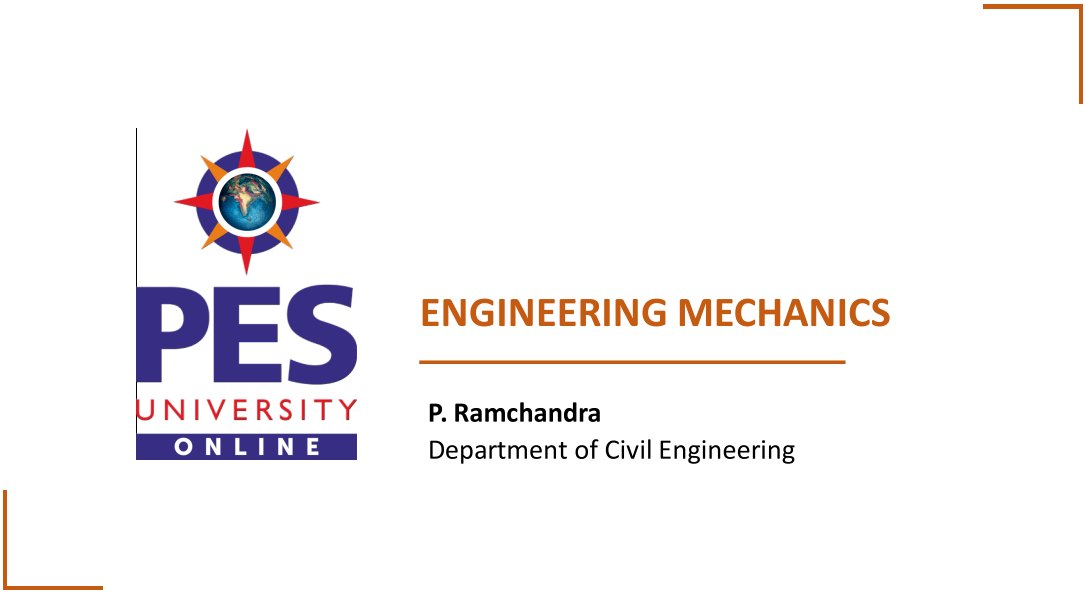


**Figure G000009.** Moment of Inertia · source slide page 1.

EV0000211

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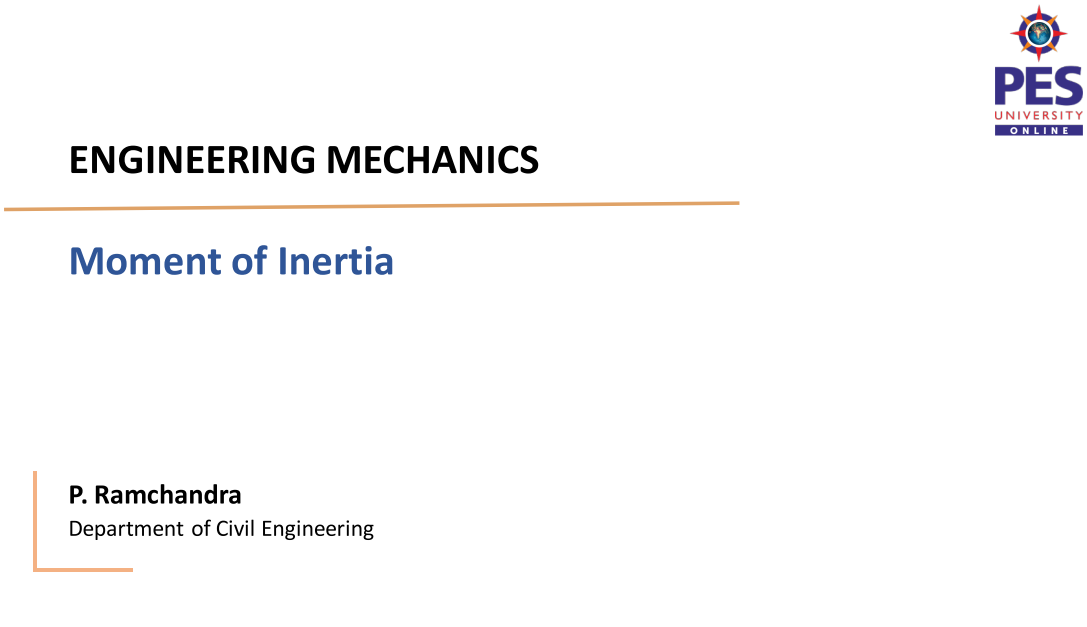


**Figure G000010.** Moment of Inertia · source slide page 2.

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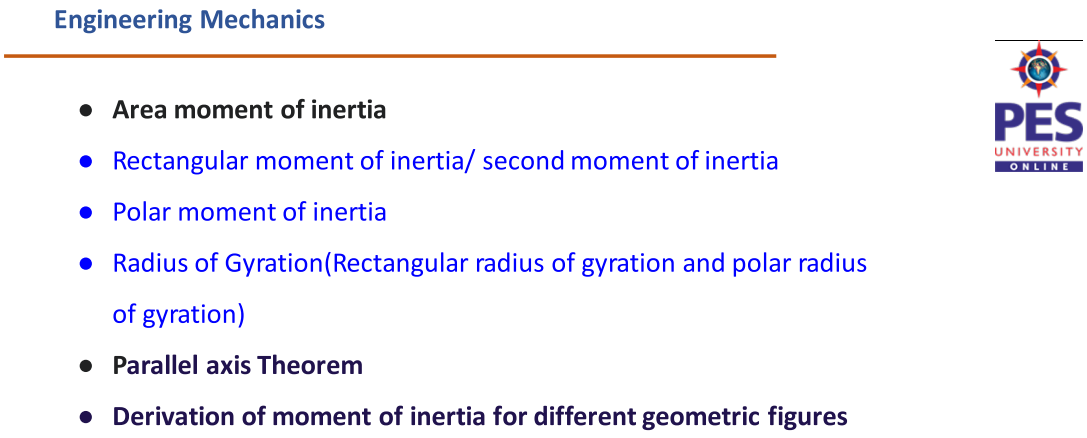


**Figure G000011.** Moment of Inertia · source slide page 3.

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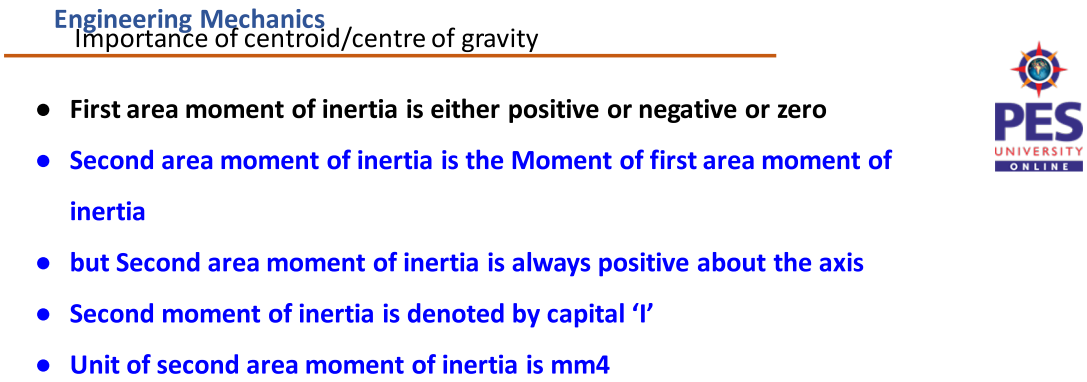


**Figure G000012.** Moment of Inertia · source slide page 4.

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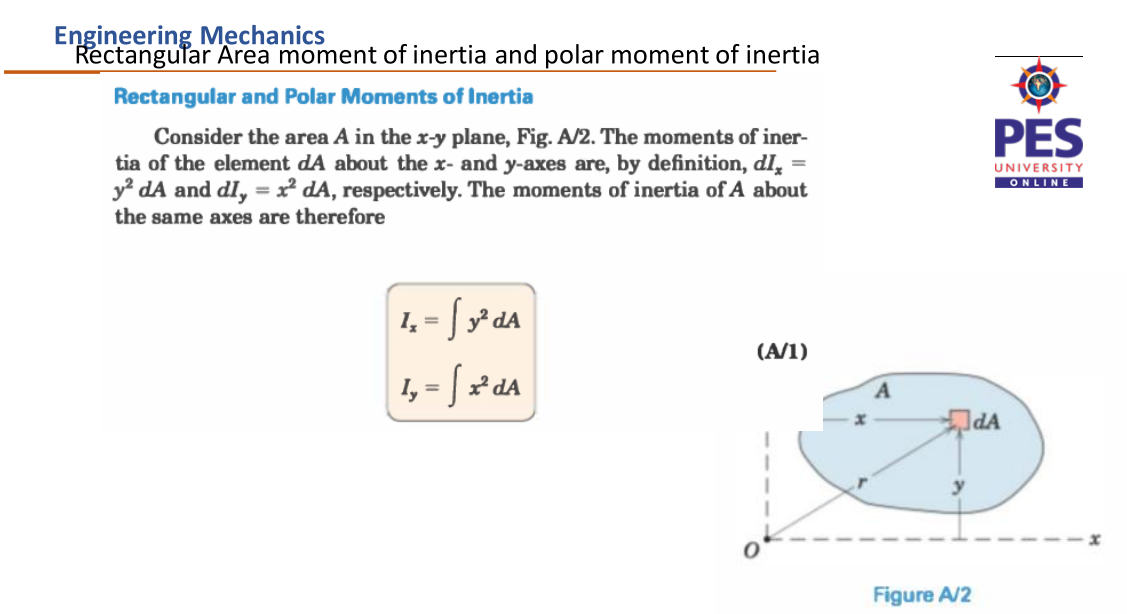


**Figure G000013.** Moment of Inertia · source slide page 5.

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Engineering Mechanics
Rectangular Area moment of inertia and polar moment of inertia

The moment of inertia of $dA$ about the pole $O$ ($z$-axis) is, by similar definition, $dI_z = r^2\,dA$. The moment of inertia of the entire area about $O$ is

$$I_z = \int r^2\,dA \qquad \text{(A/2)}$$

The expressions defined by Eqs. A/1 are called *rectangular* moments of inertia, whereas the expression of Eq. A/2 is called the *polar* moment of inertia.* Because $x^2 + y^2 = r^2$, it is clear that

$$I_z = I_x + I_y \qquad \text{(A/3)}$$

**Figure G000014.** Moment of Inertia · source slide page 6.

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Engineering Mechanics
Radius of gyration

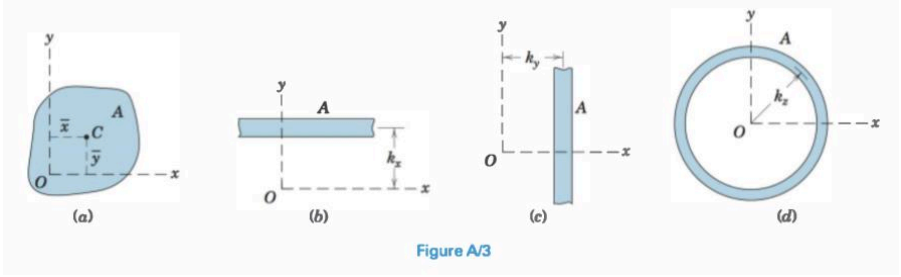


Figure A/3

$$I_x = k_x^2 A \qquad I_y = k_y^2 A \qquad I_z = k_z^2 A$$

or

$$k_x = \sqrt{I_x/A} \qquad k_y = \sqrt{I_y/A} \qquad k_z = \sqrt{I_z/A}$$

$$k_z^2 = k_x^2 + k_y^2$$

**Figure G000015.** Moment of Inertia · source slide page 7.

EV0000229

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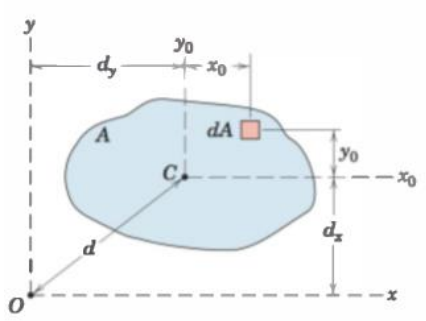
Engineering Mechanics
Transfer axis theorem




- The moment of inertia of an area about a non centroidal axis may be easily expressed in terms of the moment of inertia about a parallel centroidal axis.

**Figure G000016.** Moment of Inertia · source slide page 8.

EV0000232

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Engineering Mechanics
Transfer axis theorem

- By definition, the moment of inertia of the element dA about the x-axis is given by

$$dI_x = (y_0 + d_x)^2\, dA$$

**Expanding and integrating give us**

$$I_x = \int y_0^2\, dA + 2d_x \int y_0\, dA + d_x^2 \int dA$$

**Figure G000017.** Moment of Inertia · source slide page 9.

EV0000235

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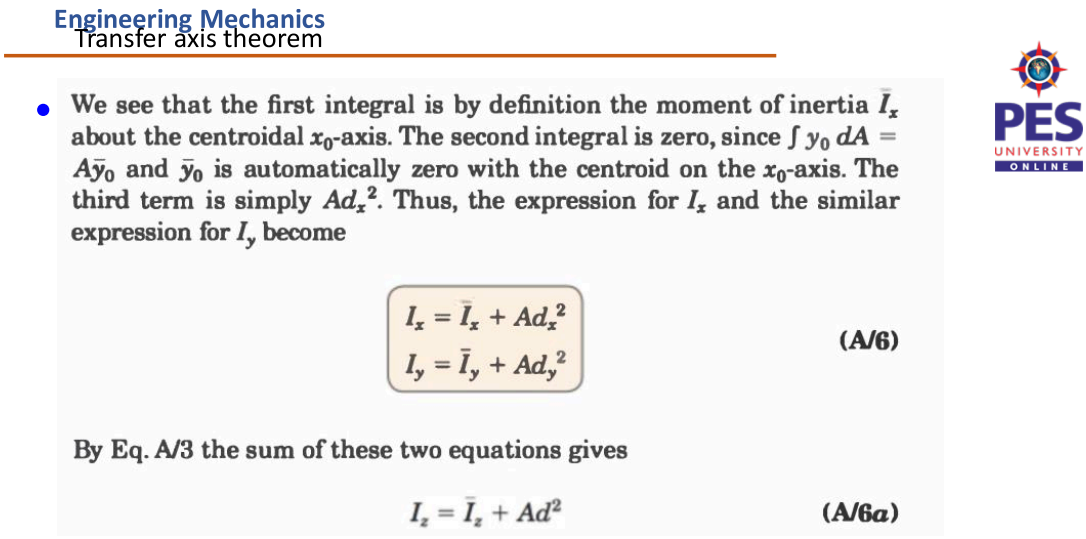


**Figure G000018.** Moment of Inertia · source slide page 10.

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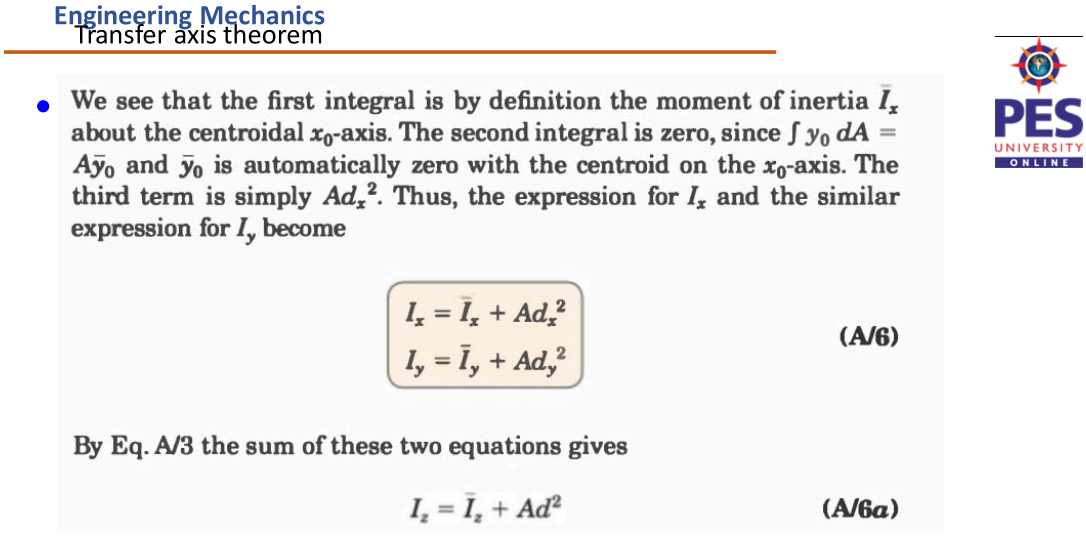


**Figure G000019.** Moment of Inertia · source slide page 11.

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Engineering Mechanics

Transfer axis theorem

- **Conditions for parallel axis theorem**

1. **two axis should be there and two axis must be parallel to each other**
2. **Between two axis, one axis has to pass through the centroidal axis**

**Figure G000020.** Moment of Inertia · source slide page 12.

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Engineering Mechanics

Transfer axis theorem

$[I_x = \int y^2\, dA]$ $\quad \bar{I}_x = \int_{-h/2}^{h/2} y^2 b\, dy = \frac{1}{12} bh^3$ *Ans.*

**By interchange of symbols, the moment of inertia about the centroidal $y_0$-axis is**

$$\bar{I}_y = \frac{1}{12} hb^3 \quad \textit{Ans.}$$

**The centroidal polar moment of inertia is**

$[\bar{I}_z = \bar{I}_x + \bar{I}_y]$ $\quad \bar{I}_z = \frac{1}{12}(bh^3 + hb^3) = \frac{1}{12} A(b^2 + h^2)$ *Ans.*

**By the parallel-axis theorem, the moment of inertia about the $x$-axis is**

$[I_x = \bar{I}_x + Ad_x^2]$ $\quad I_x = \frac{1}{12} bh^3 + bh\left(\frac{h}{2}\right)^2 = \frac{1}{3} bh^3 = \frac{1}{3} Ah^2$ *Ans.*

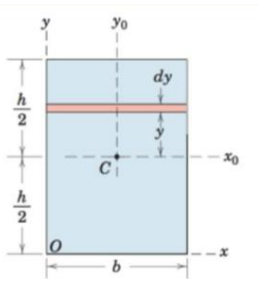


**Figure G000021.** Moment of Inertia · source slide page 13.

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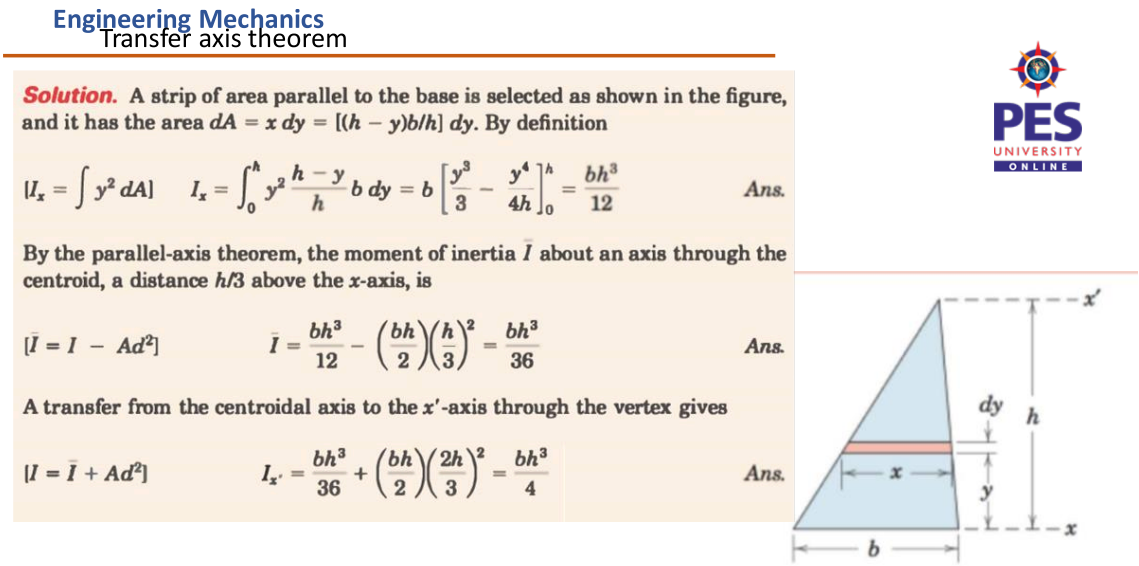


**Figure G000022.** Moment of Inertia · source slide page 14.

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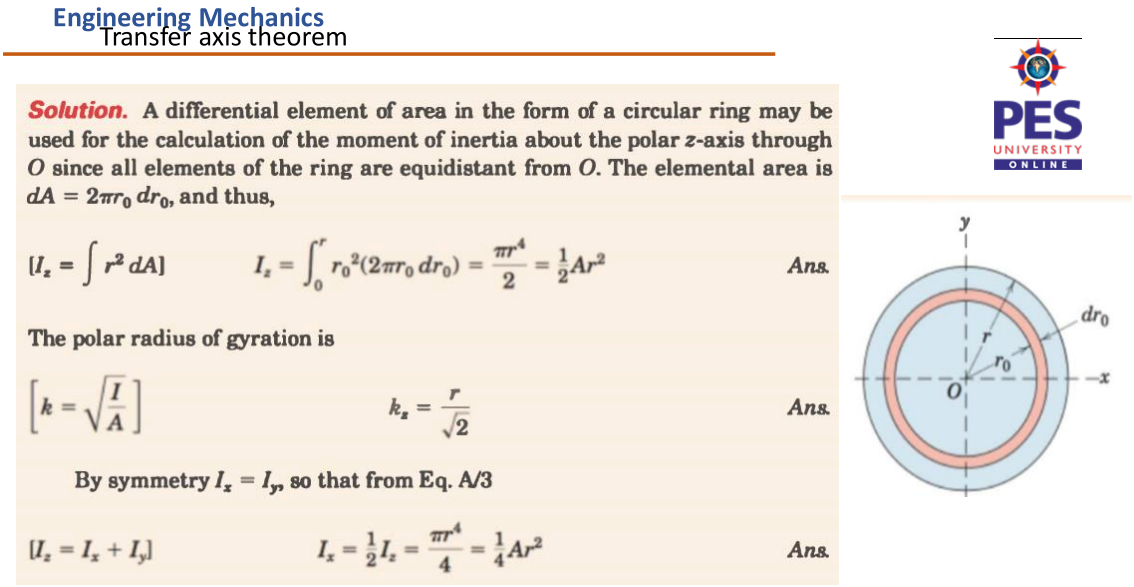


**Figure G000023.** Moment of Inertia · source slide page 15.

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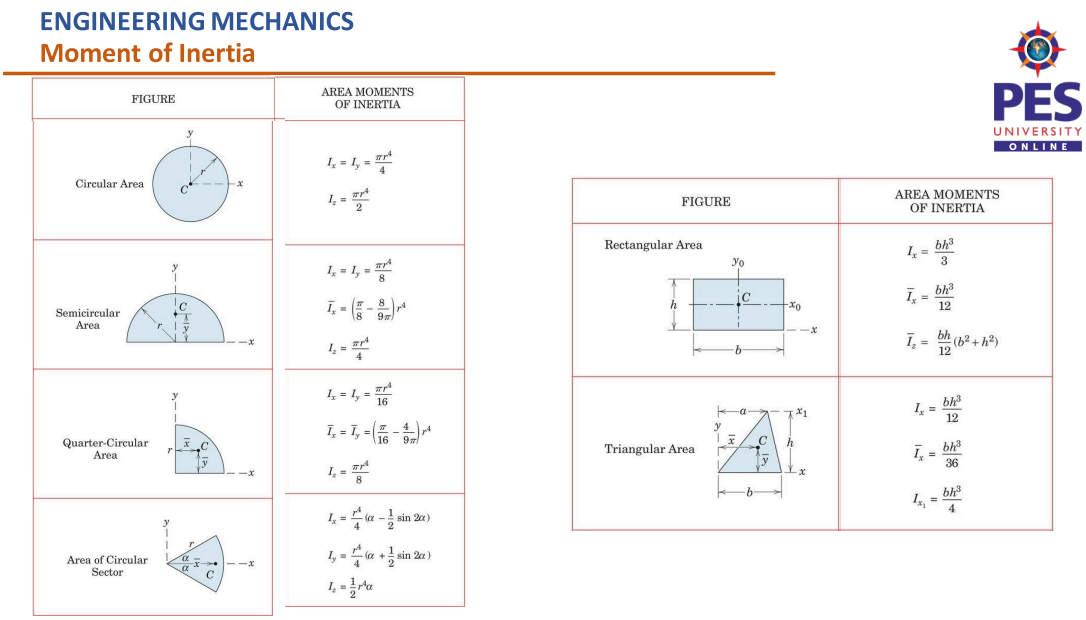


**Figure G000024.** Moment of Inertia · source slide page 16.

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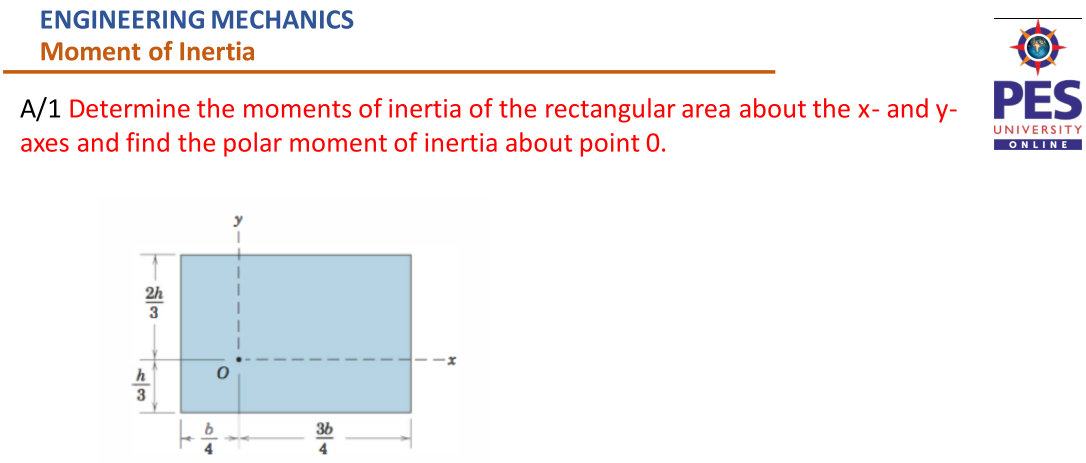


**Figure G000025.** Moment of Inertia · source slide page 17.

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ENGINEERING MECHANICS
Moment of Inertia

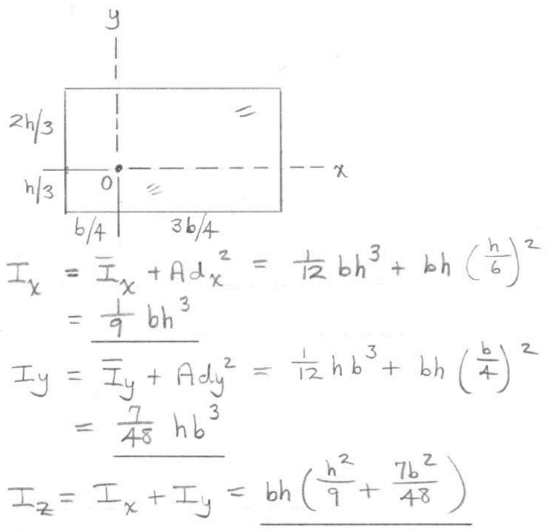


**Figure G000026.** Moment of Inertia · source slide page 18.

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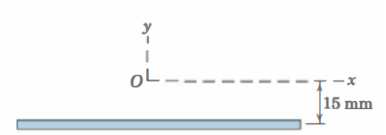
ENGINEERING MECHANICS
Moment of Inertia

A/3 The narrow rectangular strip has an area of 300 mm$^2$, and its moment of inertia about the y-axis is 35(10$^3$) mm$^4$. Obtain a close approximation to the polar radius of gyration about point 0.




$I_x \approx Ad^2 = 300(15)^2 = 67.5(10^3)\ mm^4$

$J_O = I_x + I_y = 67.5(10^3) + 35(10^3) = 102.5(10^3)\ mm^4$

$k_O = \sqrt{J_O/A} = \sqrt{\frac{102.5(10^3)}{300}} = 18.48\ mm$

**Figure G000027.** Moment of Inertia · source slide page 19.

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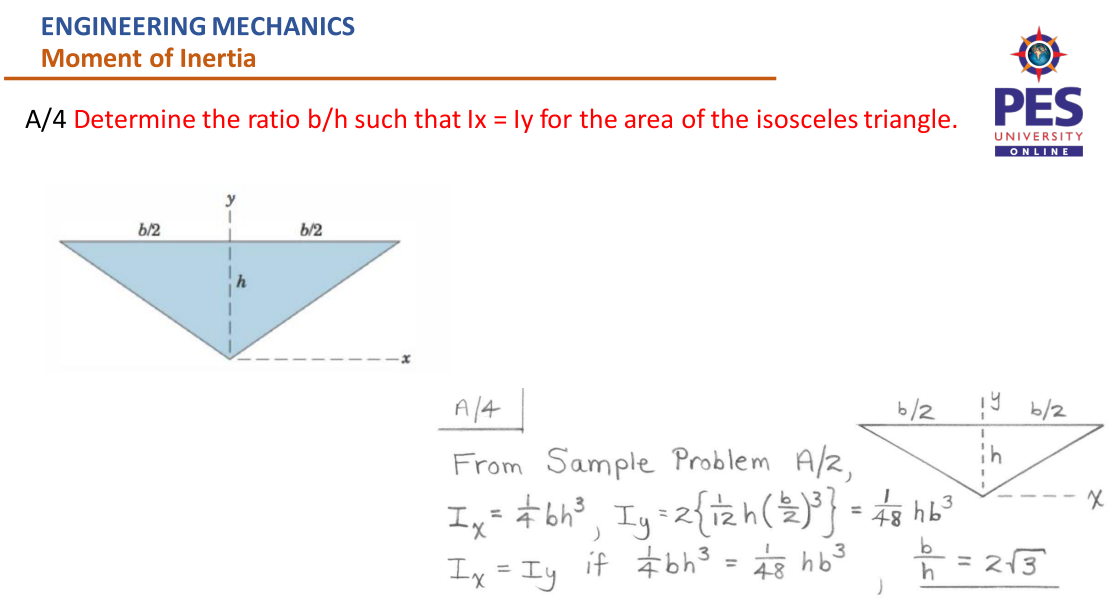


**Figure G000028.** Moment of Inertia · source slide page 20.

EV0000268

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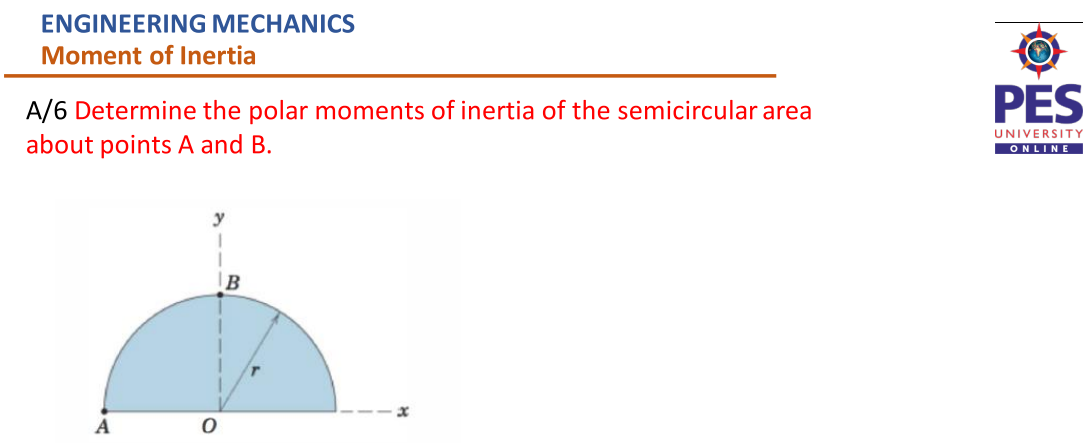


**Figure G000029.** Moment of Inertia · source slide page 21.

EV0000271

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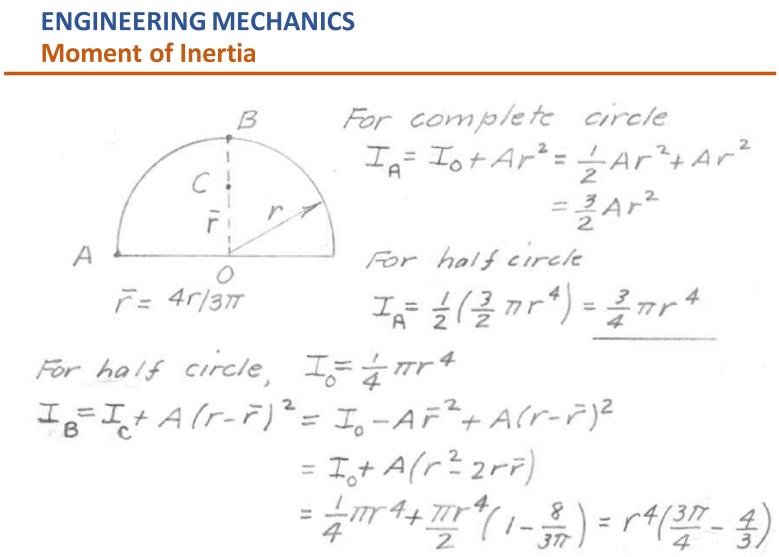


**Figure G000030.** Moment of Inertia · source slide page 22.

EV0000274

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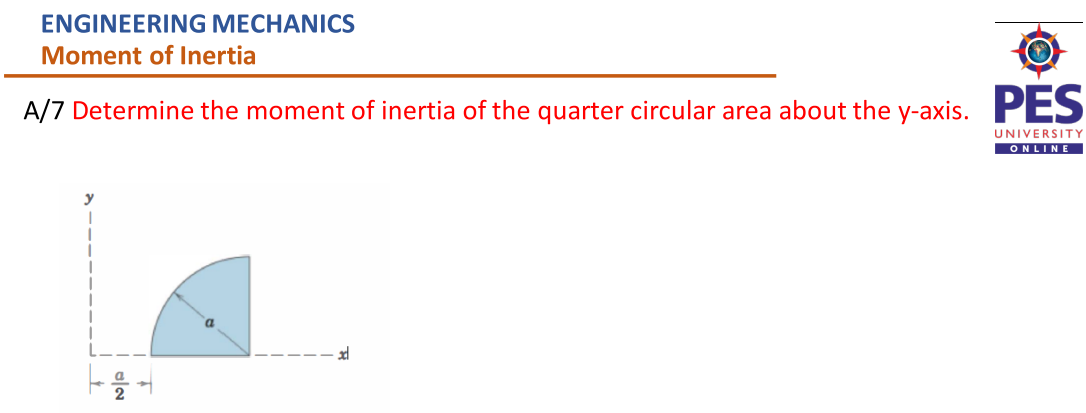


**Figure G000031.** Moment of Inertia · source slide page 23.

EV0000277

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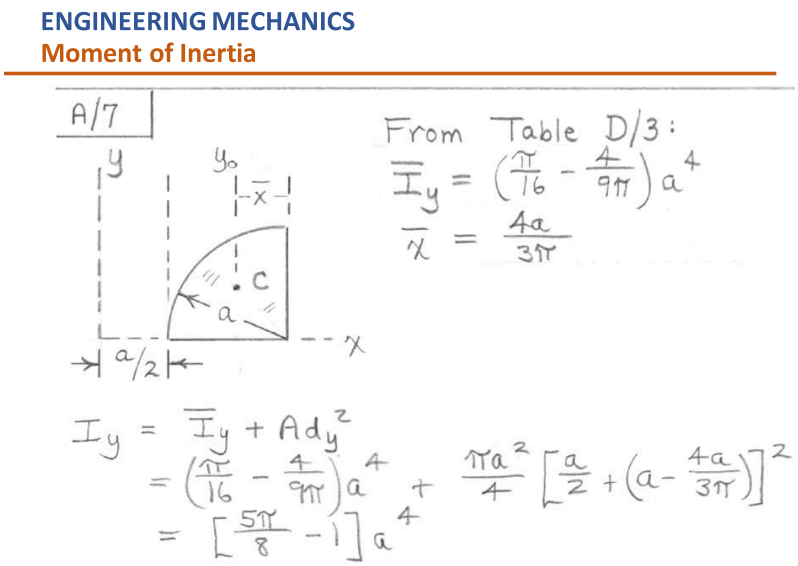


**Figure G000032.** Moment of Inertia · source slide page 24.

EV0000280

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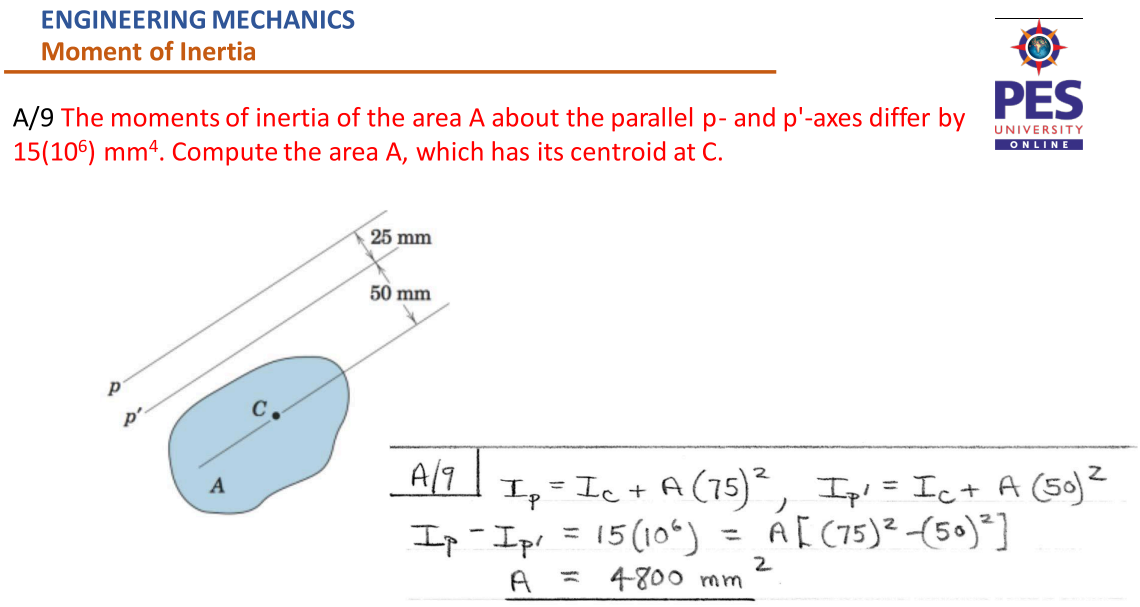


**Figure G000033.** Moment of Inertia · source slide page 25.

EV0000283

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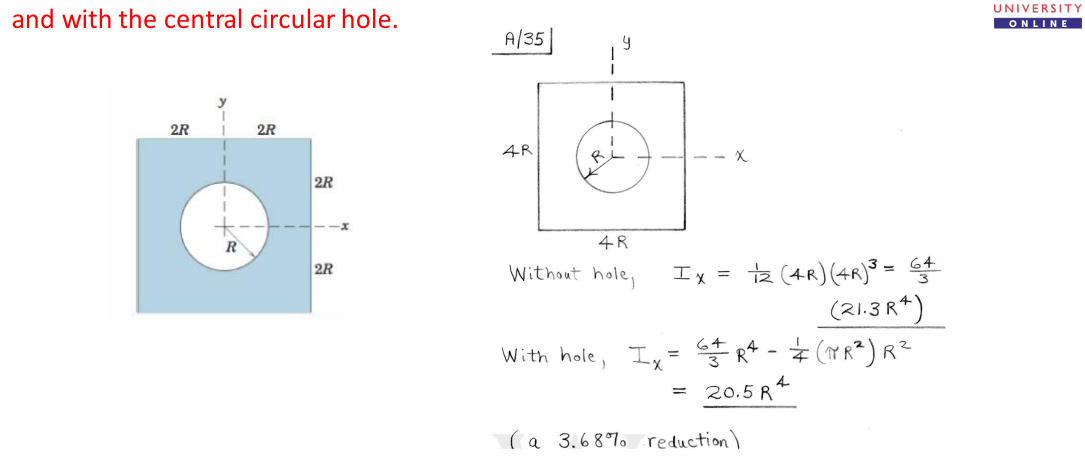


**Figure G000034.** Moment of Inertia · source slide page 26.

EV0000286

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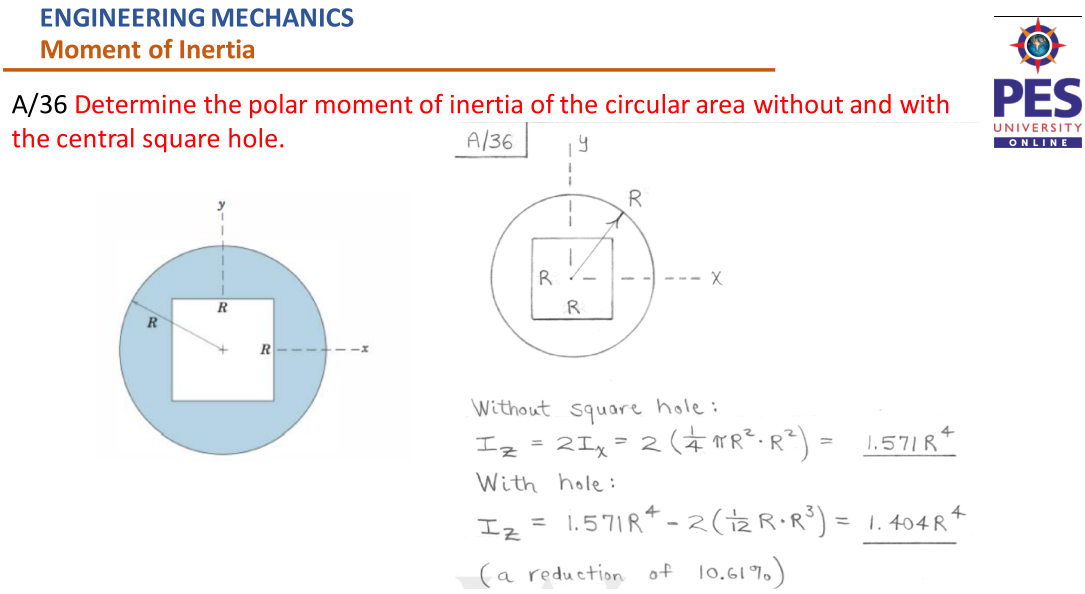


**Figure G000035.** Moment of Inertia · source slide page 27.

EV0000289

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ENGINEERING MECHANICS
Moment of Inertia

A/37 Calculate the polar radius of gyration of the area of the angle section about point A Note that the width of the legs is small compared with the length of each leg.

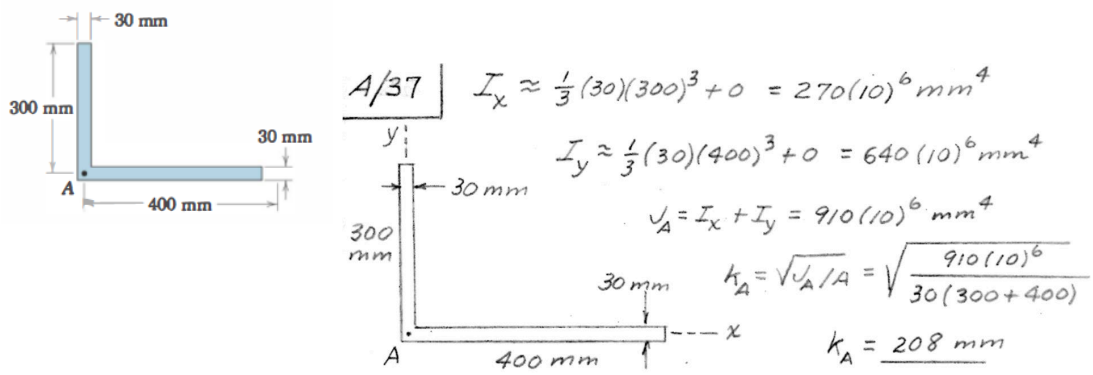


**Figure G000036.** Moment of Inertia · source slide page 28.

EV0000292

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ENGINEERING MECHANICS
Moment of Inertia

A/38 Calculate the polar radius of gyration of the area of the angle section about point A Note that the width of the legs is small compared with the length of each leg.

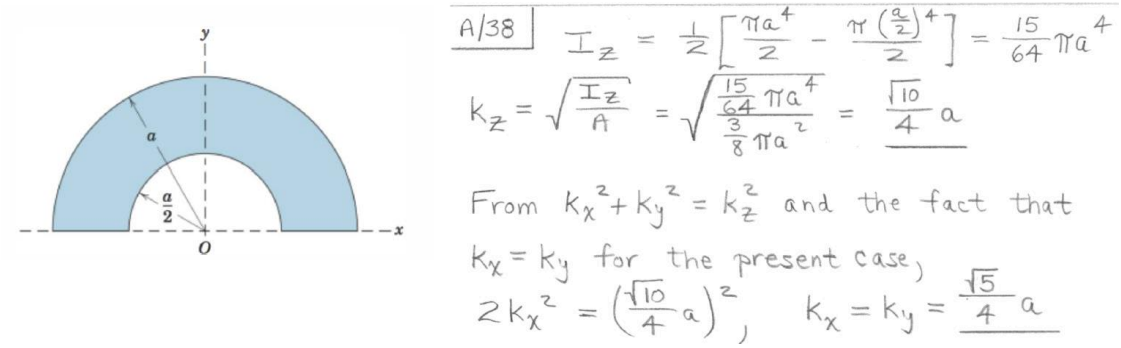


**Figure G000037.** Moment of Inertia · source slide page 29.

EV0000295

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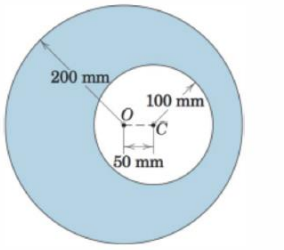
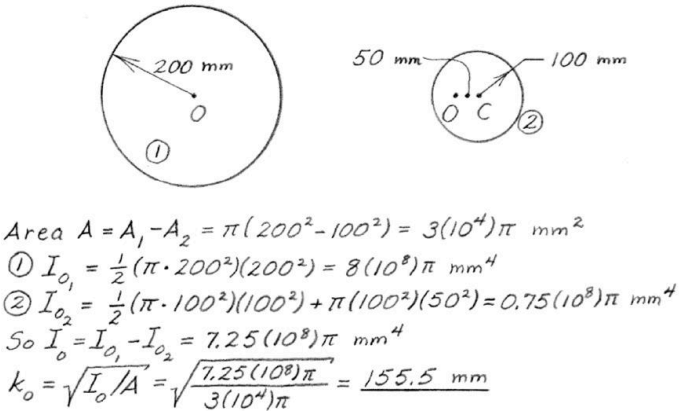
ENGINEERING MECHANICS
Moment of Inertia

A/39 Calculate the polar radius of gyration of the shaded area about the center 0 of the larger circle.




Area $A = A_1 - A_2 = \pi(200^2 - 100^2) = 3(10^4)\pi \text{ mm}^2$

① $I_{O_1} = \frac{1}{2}(\pi \cdot 200^2)(200^2) = 8(10^8)\pi \text{ mm}^4$

② $I_{O_2} = \frac{1}{2}(\pi \cdot 100^2)(100^2) + \pi(100^2)(50^2) = 0.75(10^8)\pi \text{ mm}^4$

So $I_O = I_{O_1} - I_{O_2} = 7.25(10^8)\pi \text{ mm}^4$

$k_O = \sqrt{I_O/A} = \sqrt{\frac{7.25(10^8)\pi}{3(10^4)\pi}} = 155.5 \text{ mm}$

**Figure G000038.** Moment of Inertia · source slide page 30.

EV0000298

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
Moment of Inertia

A/40 Determine the percent reductions in both area and area moment of inertia about the y-axis caused by removal of the rectangular cutout from the rectangular plate of baseband height h.

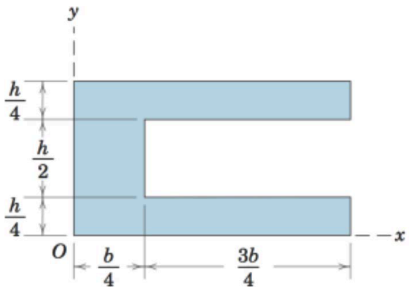



**Figure G000039.** Moment of Inertia · source slide page 31.

EV0000301

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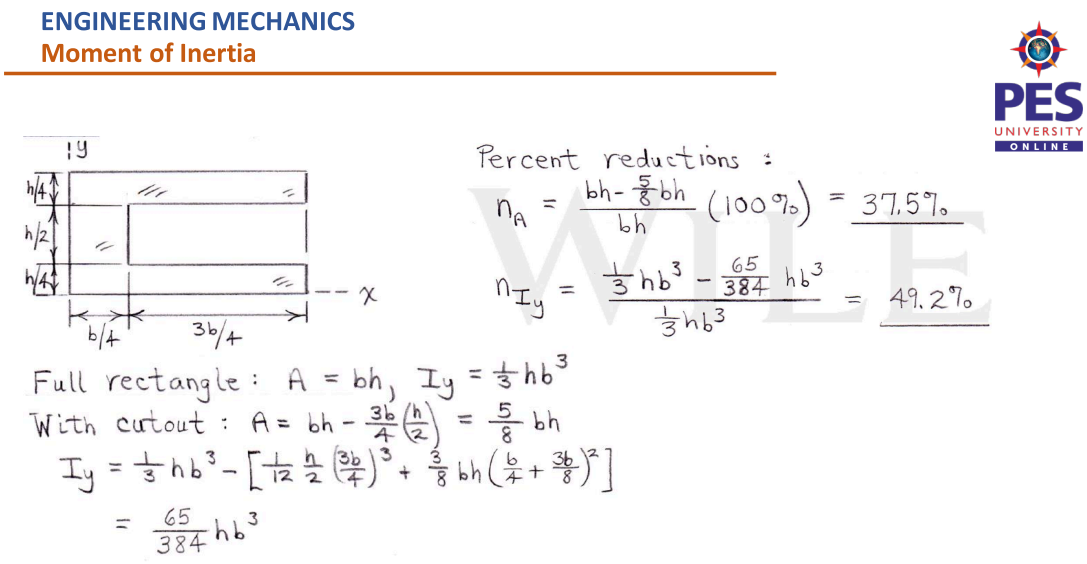


**Figure G000040.** Moment of Inertia · source slide page 32.

EV0000304

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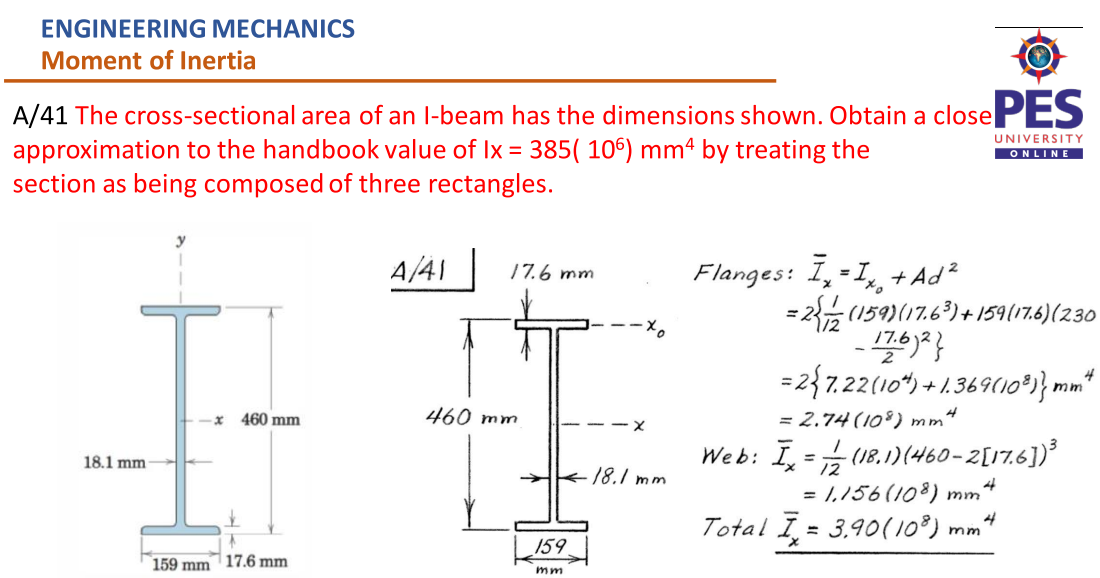


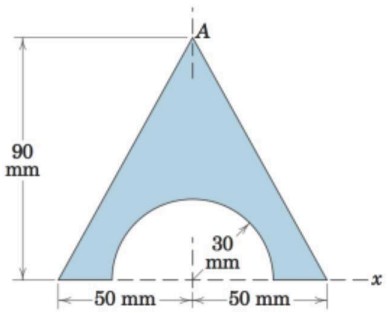

**Figure G000041.** Moment of Inertia · source slide page 33.

EV0000307

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
Moment of Inertia

A/42 Calculate the moment of inertia of the shaded area about the x-axis.



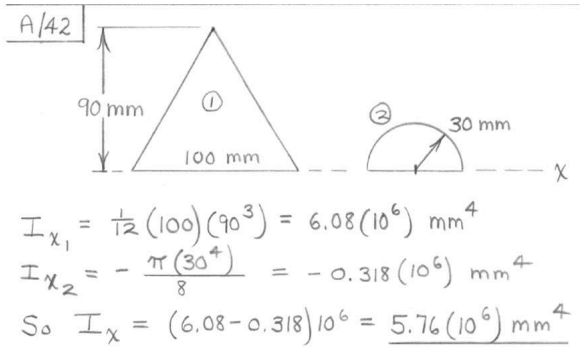




**Figure G000042.** Moment of Inertia · source slide page 34.

EV0000310

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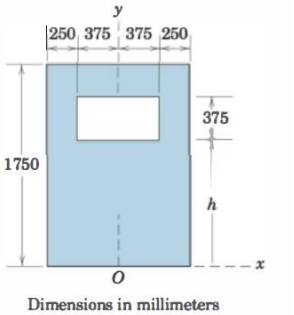
ENGINEERING MECHANICS
Moment of Inertia

A/43 The variable h designates the arbitrary vertical location of the bottom of the rectangular cutout within the rectangular area. Determine the area moment of inertia about the x-axis for (a) h = 1000 mm and (b) h = 1500 mm.



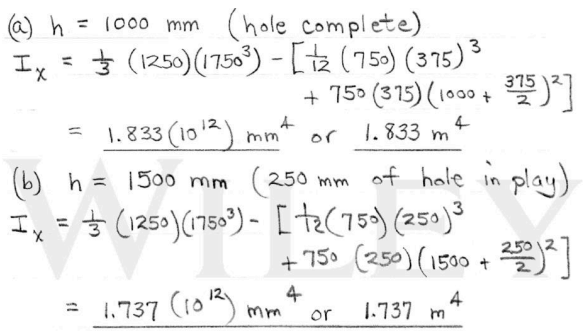



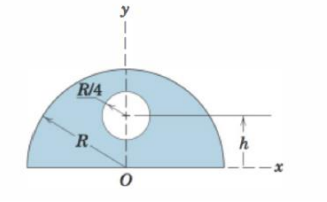

**Figure G000043.** Moment of Inertia · source slide page 35.

EV0000313

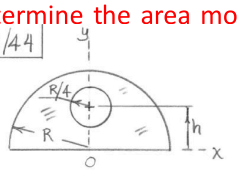
***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
Moment of Inertia

A/44 The variable h designates the arbitrary vertical location of the center of the circular cutout within the semicircular area. Determine the area moment of inertia about the x-axis for (a) h = 0 and (b) h = R/2.




A/44




(a) $h = 0$ (One-half of hole considered)

$$I_x = \frac{\pi R^4}{8} - \frac{\pi (R/4)^4}{8} = \frac{255}{2048}\pi R^4 \quad (0.391 R^4)$$

(b) $h = \frac{R}{2}$ (Entire hole now in play)

$$I_x = \frac{\pi R^4}{8} - \left[\frac{\pi (R/4)^4}{4} + \pi\left(\frac{R}{4}\right)^2\left(\frac{R}{2}\right)^2\right]$$

$$= \frac{111}{1024}\pi R^4 \quad (0.341 R^4)$$

**Figure G000044.** Moment of Inertia · source slide page 36.

EV0000316

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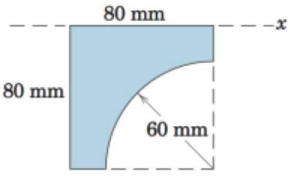
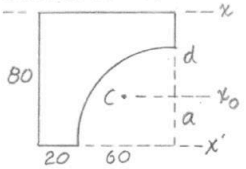
ENGINEERING MECHANICS
Moment of Inertia

A/45 Calculate the moment of inertia of the shaded area about the x-axis.




A/45

Dimen. in mm




Square: $I_x = \frac{1}{3}b^4 = \frac{1}{3}(80)^4 = 13.65(10^6)\ mm^4$

Quarter-circle: $a = \frac{4r}{3\pi} = \frac{4(60)}{3\pi} = 25.46\ mm$

$d = 80 - 25.46 = 54.54\ mm$

$$I_x = I_{x_0} + Ad^2 = I_{x'} - Aa^2 + Ad^2$$

$$= \frac{-1}{4}\frac{\pi r^4}{4} - \frac{\pi r^2}{4}(d^2 - a^2) = -\frac{\pi r^2}{4}\left(\frac{r^2}{4} + d^2 - a^2\right)$$

$$= -\frac{\pi (60)^2}{4}\left[\frac{60^2}{4} + (54.54)^2 - (25.46)^2\right]$$

$$= -9.120(10^6)\ mm^4$$

Total $I_x = (13.65 - 9.120)(10^6) = 4.53(10^6)\ mm^4$

**Figure G000045.** Moment of Inertia · source slide page 37.

EV0000319

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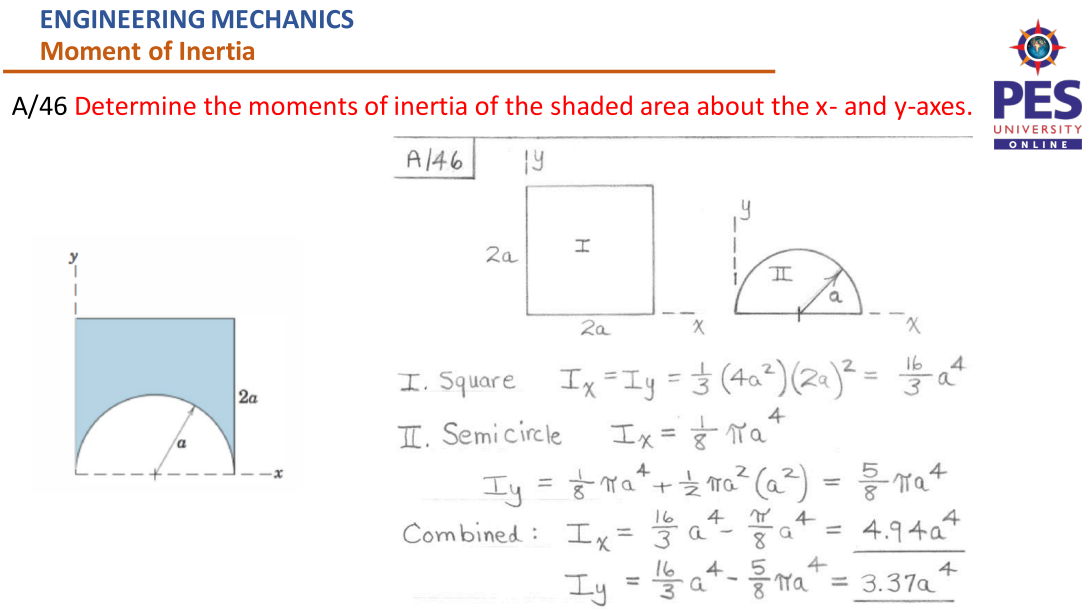


**Figure G000046.** Moment of Inertia · source slide page 38.

EV0000322

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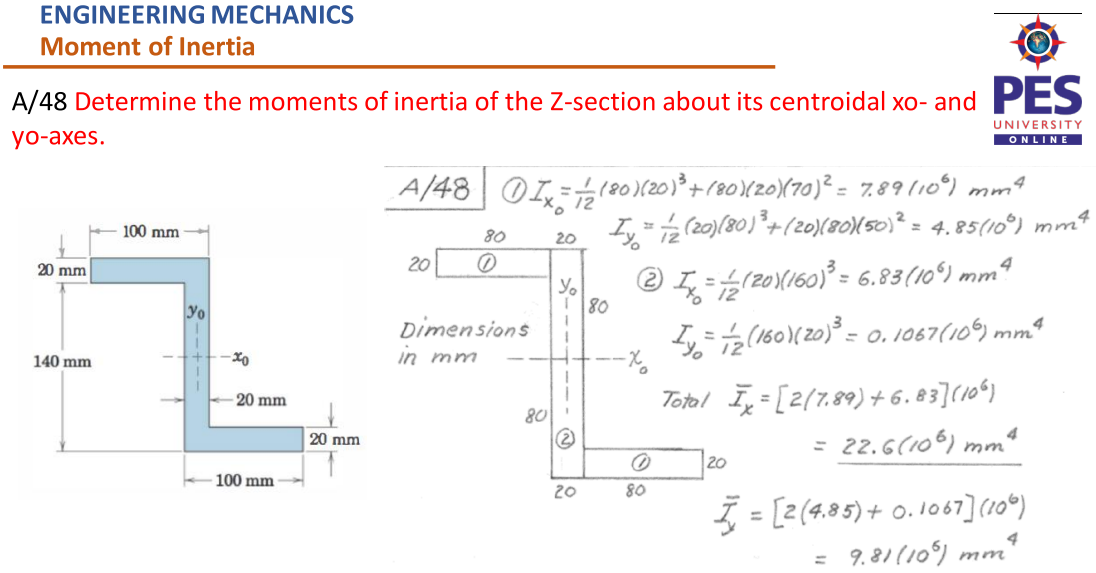


**Figure G000047.** Moment of Inertia · source slide page 39.

EV0000325

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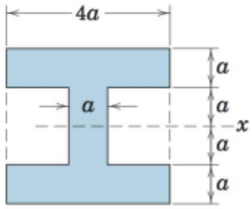
ENGINEERING MECHANICS
Moment of Inertia

A/49 Determine the moment of inertia of the shaded area about the x-axis in two different ways.



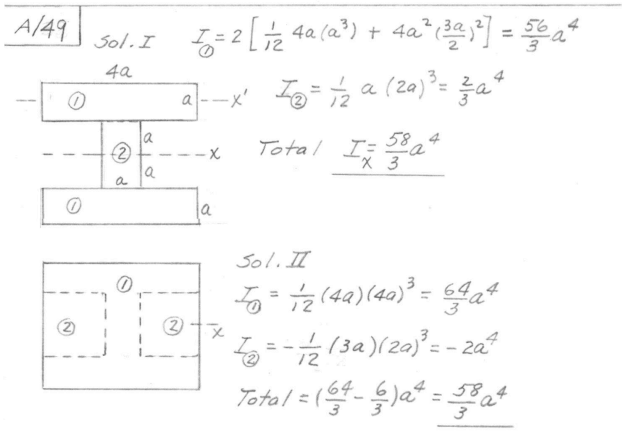




**Figure G000048.** Moment of Inertia · source slide page 40.

EV0000328

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

THANK YOU

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**Figure G000049.** Moment of Inertia · source slide page 41.

EV0000331

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS P. Ramchandra Department of Civil Engineering EV0000210 · EV0000211 · EV0000212
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS Moment of Inertia P. Ramchandra Department of Civil Engineering EV0000213 · EV0000214 · EV0000215
3. Source slide page 3 (extracted text): Engineering Mechanics ●Area moment of inertia ●Rectangular moment of inertia/ second moment of inertia ●Polar moment of inertia ●Radius of Gyration(Rectangular radius of gyration and polar radius of gyration) ●Parallel axis Theorem ●Derivation of moment of inertia for different geometric figures EV0000216 · EV0000217 · EV0000218
4. Source slide page 4 (extracted text): Engineering Mechanics Importance of centroid/centre of gravity ●First area moment of inertia is either positive or negative or zero ●Second area moment of inertia is the Moment of first area moment of inertia ●but Second area moment of inertia is always positive about the axis ●Second moment of inertia is denoted by capital 'I' ●Unit of second area moment of inertia is mm4 EV0000219 · EV0000220 · EV0000221
5. Source slide page 5 (extracted text): Engineering Mechanics Rectangular Area moment of inertia and polar moment of inertia EV0000222 · EV0000223 · EV0000224
6. Source slide page 6 (extracted text): Engineering Mechanics Rectangular Area moment of inertia and polar moment of inertia EV0000225 · EV0000226 · EV0000227
7. Source slide page 7 (extracted text): Engineering Mechanics Radius of gyration EV0000228 · EV0000229 · EV0000230
8. Source slide page 8 (extracted text): Engineering Mechanics Transfer axis theorem ●The moment of inertia of an area about a non centroidal axis may be easily expressed in terms of the moment of inertia about a parallel centroidal axis. EV0000231 · EV0000232 · EV0000233
9. Source slide page 9 (extracted text): Engineering Mechanics Transfer axis theorem ● By definition, the moment of inertia of the element dA about the x- axis is given by EV0000234 · EV0000235 · EV0000236
10. Source slide page 10 (extracted text): Engineering Mechanics Transfer axis theorem ● EV0000237 · EV0000238 · EV0000239
11. Source slide page 11 (extracted text): Engineering Mechanics Transfer axis theorem ● EV0000240 · EV0000241 · EV0000242
12. Source slide page 12 (extracted text): Engineering Mechanics Transfer axis theorem ●Conditions for parallel axis theorem 1. two axis should be there and two axis must be parallel to each other 2. Between two axis, one axis has to pass through the centroidal axis EV0000243 · EV0000244 · EV0000245
13. Source slide page 13 (extracted text): Engineering Mechanics Transfer axis theorem EV0000246 · EV0000247 · EV0000248
14. Source slide page 14 (extracted text): Engineering Mechanics Transfer axis theorem EV0000249 · EV0000250 · EV0000251
15. Source slide page 15 (extracted text): Engineering Mechanics Transfer axis theorem EV0000252 · EV0000253 · EV0000254
16. Source slide page 16 (extracted text): ENGINEERING MECHANICS Moment of Inertia EV0000255 · EV0000256 · EV0000257

17. Source slide page 17 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/1 Determine the moments of inertia of the rectangular area about the x- and y- axes and find the polar moment of inertia about point 0. EV0000258 · EV0000259 · EV0000260
18. Source slide page 18 (extracted text): ENGINEERING MECHANICS Moment of Inertia EV0000261 · EV0000262 · EV0000263
19. Source slide page 19 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/3 The narrow rectangular strip has an area of 300 mm2, and its moment of inertia about the y-axis is 35(103) mm4. Obtain a close approximation to the polar radius of gyration about point 0. EV0000264 · EV0000265 · EV0000266
20. Source slide page 20 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/4 Determine the ratio b/h such that Ix = Iy for the area of the isosceles triangle. EV0000267 · EV0000268 · EV0000269
21. Source slide page 21 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/6 Determine the polar moments of inertia of the semicircular area about points A and B. EV0000270 · EV0000271 · EV0000272
22. Source slide page 22 (extracted text): ENGINEERING MECHANICS Moment of Inertia EV0000273 · EV0000274 · EV0000275
23. Source slide page 23 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/7 Determine the moment of inertia of the quarter circular area about the y-axis. EV0000276 · EV0000277 · EV0000278
24. Source slide page 24 (extracted text): ENGINEERING MECHANICS Moment of Inertia EV0000279 · EV0000280 · EV0000281
25. Source slide page 25 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/9 The moments of inertia of the area A about the parallel p- and p'-axes differ by 15(106) mm4. Compute the area A, which has its centroid at C. EV0000282 · EV0000283 · EV0000284
26. Source slide page 26 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/35 Determine the moment of inertia about the x-axis of the square area without and with the central circular hole. EV0000285 · EV0000286 · EV0000287
27. Source slide page 27 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/36 Determine the polar moment of inertia of the circular area without and with the central square hole. EV0000288 · EV0000289 · EV0000290
28. Source slide page 28 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/37 Calculate the polar radius of gyration of the area of the angle section about point A Note that the width of the legs is small compared with the length of each leg. EV0000291 · EV0000292 · EV0000293
29. Source slide page 29 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/38 Calculate the polar radius of gyration of the area of the angle section about point A Note that the width of the legs is small compared with the length of each leg. EV0000294 · EV0000295 · EV0000296
30. Source slide page 30 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/39 Calculate the polar radius of gyration of the shaded area about the center 0 of the larger circle. EV0000297 · EV0000298 · EV0000299

31. Source slide page 31 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/40 Determine the percent reductions in both area and area moment of inertia about the y-axis caused by removal of the rectangular cutout from the rectangular plate of baseband height h. EV0000300 · EV0000301 · EV0000302
32. Source slide page 32 (extracted text): ENGINEERING MECHANICS Moment of Inertia EV0000303 · EV0000304 · EV0000305
33. Source slide page 33 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/41 The cross-sectional area of an I-beam has the dimensions shown. Obtain a close approximation to the handbook value of Ix = 385( 106) mm4 by treating the section as being composed of three rectangles. EV0000306 · EV0000307 · EV0000308
34. Source slide page 34 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/42 Calculate the moment of inertia of the shaded area about the x-axis. EV0000309 · EV0000310 · EV0000311
35. Source slide page 35 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/43 The variable h designates the arbitrary vertical location of the bottom of the rectangular cutout within the rectangular area. Determine the area moment of inertia about the x-axis for (a) h = 1000 mm and (b) h = 1500 mm. EV0000312 · EV0000313 · EV0000314
36. Source slide page 36 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/44 The variable h designates the arbitrary vertical location of the center of the circular cutout within the semicircular area. Determine the area moment of inertia about the x-axis for (a) h = 0 and (b) h = R/2. EV0000315 · EV0000316 · EV0000317
37. Source slide page 37 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/45 Calculate the moment of inertia of the shaded area about the x-axis. EV0000318 · EV0000319 · EV0000320
38. Source slide page 38 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/46 Determine the moments of inertia of the shaded area about the x- and y-axes. EV0000321 · EV0000322 · EV0000323
39. Source slide page 39 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/48 Determine the moments of inertia of the Z-section about its centroidal xo- and yo-axes. EV0000324 · EV0000325 · EV0000326
40. Source slide page 40 (extracted text): ENGINEERING MECHANICS Moment of Inertia A/49 Determine the moment of inertia of the shaded area about the x-axis in two different ways. EV0000327 · EV0000328 · EV0000329
41. Source slide page 41 (extracted text): THANK YOU P. Ramchandra Department of Civil Engineering ramachandrap@pes.edu +91 9845347257 Extn 736 EV0000330 · EV0000331 · EV0000332

SHORT-NOTE RECAP

Source deck preserved: Moment of Inertia. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000210

# 15 Source slide deck: UE20CV101 UNIT 4 SESSION 6

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## Graphs and diagrams

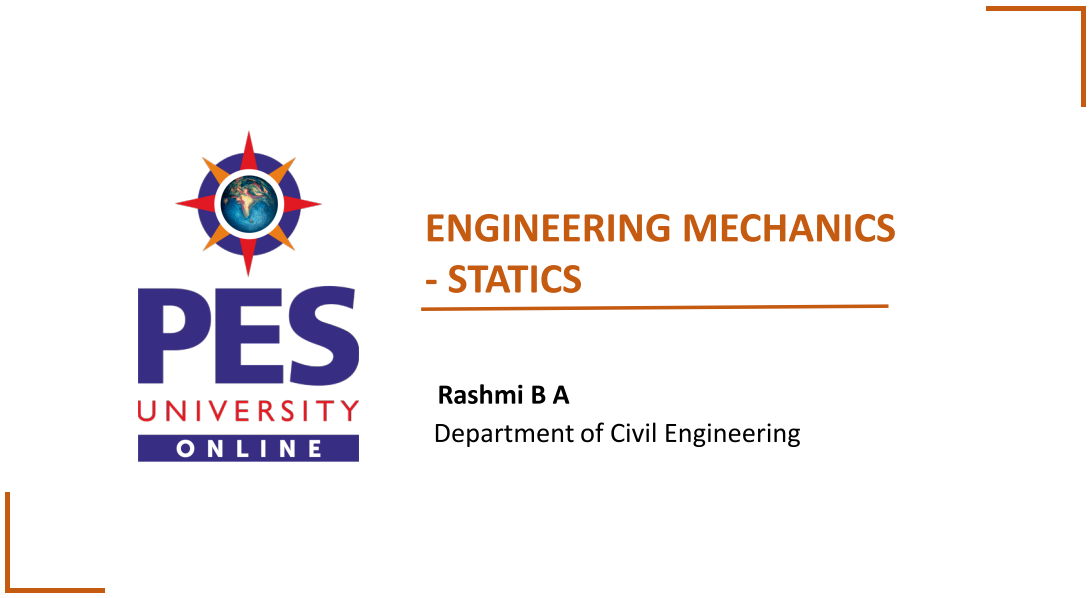


**Figure G000073.** UE20CV101 UNIT 4 SESSION 6 · source slide page 1.

EV0000403

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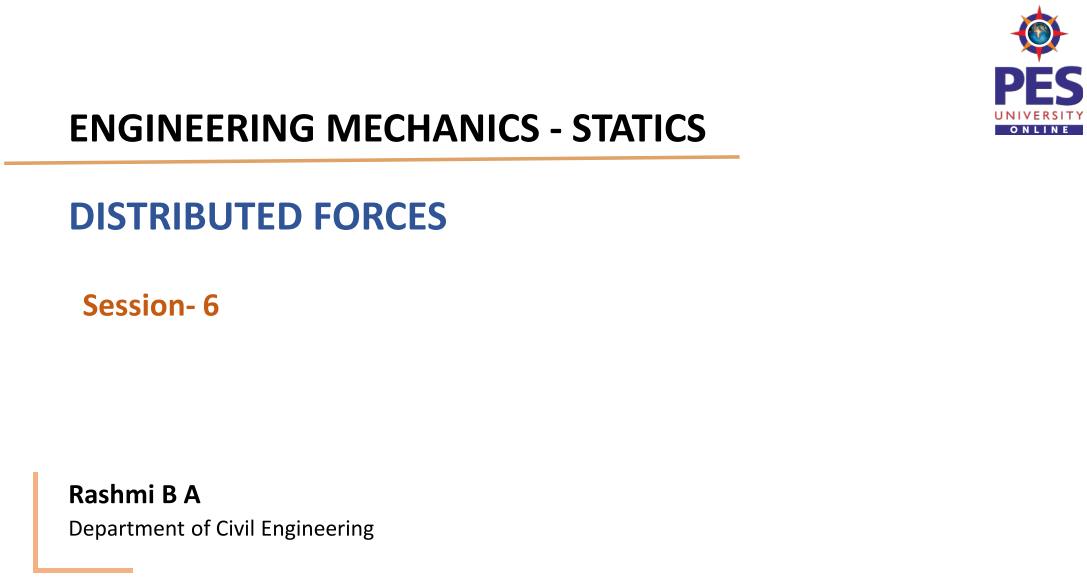


**Figure G000074.** UE20CV101 UNIT 4 SESSION 6 · source slide page 2.

EV0000406

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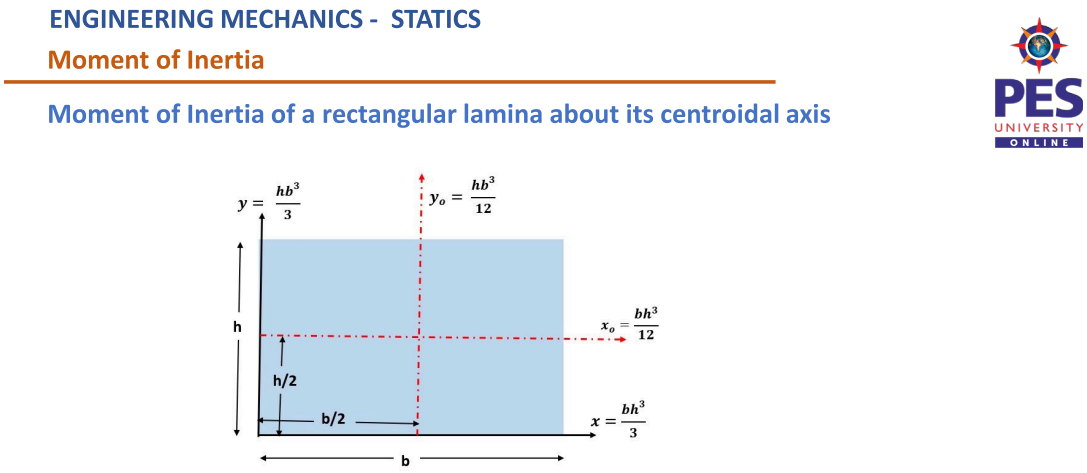


**Figure G000075.** UE20CV101 UNIT 4 SESSION 6 · source slide page 3.

EV0000409

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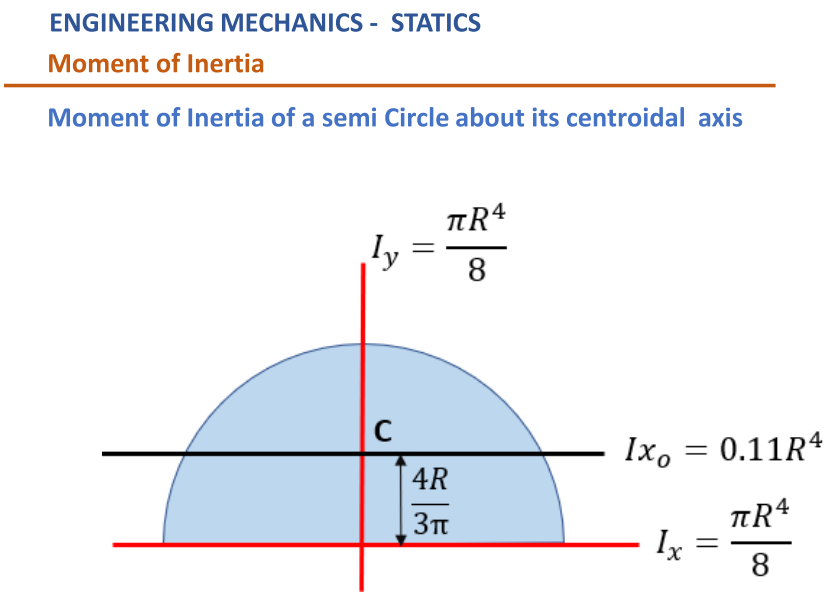


**Figure G000076.** UE20CV101 UNIT 4 SESSION 6 · source slide page 4.

EV0000412

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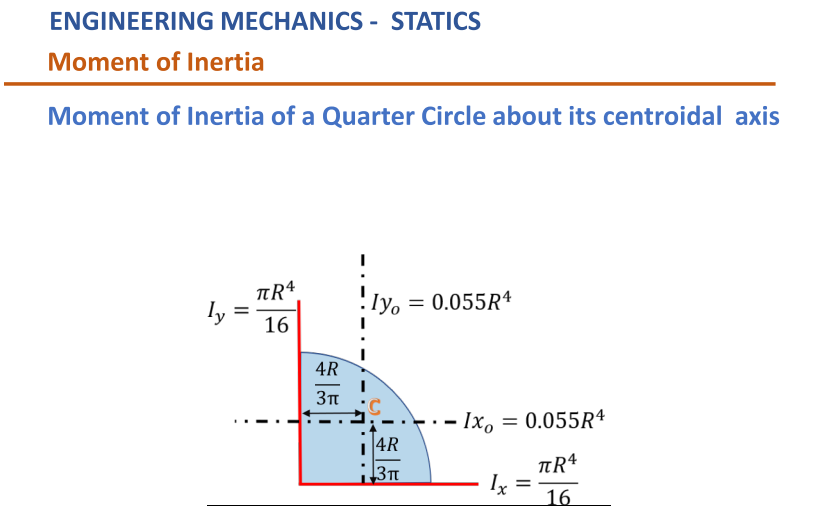


**Figure G000077.** UE20CV101 UNIT 4 SESSION 6 · source slide page 5.

EV0000415

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**Figure G000078.** UE20CV101 UNIT 4 SESSION 6 · source slide page 6.

EV0000418

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2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 6 Rashmi B A Department of Civil Engineering EV0000405 · EV0000406 · EV0000407
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia Moment of Inertia of a rectangular lamina about its centroidal axis EV0000408 · EV0000409 · EV0000410
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia Moment of Inertia of a semi Circle about its centroidal axis EV0000411 · EV0000412 · EV0000413
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia Moment of Inertia of a Quarter Circle about its centroidal axis EV0000414 · EV0000415 · EV0000416
6. Source slide page 6 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000417 · EV0000418 · EV0000419

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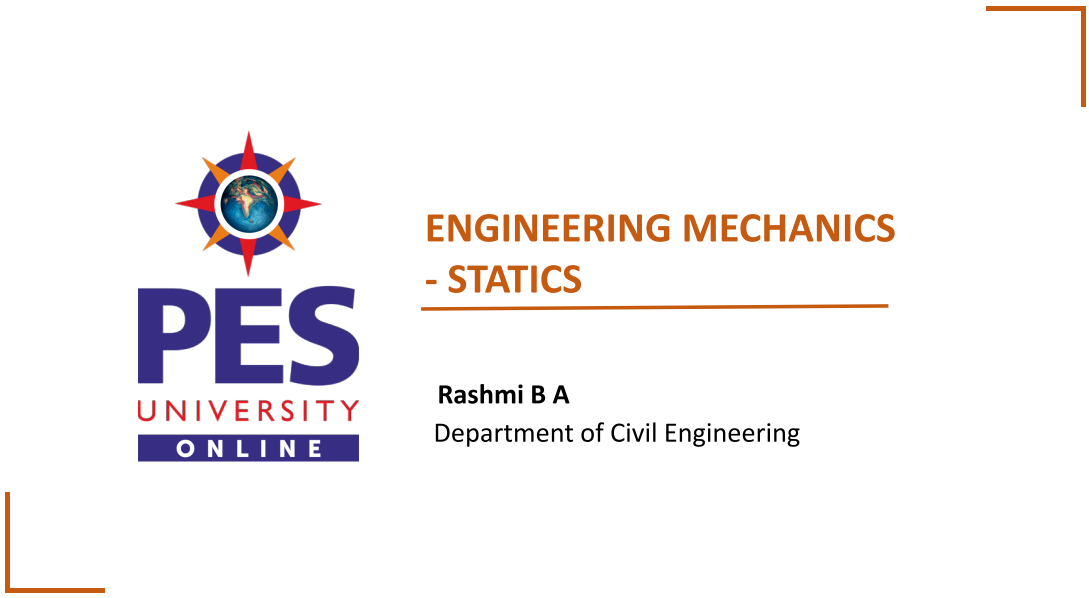


**Figure G000087.** UE20CV101 UNIT 4 SESSION 11 · source slide page 1.

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# ENGINEERING MECHANICS - STATICS

## DISTRIBUTED FORCES

Session- 11

**Rashmi B A**
Department of Civil Engineering

**Figure G000088.** UE20CV101 UNIT 4 SESSION 11 · source slide page 2.

EV0000448

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## ENGINEERING MECHANICS - STATICS
## Moment of Inertia: Numerical

**Problem A/44.** Determine the moments of inertia of the Z-section about its centroidal $x_o$- and $y_o$ -axes.

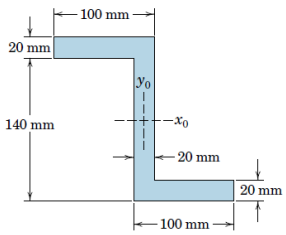


**Figure G000089.** UE20CV101 UNIT 4 SESSION 11 · source slide page 3.

EV0000451

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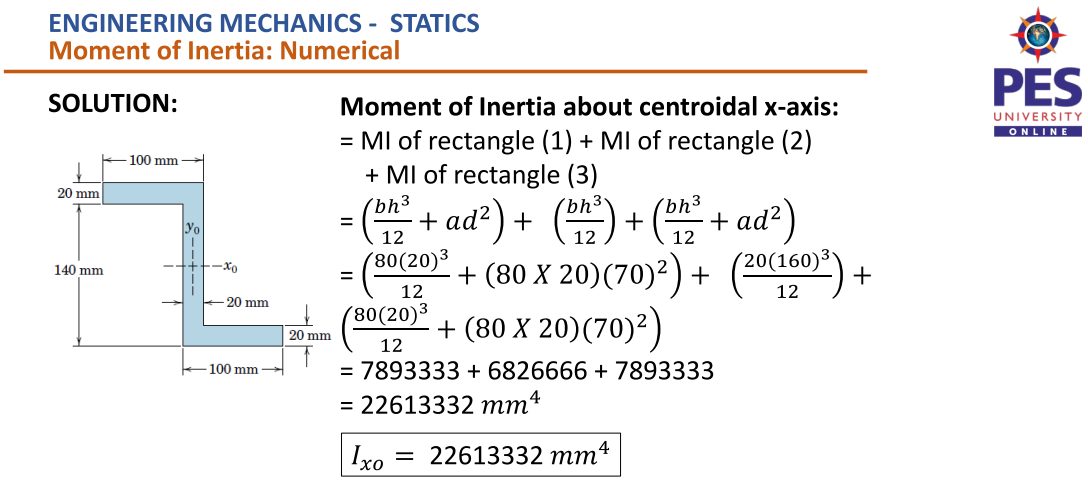


**Figure G000090.** UE20CV101 UNIT 4 SESSION 11 · source slide page 4.

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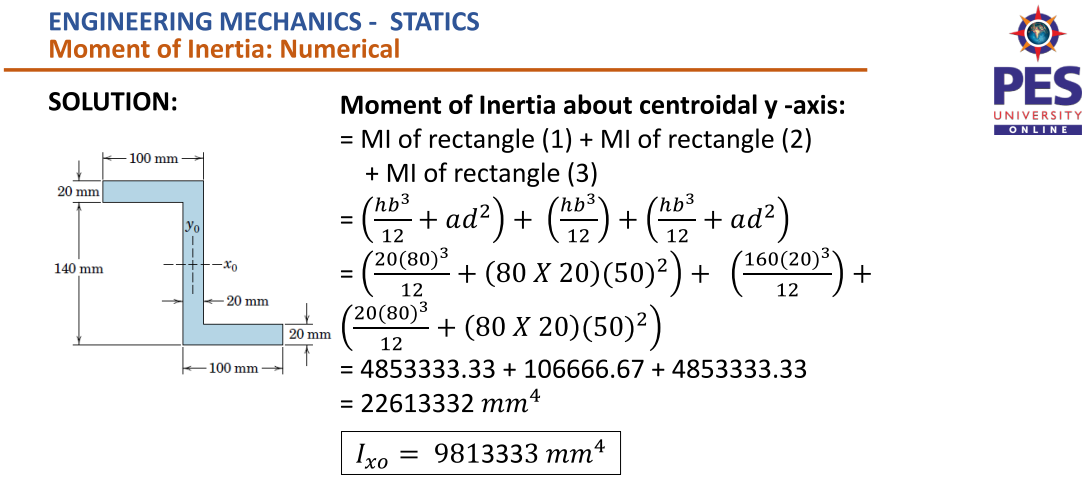


**Figure G000091.** UE20CV101 UNIT 4 SESSION 11 · source slide page 5.

EV0000457

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**Figure G000092.** UE20CV101 UNIT 4 SESSION 11 · source slide page 6.

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2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 11 Rashmi B A Department of Civil Engineering EV0000447 · EV0000448 · EV0000449
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/44. Determine the moments of inertia of the Z-section about its centroidal $xo$- and $yo$-axes. EV0000450 · EV0000451 · EV0000452
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Moment of Inertia about centroidal x-axis: = MI of rectangle (1) + MI of rectangle (2) + MI of rectangle (3) $bh3$ $bh3$ $bh3$ = + $ad2$ + + + $ad2$ 12 12 12 80(20)3 20(160)3 = + 80 $X20$ (70)2 + + 12 12 80(20)3 + 80 $X20$ (70)2 12 = 7893333 + 6826666 + 7893333 = 22613332 $mm4$ $mm4$ $Ixo$= 22613332 EV0000453 · EV0000454 · EV0000455
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical SOLUTION: Moment of Inertia about centroidal y -axis: = MI of rectangle (1) + MI of rectangle (2) + MI of rectangle (3) $hb3$ $hb3$ $hb3$ = + $ad2$ + + + $ad2$ 12 12 12 20(80)3 160(20)3 = + 80 $X20$ (50)2 + + 12 12 20(80)3 + 80 $X20$ (50)2 12 = 4853333.33 + 106666.67 + 4853333.33 = 22613332 $mm4$ $mm4$ $Ixo$= 9813333 EV0000456 · EV0000457 · EV0000458
6. Source slide page 6 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000459 · EV0000460 · EV0000461

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# 17 Source slide deck: UE20CV101 UNIT 4 SESSION 12

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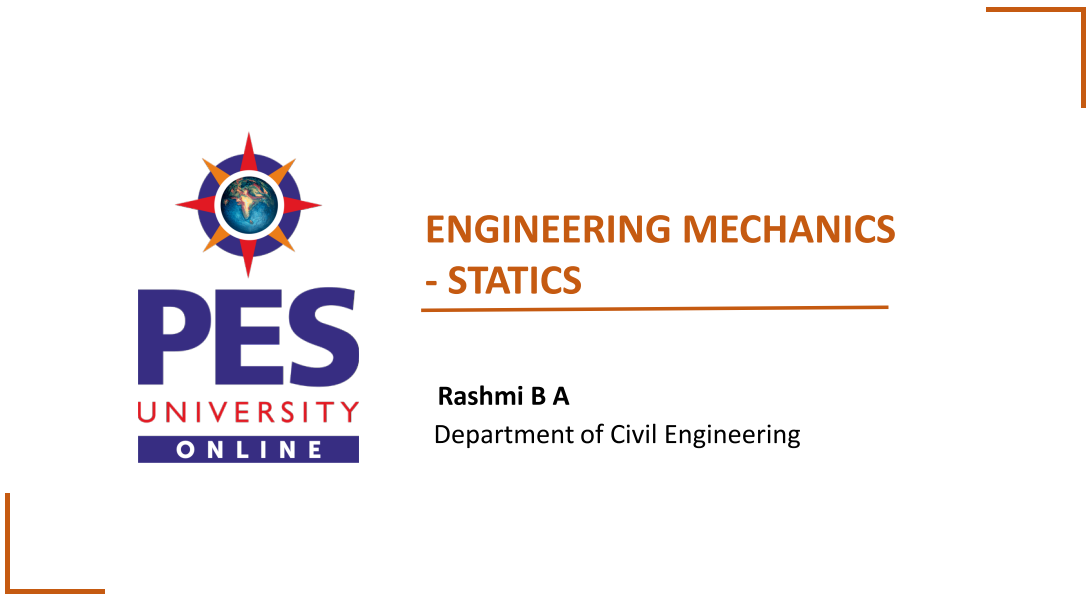


**Figure G000093.** UE20CV101 UNIT 4 SESSION 12 · source slide page 1.

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# ENGINEERING MECHANICS - STATICS

## DISTRIBUTED FORCES

Session- 12

**Rashmi B A**
Department of Civil Engineering

**Figure G000094.** UE20CV101 UNIT 4 SESSION 12 · source slide page 2.

EV0000466

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## ENGINEERING MECHANICS - STATICS

### Moment of Inertia: Numerical

**Problem A/60.** Calculate the polar radius of gyration of the shaded area about its centroid C.

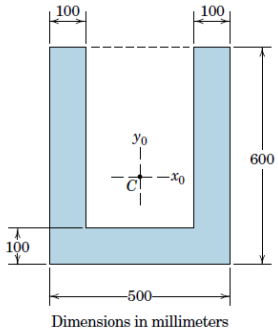


**Figure G000095.** UE20CV101 UNIT 4 SESSION 12 · source slide page 3.

EV0000469

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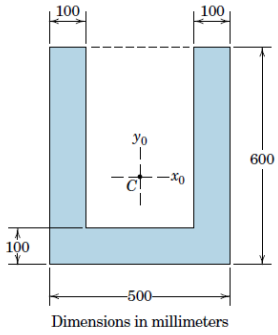
## ENGINEERING MECHANICS - STATICS

### Moment of Inertia: Numerical

**Problem A/60.** Calculate the polar radius of gyration of the shaded area about its centroid C.




Solution:

Due to symmetry $\bar{x} = 250\ mm$

$$\bar{y} = \frac{\sum a_i y_i}{A} = 250\ mm$$

$$I_{x_o} = 51.26 \times 10^8\ mm^4$$

$$I_{y_o} = 51.26 \times 10^8\ mm^4$$

$$I_{z_o} = 102.52 \times 10^8\ mm^4$$

$$A = 150000\ mm^2$$

$$\boxed{K_z = \sqrt{I_z/A} = 261.43\ mm}$$

**Figure G000096.** UE20CV101 UNIT 4 SESSION 12 · source slide page 4.

EV0000472

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## THANK YOU

**Rashmi B A**
Department of Civil Engineering
**rashmiba@pes.edu**

**Figure G000097.** UE20CV101 UNIT 4 SESSION 12 · source slide page 5.

EV0000475

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2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 12 Rashmi B A Department of Civil Engineering EV0000465 · EV0000466 · EV0000467
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/60. Calculate the polar radius of gyration of the shaded area about its centroid C. EV0000468 · EV0000469 · EV0000470
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Moment of Inertia: Numerical Problem A/60. Calculate the polar radius of gyration of the shaded area about its centroid C. Solution: Due to symmetry ıƀx= 250 $mm$ σ $aiyi$ ϖy= = 250 $mm$ $A$ 51.26 × 108 $mm4$ $Ixo$= 51.26 × 108 $mm4$ $Iyo$= 102.52 × 108 $mm4$ $Izo$= $A$= 150000 $mm2$ $Kz$= $Iz/A$= 261.43 $mm$ EV0000471 · EV0000472 · EV0000473
5. Source slide page 5 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000474 · EV0000475 · EV0000476

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# 18 Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT

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## Graphs and diagrams

**Figure G000098.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 1.

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# Engineering Mechanics

# Statics

**EIGHTH Edition**
**SI Version**

J. L. Meriam and L. G. Kraige

## Unit 4
## Moment of Inertia & Friction
## Chapter 6
Friction

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**Figure G000099.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 2.

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Engineering Mechanics Statics

**Course Contents:**

**UNIT 4 : Friction**

Introduction, Frictional Phenomena - Types of Friction, Dry Friction, Fluid Friction, Internal Friction, Mechanism of Dry Friction, Static Friction, Kinetic Friction, Friction Angles, Factors Affecting Friction, Numerical involving bodies placed on Horizontal surfaces and inclined Surface,



**Figure G000100.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 3.

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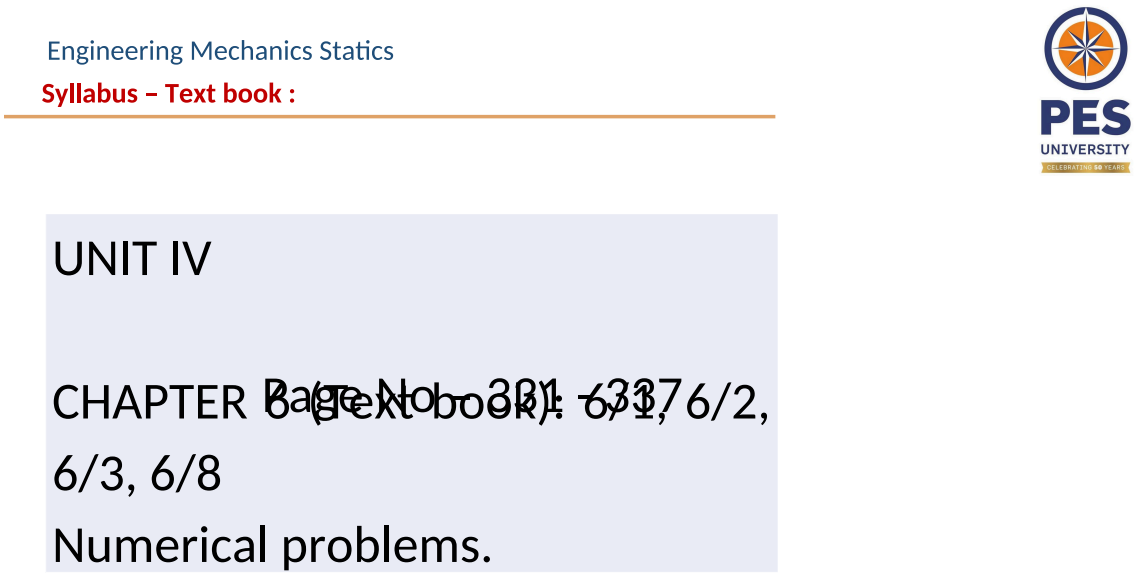


**Figure G000101.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 4.

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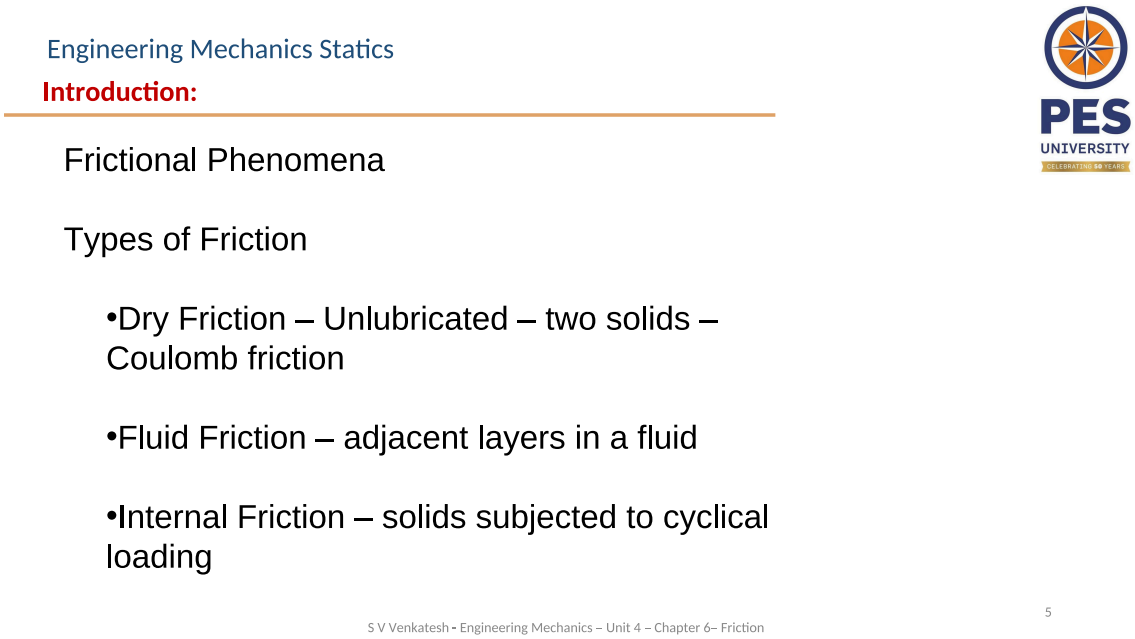


**Figure G000102.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 5.

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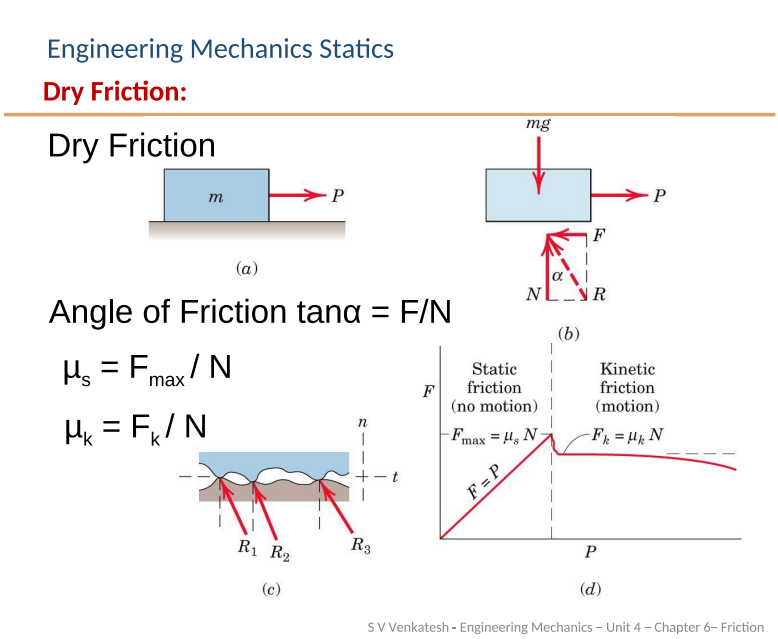


**Figure G000103.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 6.

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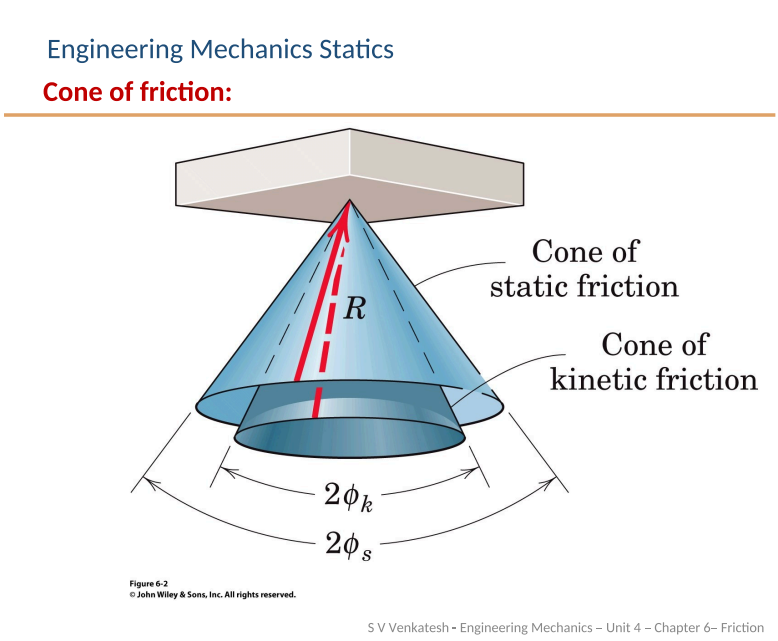


**Figure G000104.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 7.

EV0000496

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Engineering Mechanics Statics

**Types of Friction Problems:**

## Types of Friction Problems – 3 types

1. **In the *first type of problem*,**

The condition of impending motion is known to exist. Here a body which is in equilibrium is on the verge of slipping, and the friction force equals the limiting static friction $Fmax = \mu_s N$. *The equations of equilibrium will, of course, also hold.*



**Figure G000105.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 8.

EV0000499

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Engineering Mechanics Statics

**Types of Friction Problems:**

**2. In the *second type of problem*,**

Neither the condition of impending motion nor the condition of motion is known to exist. To determine the actual friction conditions, we first assume static equilibrium and then solve for the friction force *F necessary for equilibrium. Three* outcomes are possible:

*(a) F < (Fmax =* $\mu_s N$*): Here the friction force necessary for* equilibrium can be supported, and therefore the body is in static equilibrium as assumed. One has to emphasize that the *actual* friction force *F is less than the limiting value Fmax =* $\mu_s N$ and that *F is determined solely by the equations of* equilibrium.



**Figure G000106.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 9.

EV0000502

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Engineering Mechanics Statics

**Types of Friction Problems:**

**2. In the *second type of problem (Cont...)***

*(b) F = (Fmax = $\mu_s N$): Since the friction force F is at its maximum* value *Fmax, motion impends, as discussed in problem type (1).* The assumption of static equilibrium is valid.

*(c) F > (Fmax = $\mu_s N$): Clearly this condition is impossible, because* the surfaces cannot support more force than the maximum $\mu_s N$. The assumption of equilibrium is therefore invalid, and motion occurs. The friction force *F is equal to* $\mu_k N$.

S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction 10

**Figure G000107.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 10.

EV0000505

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Engineering Mechanics Statics

**Types of Friction Problems:**

**3. In the *third type of problem*,**

Relative motion is known to exist between the contacting surfaces, and thus the kinetic coefficient of friction clearly applies. For this problem type, Fk = $\mu_k N$ always gives the kinetic friction force directly.

S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction 11

**Figure G000108.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 11.

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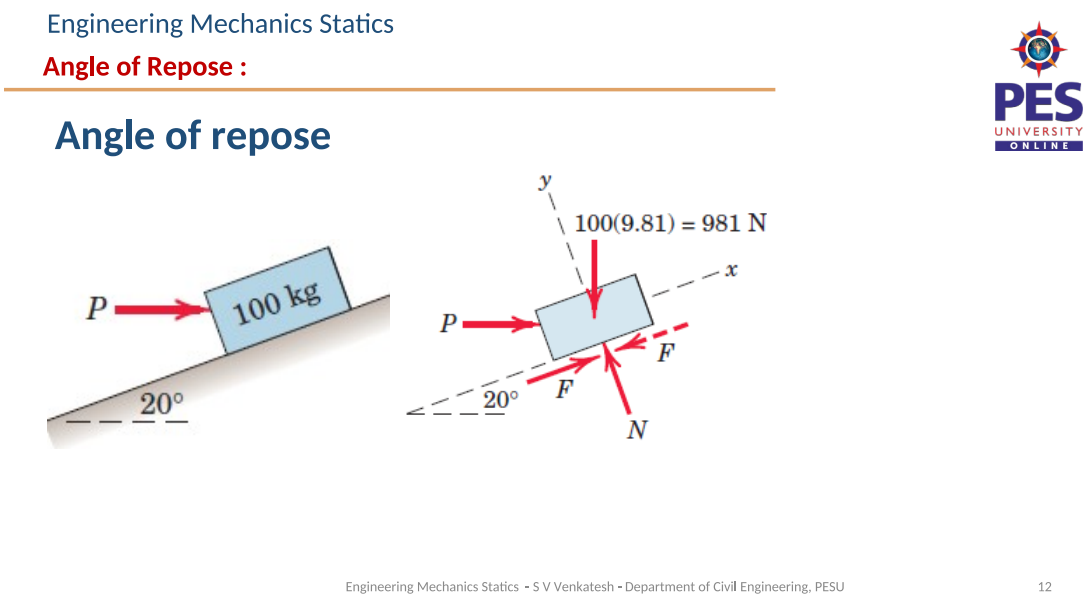


**Figure G000109.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 12.

EV0000510

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**Figure G000110.** UE26ME142A Unit 4 class 53 Lecture PPT · source slide page 13.

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## Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000477 · EV0000478 · EV0000479
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 4 Moment of Inertia & Friction Chapter 6 Friction Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000480 · EV0000481 · EV0000482
3. Source slide page 3 (extracted text): Engineering Mechanics Statics Course Contents: UNIT 4 : Friction Introduction, Frictional Phenomena - Types of Friction, Dry Friction, Fluid Friction, Internal Friction, Mechanism of Dry Friction, Static Friction, Kinetic Friction, Friction Angles, Factors Affecting Friction, Numerical involving bodies placed on Horizontal surfaces and inclined Surface, 3 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000483 · EV0000484 · EV0000485
4. Source slide page 4 (extracted text): Engineering Mechanics Statics Syllabus – Text book : UNIT IV Page No – 331 - 337 CHAPTER 6 (Text book): 6/1, 6/2, 6/3, 6/8 4 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction Numerical problems. EV0000486 · EV0000487 · EV0000488
5. Source slide page 5 (extracted text): Engineering Mechanics Statics Introduction: Frictional Phenomena Types of Friction •Dry Friction – Unlubricated – two solids –Coulomb friction •Fluid Friction – adjacent layers in a fluid •Internal Friction – solids subjected to cyclical loading 5 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000489 · EV0000490 · EV0000491
6. Source slide page 6 (extracted text): Engineering Mechanics Statics Dry Friction: Dry Friction Angle of Friction tanα = F/N μs = Fmax / N μk = Fk / N 6 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000492 · EV0000493 · EV0000494
7. Source slide page 7 (extracted text): Engineering Mechanics Statics Cone of friction: 7 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000495 · EV0000496 · EV0000497
8. Source slide page 8 (extracted text): Engineering Mechanics Statics Types of Friction Problems: Types of Friction Problems – 3 types 1.In the firsttypeofproblem, The condition of impending motion is known to exist. Here a body which is in equilibrium is on the verge of slipping, and the friction force equals the limiting static friction Fmax = μsN. The equations of equilibrium will, of course, alsohold. 8 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000498 · EV0000499 · EV0000500
9. Source slide page 9 (extracted text): Engineering Mechanics Statics Types of Friction Problems: 2. In the secondtypeofproblem, Neither the condition of impending motion nor the condition of motion is known to exist. To determine the actual friction conditions, we first assume static equilibrium and then solve for the friction force F necessary for equilibrium. Three outcomes are possible: (a) F < (Fmax = μsN): Here the friction force necessary for equilibrium can be supported, and therefore the body is in static equilibrium as assumed. One has to emphasize that the actual 9 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction friction force FislessthanthelimitingvalueFmax EV0000501 · EV0000502 · EV0000503

10. Source slide page 10 (extracted text): Engineering Mechanics Statics Types of Friction Problems: 2. In the second type of problem (Cont...) (b)F =(Fmax=µsN):Sincethefrictionforce F is at its maximum value Fmax, motion impends, as discussed in problem type (1). The assumption of static equilibrium is valid. (c)F>(Fmax=µsN):Clearlythisconditionis impossible, because the surfaces cannot support more force than the maximum µsN. The assumption of equilibrium is therefore 10 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction invalid and motion occurs The friction force EV0000504 · EV0000505 · EV0000506
11. Source slide page 11 (extracted text): Engineering Mechanics Statics Types of Friction Problems: 3. In the thirdtypeofproblem, Relative motion is known to exist between the contacting surfaces, and thus the kinetic coefficient of friction clearly applies. For this problem type, Fk = µkN always gives the kinetic friction force directly. 11 S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction EV0000507 · EV0000508 · EV0000509
12. Source slide page 12 (extracted text): Engineering Mechanics Statics Angle of Repose : Angle of repose Engineering Mechanics Statics - S V Venkatesh - Department of Civil 12 Engineering, PESU ## Page / slide 13 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000510
13. Source slide page 13: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000511

SHORT-NOTE RECAP

Source deck preserved: UE26ME142A Unit 4 class 53 Lecture PPT. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000477

# 19 Source slide deck: UE20CV101 UNIT 5 SESSION 2

This entry preserves the complete slide deck titled UE20CV101 UNIT 5 SESSION 2 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_04, to avoid an unsupported match. EV0000512 · EV0000513

## Graphs and diagrams

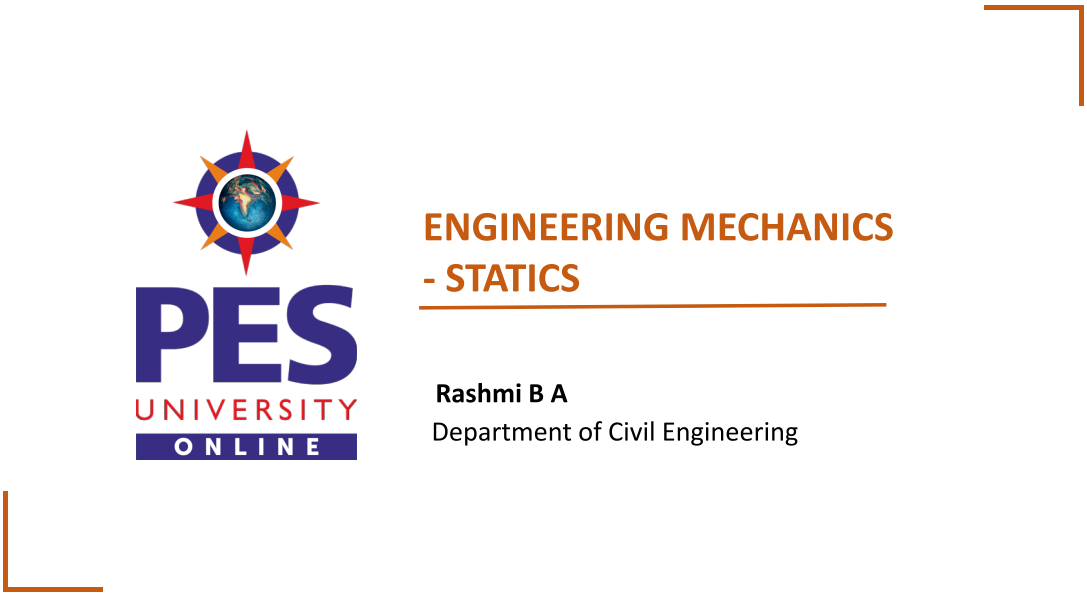


**Figure G000111.** UE20CV101 UNIT 5 SESSION 2 · source slide page 1.

EV0000513

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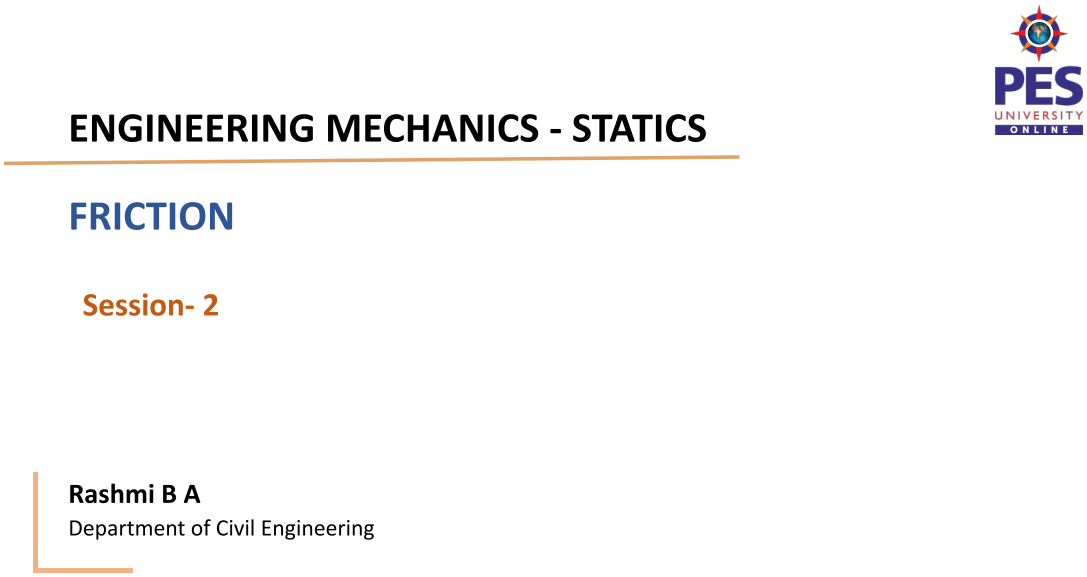


**Figure G000112.** UE20CV101 UNIT 5 SESSION 2 · source slide page 2.

EV0000516

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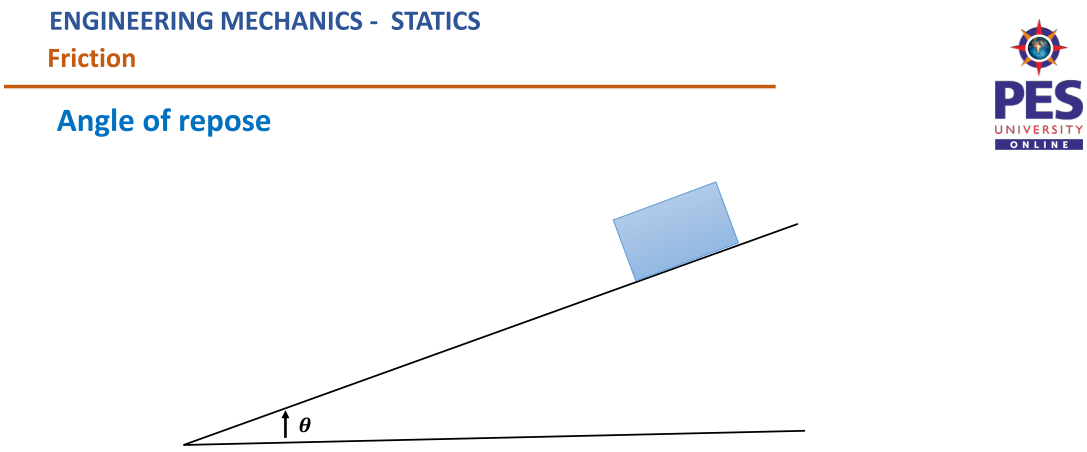


**Figure G000113.** UE20CV101 UNIT 5 SESSION 2 · source slide page 3.

EV0000519

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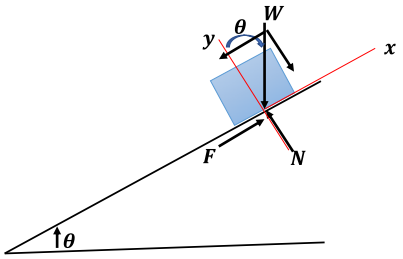
ENGINEERING MECHANICS - STATICS

Friction

## Angle of repose

***Applying the conditions of equilibrium:***




$\sum F_x = 0 \qquad -W\sin(\theta) + F = 0$

$F = W\sin(\theta) \text{ -- } -(1)$

$\sum F_y = 0 \qquad -W\cos(\theta) + N = 0$

$N = W\cos(\theta) \text{ -- } -(2)$

*when the motion is impending Frictional force will be*

$F_{max} = \mu_s N$ ( *substituting in* (1)

$\mu_s N = W\sin(\theta)$ from (2)

$\mu_s W\cos(\theta) = W\sin(\theta)$ $\boxed{\mu_s = \tan(\theta_{max})}$

$\boxed{\theta_{max} = \tan^{-1}(\mu_s)}$

**Figure G000114.** UE20CV101 UNIT 5 SESSION 2 · source slide page 4.

EV0000522

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ENGINEERING MECHANICS - STATICS

Friction

Problem 6/1: The force *P* is applied to the 90-kg crate, which is stationary before the force is applied. Determine the magnitude and direction of the friction force *F* exerted by the horizontal surface on the crate if (*a*) *P* = 300 N, (*b*) *P* = 400 N, and (*c*) *P* = 500 N.

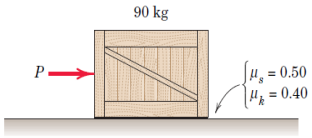


**Figure G000115.** UE20CV101 UNIT 5 SESSION 2 · source slide page 5.

EV0000525

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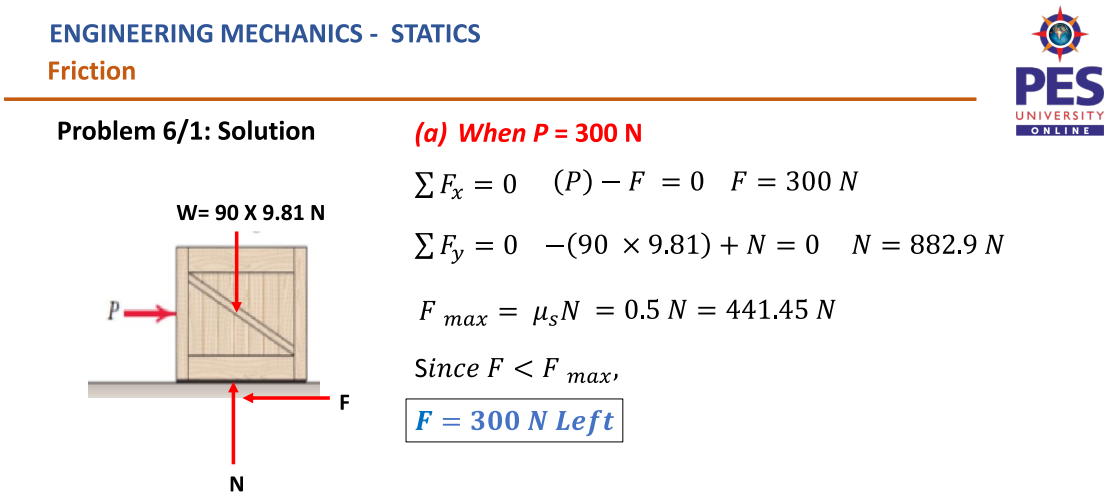


**Figure G000116.** UE20CV101 UNIT 5 SESSION 2 · source slide page 6.

EV0000528

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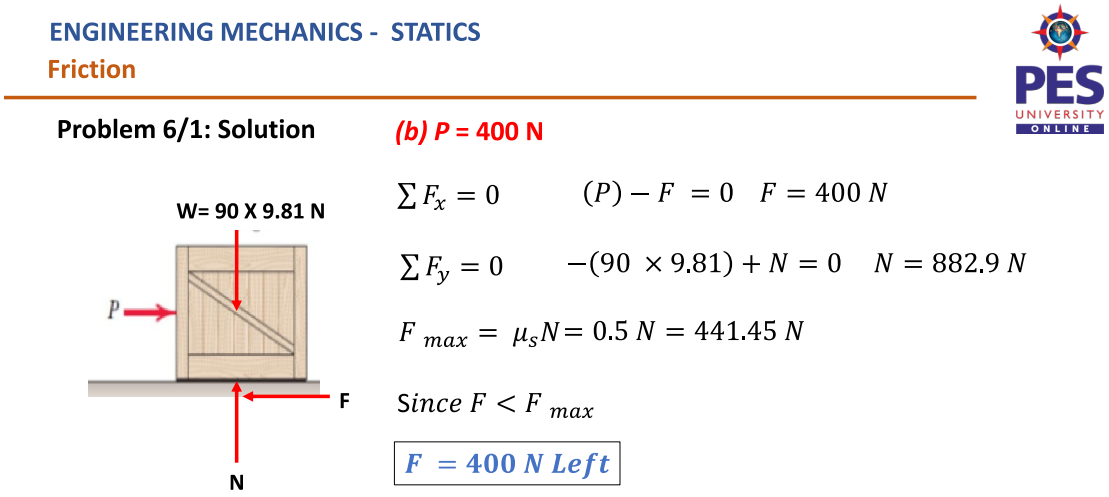


**Figure G000117.** UE20CV101 UNIT 5 SESSION 2 · source slide page 7.

EV0000531

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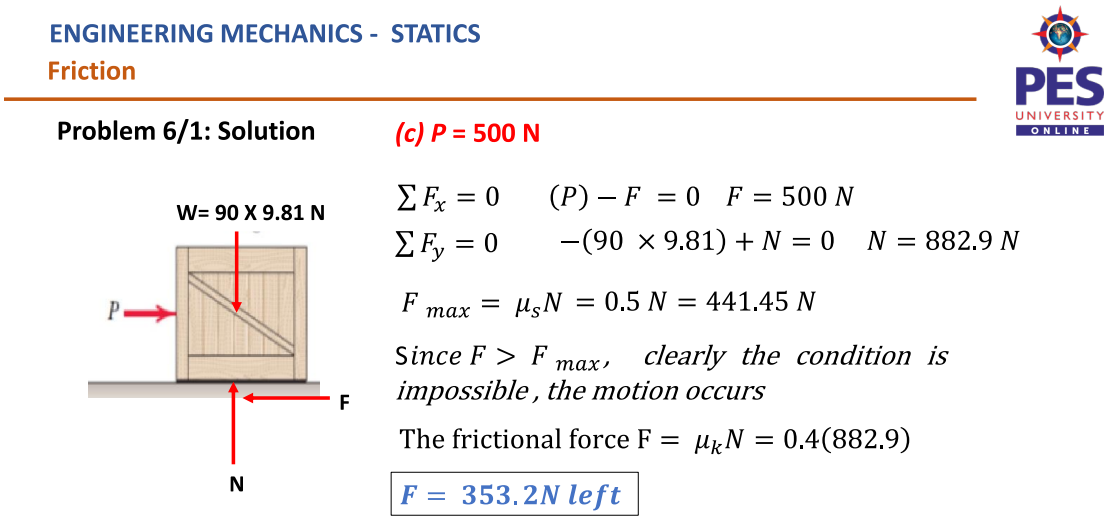


**Figure G000118.** UE20CV101 UNIT 5 SESSION 2 · source slide page 8.

EV0000534

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**Figure G000119.** UE20CV101 UNIT 5 SESSION 2 · source slide page 9.

EV0000537

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000512 · EV0000513 · EV0000514
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS FRICTION Session- 2 Rashmi B A Department of Civil Engineering EV0000515 · EV0000516 · EV0000517
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Friction Angle of repose $\boldsymbol{\theta}$ EV0000518 · EV0000519 · EV0000520
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Friction $\boldsymbol{W\ \theta}$ Angle of repose $\boldsymbol{y\ x}$ Applying the conditions of equilibrium: $\sigma\ Fx = 0\ -W\sin\theta + F = 0\ \boldsymbol{F\ N}\ F = W\sin\theta ---(1)$ $\sigma\ 0\ -W\cos\theta + N = 0\ Fy = \boldsymbol{\theta}\ N = Wcos\theta ---(2)$ $whenthemotionisimpendingFrictionalforcewillbe\ Fmax = \mu sN($ $substitutingin1\ \mu sN = W\sin\theta$ from (2) $\mu sWcos(\theta) = W\sin\theta\ \mu s = \tan(\theta max)\ \theta max = \tan{-1}(\mu s)$ EV0000521 · EV0000522 · EV0000523
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Friction Problem 6/1: The force P is applied to the 90-kg crate, which is stationary before the force is applied. Determine the magnitude and direction of the friction force F exerted by the horizontal surface on the crate if (a) P = 300 N, (b) P = 400 N, and (c) P = 500 N. EV0000524 · EV0000525 · EV0000526
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Friction Problem 6/1: Solution (a) When P = 300 N $\sigma\ Fx = 0\ P - F = 0\ F = 300\ N$ W= 90 X 9.81 N $\sigma\ Fy = 0\ -90 \times 9.81 + N = 0\ N = 882.9\ N\ Fmax = \mu sN = 0.5\ N = 441.45\ N\ SinceF < Fmax$, F $\boldsymbol{F = 300NLeft}$ N EV0000527 · EV0000528 · EV0000529
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Friction Problem 6/1: Solution (b) P = 400 N $\sigma\ Fx = 0\ P - F = 0\ F = 400\ N$ W= 90 X 9.81 N $\sigma\ Fy = 0\ -90 \times 9.81 + N = 0\ N = 882.9\ N\ Fmax = \mu sN = 0.5\ N = 441.45\ N$ F $SinceF < Fmax\ \boldsymbol{F = 400NLeft}$ N EV0000530 · EV0000531 · EV0000532
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS - STATICS Friction Problem 6/1: Solution (c) P = 500 N $\sigma\ Fx = 0\ P - F = 0\ F = 500\ N$ W= 90 X 9.81 N $\sigma\ Fy = 0\ -90 \times 9.81 + N = 0\ N = 882.9\ N\ Fmax = \mu sN = 0.5\ N = 441.45\ N\ SinceF > Fmax$, clearly the condition is impossible,themotionoccurs F The frictional force F = $\mu kN$= 0.4 882.9 N $\boldsymbol{F = 353.2Nleft}$ EV0000533 · EV0000534 · EV0000535
9. Source slide page 9 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000536 · EV0000537 · EV0000538

SHORT-NOTE RECAP

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EV0000512

# 20 Source slide deck: UE20CV101 UNIT 5 SESSION 3

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## Graphs and diagrams

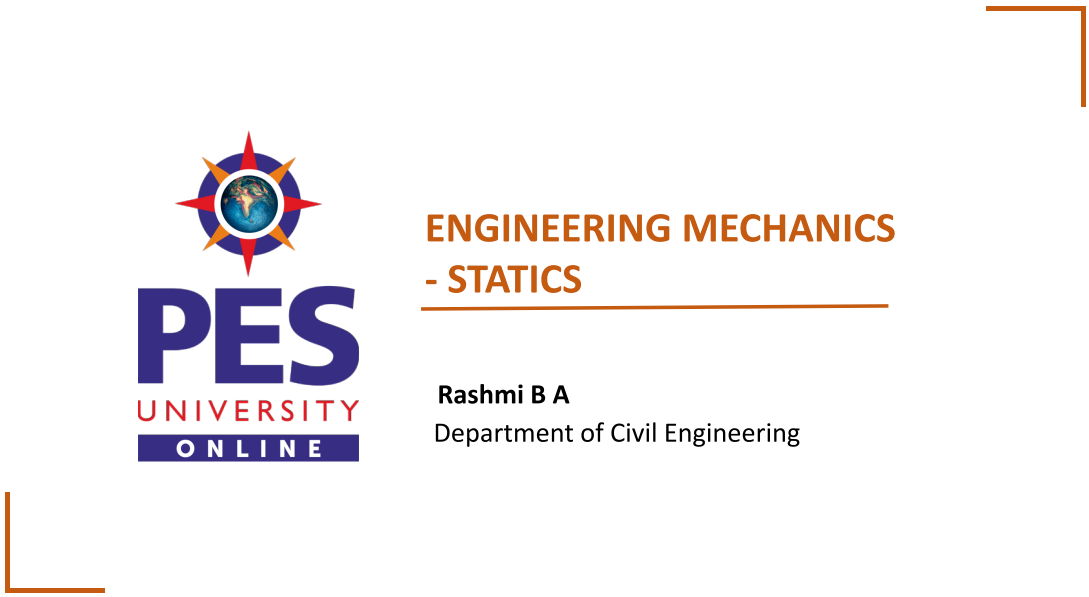


**Figure G000128.** UE20CV101 UNIT 5 SESSION 3 · source slide page 1.

EV0000564

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# ENGINEERING MECHANICS - STATICS

## FRICTION

### Session- 3

**Rashmi B A**
Department of Civil Engineering

**Figure G000129.** UE20CV101 UNIT 5 SESSION 3 · source slide page 2.

EV0000567

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### ENGINEERING MECHANICS - STATICS
### Friction- Numerical

Problem 6/3: The force *P* is applied to the 50-kg block when it is at rest. Determine the magnitude and direction of the friction force exerted by the surface on the block if (*a*) *P* = 0, (*b*) *P* = 200 N, and (*c*) *P = 250 N*. (*d*) What value of *P* is required to initiate motion up the incline? The coefficients of static and kinetic friction between the block and the incline are $\mu_s = 0.25$ and $\mu_K = 0.20$, respectively.

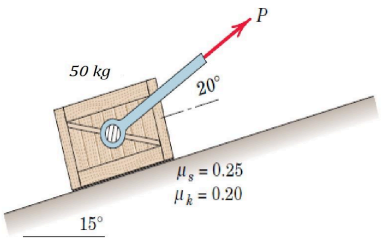


**Figure G000130.** UE20CV101 UNIT 5 SESSION 3 · source slide page 3.

EV0000570

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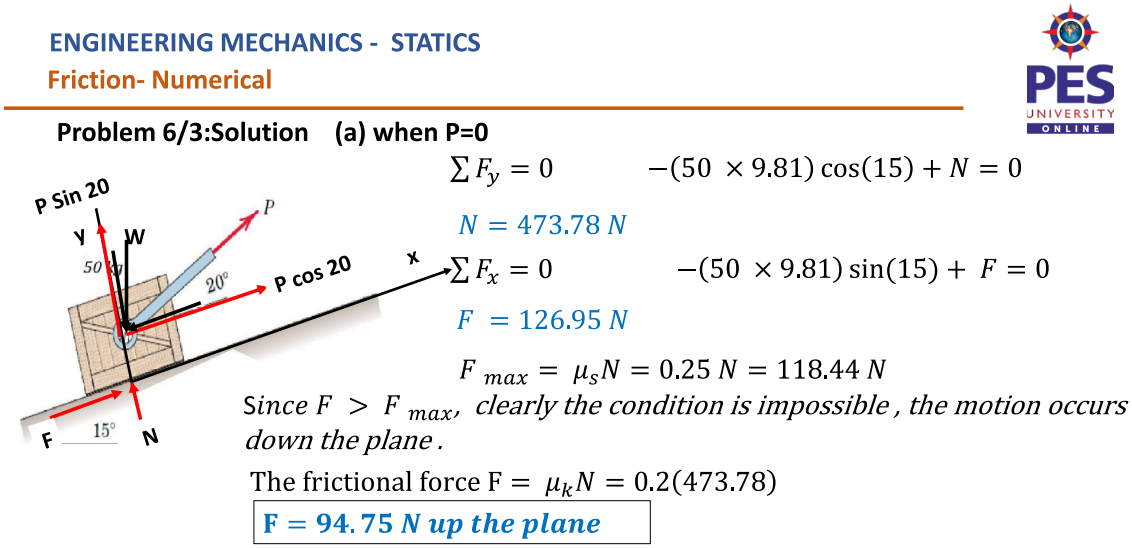


**Figure G000131.** UE20CV101 UNIT 5 SESSION 3 · source slide page 4.

EV0000573

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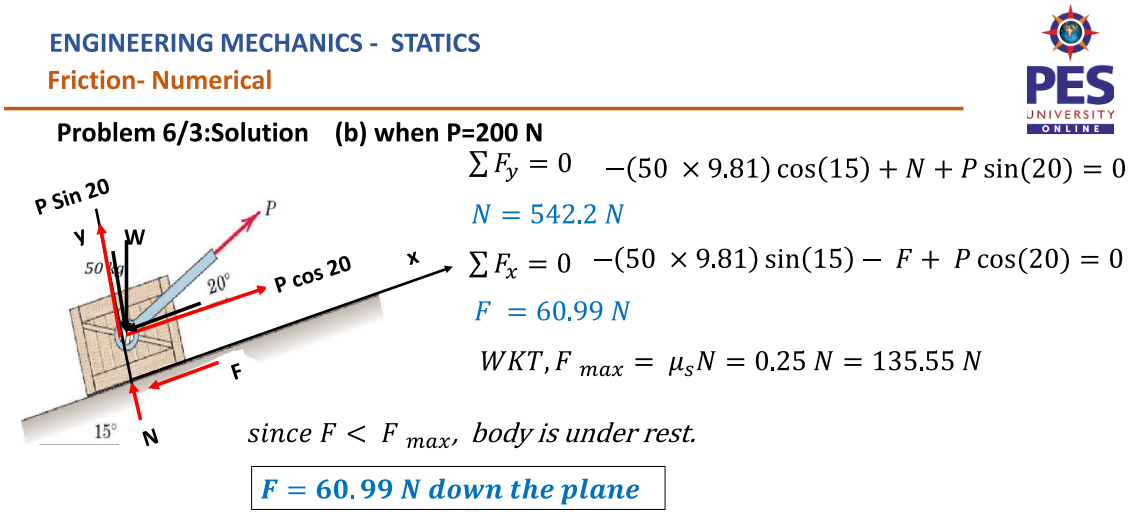


**Figure G000132.** UE20CV101 UNIT 5 SESSION 3 · source slide page 5.

EV0000576

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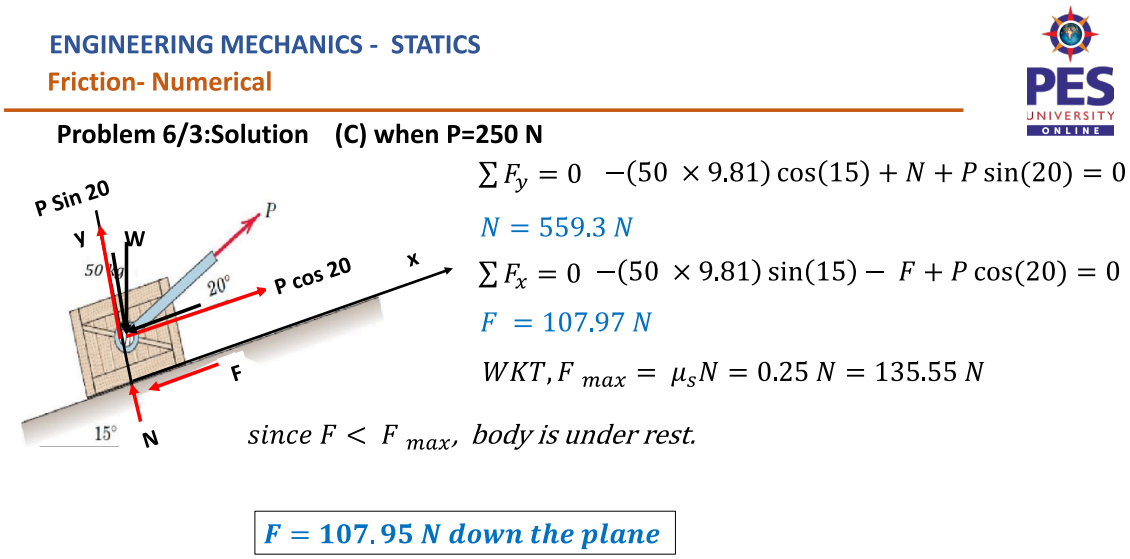


**Figure G000133.** UE20CV101 UNIT 5 SESSION 3 · source slide page 6.

EV0000579

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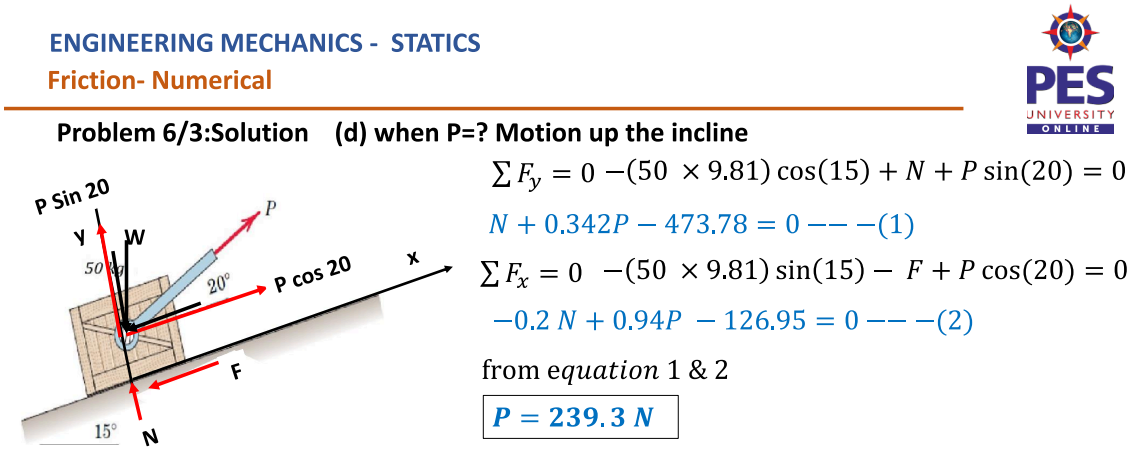


**Figure G000134.** UE20CV101 UNIT 5 SESSION 3 · source slide page 7.

EV0000582

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ENGINEERING MECHANICS - STATICS

Friction- Numerical

- Problem 6/6: Determine the minimum coefficient of static friction $\mu_s$ which will allow the drum with fixed inner hub to be rolled up the 15° incline at a steady speed without slipping. What are the corresponding values of the force *P* and the friction force *F*?

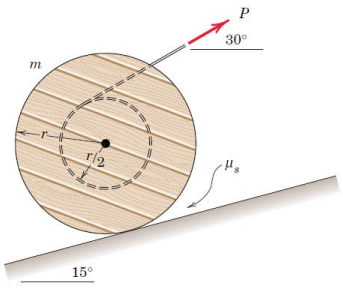


**Figure G000135.** UE20CV101 UNIT 5 SESSION 3 · source slide page 8.

EV0000585

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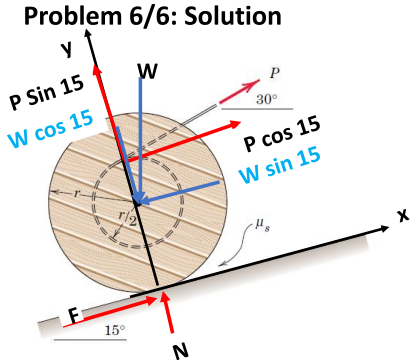
ENGINEERING MECHANICS - STATICS

Friction- Numerical

Problem 6/6: Solution




***Applying the conditions of equilibrium***

$\sum M_o = 0 \qquad +F(r) - P\cos(15)\left(\frac{r}{2}\right) = 0$

$F = 0.483\,P$

$\sum F_x = 0 \qquad +F - W\sin(15) + P\cos(15) = 0$

$P = 0.1786\,mg$

$\sum F_y = 0 \qquad -W\cos(15) + N + P\sin(15) = 0$

$N = 0.919\,mg$

$$\boxed{\mu_s = \frac{F}{N} = 0.094}$$

**Figure G000136.** UE20CV101 UNIT 5 SESSION 3 · source slide page 9.

EV0000588

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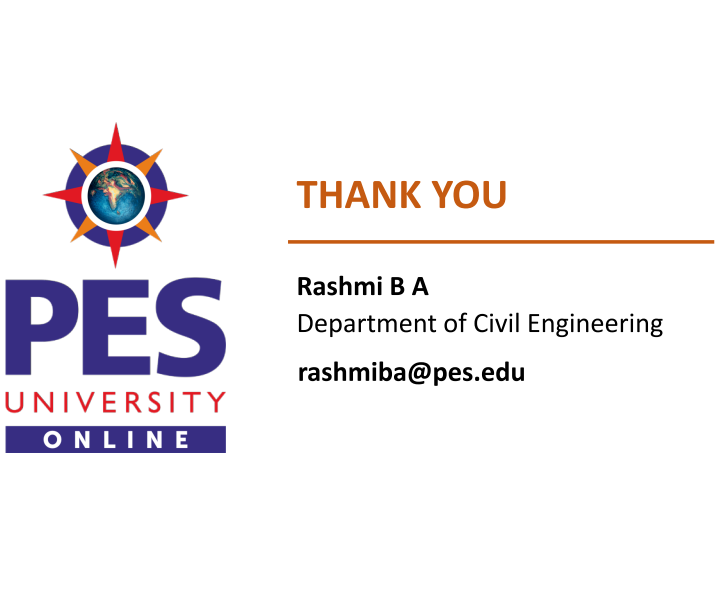


**Figure G000137.** UE20CV101 UNIT 5 SESSION 3 · source slide page 10.

EV0000591

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## Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000563 · EV0000564 · EV0000565
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS FRICTION Session- 3 Rashmi B A Department of Civil Engineering EV0000566 · EV0000567 · EV0000568
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/3: The force P is applied to the 50-kg block when it is at rest. Determine the magnitude and direction of the friction force exerted by the surface on the block if (a) P = 0, (b) P = 200 N, and (c) P = 250 N. (d) What value of P is required to initiate motion up the incline? The coefficients of static and kinetic friction between the block and the incline are $\mu s$= 0.25 and $\mu K$= 0.20, respectively. EV0000569 · EV0000570 · EV0000571
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/3:Solution (a) when P=0 σ $Fy$= 0 −50 × 9.81 cos 15 + $N$= 0 $N$= 473.78 $N$ W σ $Fx$= 0 −50 × 9.81 sin(15) + $F$= 0 $F$= 126.95 $N$ $Fmax$= $\mu sN$= 0.25 $N$= 118.44 $N$ $SinceF >$ $Fmax$, clearlytheconditionisimpossible,themotionoccurs downtheplane. The frictional force F = $\mu kN$= 0.2 473.78 **F= 94. 75*Nuptheplane*** EV0000572 · EV0000573 · EV0000574
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/3:Solution (b) when P=200 N σ $Fy$= 0 −50 × 9.81 cos 15 + $N$+ $P$sin(20) = 0 $N$= 542.2 $N$ W −50 × 9.81 sin 15 −$F$+ $P$cos(20) = 0 σ $Fx$= 0 $F$= 60.99 $N$ $WKT$, $Fmax$= $\mu sN$= 0.25 $N$= 135.55 $N$ $sinceF <$ $Fmax$, bodyisunderrest. $F$= **60. *99Ndowntheplane*** EV0000575 · EV0000576 · EV0000577

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/3:Solution (C) when P=250 N σ $Fy$= 0 $-50 \times 9.81 \cos 15 + N + P\sin(20) = 0$ $N$= 559.3 $N$ W σ $Fx$= 0 $-50 \times 9.81 \sin 15 - F + P\cos(20) = 0$ $F$= 107.97 $N$ $WKT$, $Fmax$= $\mu s N$= 0.25 $N$= 135.55 $N$ $since F < Fmax$, bodyisunderrest. $\boldsymbol{F}$= **107**. **95*Ndowntheplane*** EV0000578 · EV0000579 · EV0000580
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/3:Solution (d) when P=? Motion up the incline σ $Fy$= 0 $-50 \times 9.81 \cos 15 + N + P\sin(20) = 0$ $N + 0.342P - 473.78 = 0 ---(1)$ W σ $Fx$= 0 $-50 \times 9.81 \sin 15 - F + P\cos(20) = 0$ $-0.2\ N + 0.94P - 126.95 = 0 ---(2)$ from e$quation$1 & 2 $\boldsymbol{P}$= **239**. **3*N*** EV0000581 · EV0000582 · EV0000583
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • Problem 6/6: Determine the minimum coefficient of static friction $\mu s$which will allow the drum with fixed inner hub to be rolled up the 15° incline at a steady speed without slipping. What are the corresponding values of the force P and the friction force F? EV0000584 · EV0000585 · EV0000586
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/6: Solution ***Applyingtheconditionsofequilibrium*** $r$ σ 0 + $F(r)$ $-Pcos15 = 0$ W $Mo$= 2 F = 0.483 $P$ σ $Fx$= 0 + $F - W sin15 + P\cos(15) = 0$ P = 0.1786 $mg$ σ $Fy$= 0 −W cos 15 + $N + P\sin(15) = 0$ $N$= 0.919 $mg$ $\boldsymbol{F}$ $\boldsymbol{\mu s}$= $\boldsymbol{N}$= **0**. **094** EV0000587 · EV0000588 · EV0000589
10. Source slide page 10 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000590 · EV0000591 · EV0000592

SHORT-NOTE RECAP

Source deck preserved: UE20CV101 UNIT 5 SESSION 3. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000563

# 21 Source slide deck: UE20CV101 UNIT 5 SESSION 6

This entry preserves the complete slide deck titled UE20CV101 UNIT 5 SESSION 6 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_04, to avoid an unsupported match. EV0000617 · EV0000618

## Graphs and diagrams

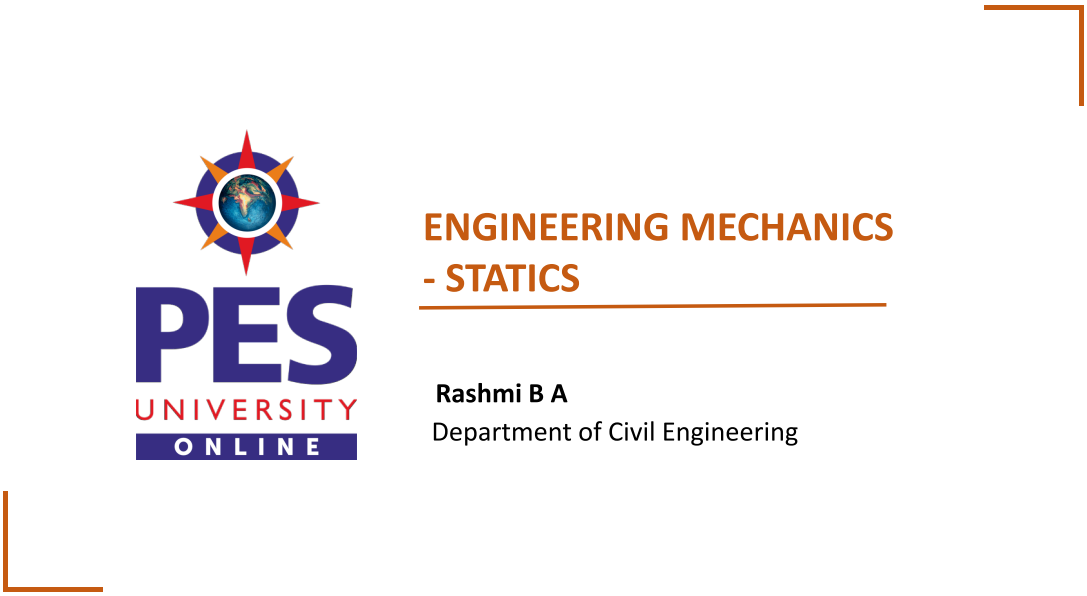



**Figure G000146.** UE20CV101 UNIT 5 SESSION 6 · source slide page 1.

EV0000618

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

# ENGINEERING MECHANICS - STATICS

## FRICTION- Numerical

### Session- 6

**Rashmi B A**
Department of Civil Engineering

**Figure G000147.** UE20CV101 UNIT 5 SESSION 6 · source slide page 2.

EV0000621

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

## ENGINEERING MECHANICS - STATICS
### Friction- Numerical

- 6/33 Determine the distance s to which the 90-kg painter can climb without causing the 4-m ladder to slip at its lower end A. The top of the 15-kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. The mass center of the painter is directly above her feet.

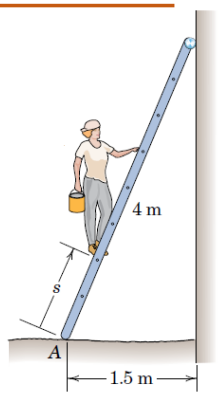


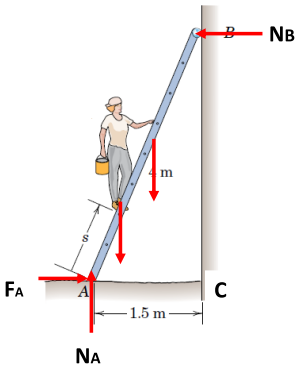

**Figure G000148.** UE20CV101 UNIT 5 SESSION 6 · source slide page 3.

EV0000624

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS - STATICS

Friction- Numerical




$$\theta = \cos^{-1}\left(\frac{1.5}{4}\right) = 67.97$$

$$\overline{CB} = 3.708\ mm$$

***Considering the FBD of ladder & Applying the conditions of equilibrium***

*weight of the ladder* $W_l = 15\ X\ 9.81 = 147.15\ N$

$\sum F_x = 0$

$0.25N_A = N_B$-------(1)

$\sum F_y = 0$

$N_A = W_l + W_m = 1030.05\ N$

*from equation* (1) $N_B = 257.512\ N$

$\sum M_A = 0$

$-W_m\ (s\cos\theta) - W_l(2\cos\theta) + N_B(\overline{BC}) = 0$

$s = 2.55\ mm$

**Figure G000149.** UE20CV101 UNIT 5 SESSION 6 · source slide page 4.

EV0000627

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS - STATICS

Friction- Numerical

- 6/36. The homogeneous square body is positioned as shown. If the coefficient of static friction at B is 0.40, determine the critical value of the angle θ below which slipping will occur. Neglect friction at A.

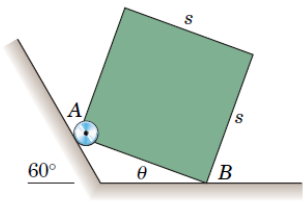



**Figure G000150.** UE20CV101 UNIT 5 SESSION 6 · source slide page 5.

EV0000630

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

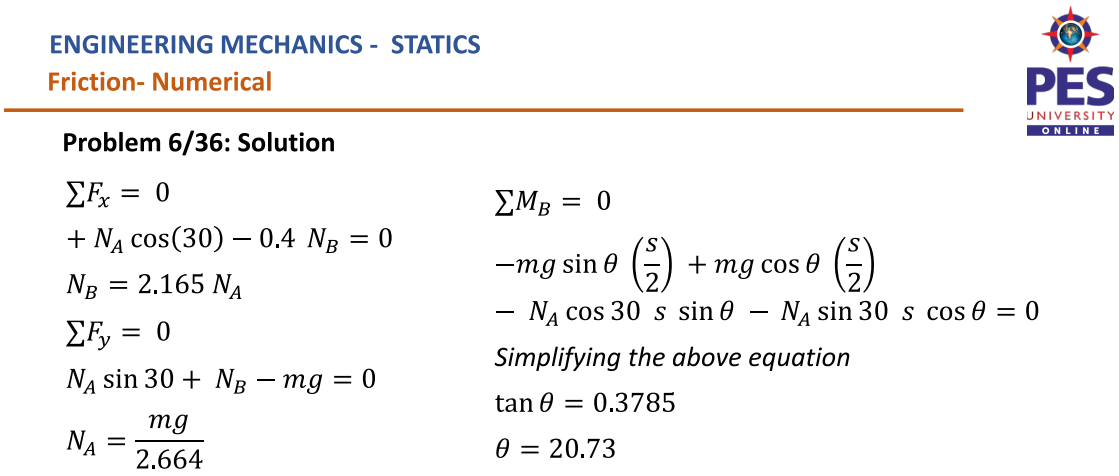



**Figure G000151.** UE20CV101 UNIT 5 SESSION 6 · source slide page 6.

EV0000633

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

**Figure G000152.** UE20CV101 UNIT 5 SESSION 6 · source slide page 7.

EV0000636

***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

### Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000617 · EV0000618 · EV0000619
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS FRICTION- Numerical Session- 6 Rashmi B A Department of Civil Engineering EV0000620 · EV0000621 · EV0000622
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • 6/33 Determine the distance s to which the 90-kg painter can climb without causing the 4-m ladder to slip at its lower end A. The top of the 15-kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. The mass center of the painter is directly above her feet. EV0000623 · EV0000624 · EV0000625
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical 1.5 $\theta = \cos -1 = 67.97$ NB 4 $CB = 3.708\ mm$ $\boldsymbol{ConsideringtheFBDofladder}$& $\boldsymbol{Applyingtheconditionsofequilibrium}$ $weightoftheladderWl = 15\ X9.81 = 147.15\ N$ $\sum Fx = 0$ $0.25NA = NB$-------(1) $\sum Fy = 0$ FA C $NA = Wl + Wm = 1030.05\ N$ $fromequation1\ NB = 257.512\ N$ NA $\sum MA = 0$ $-Wm(s\cos\theta) - Wl(\ 2\cos\theta) + NBBC = 0$ $s = 2.55\ mm$ EV0000626 · EV0000627 · EV0000628
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical • 6/36. The homogeneous square body is positioned as shown. If the coefficient of static friction at B is 0.40, determine the critical value of the angle ϑ below which slipping will occur. Neglect friction at A. EV0000629 · EV0000630 · EV0000631
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Friction- Numerical Problem 6/36: Solution $\sum Fx = 0$ $\sum MB = 0 + NA\cos 30 - 0.4\ NB = 0$ $s\ s - mg\sin\theta + mg\cos\theta$ $NB = 2.165\ NA$ 2 2 $-NA\cos 30\ s\sin\theta - NA\sin 30\ s\cos\theta = 0$ $\sum Fy = 0$ Simplifying the above equation $NA\sin 30 + NB - mg = 0$ $\tan\theta = 0.3785\ mg\ NA = \theta = 20.73$ 2.664 EV0000632 · EV0000633 · EV0000634
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000635 · EV0000636 · EV0000637

SHORT-NOTE RECAP

Source deck preserved: UE20CV101 UNIT 5 SESSION 6. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000617

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Official unit catalogue | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json | Official course/unit title and index. |
| CLASS 46 Introduction to Moment of Inertia transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 46 Introduction to Moment of Inertia transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 46 Introduction to Moment of Inertia transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 46 Introduction to Moment of Inertia transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 38 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 38 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 38 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 38 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 39 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 39 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 39 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 39 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 40 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 40 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 40 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 40 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 41 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 41 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 41 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 41 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.1 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.1 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.1 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 51.1 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 51.2 Moment of Inertia_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 51.2 Moment of Inertia_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 47 Friction_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 47 Friction_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 47 Friction_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| CLASS 47 Friction_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 48 Friction_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 48 Friction_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 48 Friction_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 48 Friction_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.1 Friction_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.1 Friction_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

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|---|---|---|
| CLASS 56.1 Friction_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.1 Friction_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.2 Friction_ Numerical transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.2 Friction_ Numerical transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.2 Friction_ Numerical transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps. |
| CLASS 56.2 Friction_ Numerical transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Slide page text: UE26ME142A Unit 4 Class 46 Lecture PPT | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |

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| Alternate extracted slide page text: UE26ME142A Unit 4 Class 46 Lecture PPT | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE26ME142A Unit 4 Class 46 Lecture PPT (converted PDF; original hash ca1b309f28b6be9ea205fed6ee7774af27f38d77c453fa0b5a65e3f2d23a897c) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: Moment of Inertia | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: Moment of Inertia | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for Moment of Inertia (converted PDF; original hash ef55e97507bbf9e623a96a1497d4f5b6794b13107ee60544f779debc752597d6) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 7 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 7 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 4 SESSION 7 (converted PDF; original hash 7665256b5e60d1a1be44d07d4530addac006f38fb4733be728f3ce43bd09dded) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 8 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 8 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE20CV101 UNIT 4 SESSION 8 (converted PDF; original hash 4dfa190774c94f173d3f82dd2e5df5780141f82e2c71c4843c044a9359c5deb1) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 9 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 9 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 4 SESSION 9 (converted PDF; original hash f14d351c141cd10666496c339542589875fdbada8cc78934218328853 25cba39) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 6 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 6 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE20CV101 UNIT 4 SESSION 6 (converted PDF; original hash d58ed7ece2c1005f874e755008057e82e8920f4ddc0764aa78dba4e65 d57fceb) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 10 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 10 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 4 SESSION 10 (converted PDF; original hash 316ff20663c7298b8309dd52cce2181452f92de8de8674062db0522946822a89) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 11 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 11 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE20CV101 UNIT 4 SESSION 11 (converted PDF; original hash a71de0e31f23e83e0787551bd98ca1bd39cb7b21d0448651b154a8e0ea2477a2) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 4 SESSION 12 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 12 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 4 SESSION 12 (converted PDF; original hash 9d36774190aac4b8a5689a0965af3aea3acd2a27165f4abaec9a21a209ce4fc0) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE26ME142A Unit 4 class 53 Lecture PPT | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE26ME142A Unit 4 class 53 Lecture PPT | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE26ME142A Unit 4 class 53 Lecture PPT (converted PDF; original hash 742906e1e1b2d3d2534379c883f30a16d6e881ca1044e876a4042effa759d096) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 5 SESSION 2 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 2 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 5 SESSION 2 (converted PDF; original hash 434a3671d2282b9341027f0a6bf8b159030c3daa324760d5cd72fbd4690b1dc6) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 5 SESSION 4 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 4 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE20CV101 UNIT 5 SESSION 4 (converted PDF; original hash defd460c19a5a7a0156848f7216b87686028436758cec4aba7f952dee1e326ff) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 5 SESSION 3 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 3 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

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| Page-render source for UE20CV101 UNIT 5 SESSION 3 (converted PDF; original hash 9e8880bf871d0e58a2b76c25bf1abdcb842bb6ee1c4e5ef947e5ed3f2c7b17fd) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 5 SESSION 5 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 5 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |
| Page-render source for UE20CV101 UNIT 5 SESSION 5 (converted PDF; original hash d128c486eada7966fd32e9a673d62895bae54293b5a6b58e496a12c2ff11e6d7) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |
| Slide page text: UE20CV101 UNIT 5 SESSION 6 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md | Complete extracted page text from the bucket's .slides.md sidecar. |
| Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 6 | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.txt | The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points. |

| Source | Path | Use |
|---|---|---|
| Page-render source for UE20CV101 UNIT 5 SESSION 6 (converted PDF; original hash 23913070dbdd60818ee3f8223fa075a73f2028509abc798885d8c546f6162a45) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf | Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

## Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

## Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00000001 | SRC00001/catalogue | covered | Unit title and index copied from the official units.json catalogue; human content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00000002 | SRC00002/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000003 | SRC00003/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000004 | SRC00004/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000005 | SRC00005/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000006 | SRC00002/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000007 | SRC00003/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000008 | SRC00004/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000009 | SRC00005/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000010 | SRC00002/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000011 | SRC00003/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000012 | SRC00004/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000013 | SRC00005/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000014 | SRC00002/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0065 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000015 | SRC00003/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0065 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000016 | SRC00004/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0065 to s0084. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000017 | SRC00005/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0065 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000018 | SRC00006/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000019 | SRC00007/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000020 | SRC00008/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000021 | SRC00009/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000022 | SRC00006/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000023 | SRC00007/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000024 | SRC00008/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000025 | SRC00009/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000026 | SRC00006/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000027 | SRC00007/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000028 | SRC00008/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000029 | SRC00009/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00000030 | SRC00010/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000031 | SRC00011/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000032 | SRC00012/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000033 | SRC00013/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000034 | SRC00010/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000035 | SRC00011/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000036 | SRC00012/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000037 | SRC00013/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000038 | SRC00010/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000039 | SRC00011/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000040 | SRC00012/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000041 | SRC00013/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000042 | SRC00014/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000043 | SRC00015/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00000044 | SRC00016/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000045 | SRC00017/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000046 | SRC00014/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000047 | SRC00015/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000048 | SRC00016/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000049 | SRC00017/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000050 | SRC00014/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000051 | SRC00015/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000052 | SRC00016/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000053 | SRC00017/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000054 | SRC00014/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0066 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000055 | SRC00015/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0066 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000056 | SRC00016/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0066 to s0067. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000057 | SRC00017/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0066 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000058 | SRC00018/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000059 | SRC00019/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000060 | SRC00020/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000061 | SRC00021/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000062 | SRC00018/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000063 | SRC00019/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000064 | SRC00020/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000065 | SRC00021/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000066 | SRC00018/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000067 | SRC00019/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000068 | SRC00020/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000069 | SRC00021/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000070 | SRC00018/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000071 | SRC00019/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000072 | SRC00020/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0064. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000073 | SRC00021/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000074 | SRC00022/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000075 | SRC00023/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000076 | SRC00024/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000077 | SRC00025/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000078 | SRC00022/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000079 | SRC00023/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000080 | SRC00024/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000081 | SRC00025/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000082 | SRC00022/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000083 | SRC00023/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000084 | SRC00024/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000085 | SRC00025/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00000086 | SRC00022/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000087 | SRC00023/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000088 | SRC00024/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000089 | SRC00025/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0064 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000090 | SRC00026/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000091 | SRC00027/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00000092 | SRC00028/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000093 | SRC00029/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000094 | SRC00026/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000095 | SRC00027/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000096 | SRC00028/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000097 | SRC00029/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000098 | SRC00026/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000099 | SRC00027/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000100 | SRC00028/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000101 | SRC00029/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000102 | SRC00026/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0058 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000103 | SRC00027/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0058 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000104 | SRC00028/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0058 to s0076. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000105 | SRC00029/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0058 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000106 | SRC00030/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000107 | SRC00031/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000108 | SRC00032/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000109 | SRC00033/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000110 | SRC00030/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000111 | SRC00031/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000112 | SRC00032/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000113 | SRC00033/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000114 | SRC00030/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000115 | SRC00031/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000116 | SRC00032/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000117 | SRC00033/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000118 | SRC00030/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000119 | SRC00031/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000120 | SRC00032/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0063. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000121 | SRC00033/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000122 | SRC00034/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000123 | SRC00035/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000124 | SRC00036/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000125 | SRC00037/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000126 | SRC00034/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000127 | SRC00035/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000128 | SRC00036/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000129 | SRC00037/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000130 | SRC00034/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000131 | SRC00035/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000132 | SRC00036/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000133 | SRC00037/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000134 | SRC00034/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00001 35 | SRC00035/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 36 | SRC00036/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0072. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 37 | SRC00037/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0062 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 38 | SRC00038/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 39 | SRC00039/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 40 | SRC00040/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV00001 41 | SRC00041/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000142 | SRC00038/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000143 | SRC00039/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000144 | SRC00040/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000145 | SRC00041/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000146 | SRC00038/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000147 | SRC00039/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000148 | SRC00040/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000149 | SRC00041/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000150 | SRC00038/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000151 | SRC00039/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000152 | SRC00040/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0081. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000153 | SRC00041/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000154 | SRC00038/t001200-001499 | covered | Timestamp interval t001200-001499; canonical segment IDs s0082 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000155 | SRC00039/t001200-001499 | covered | Timestamp interval t001200-001499; canonical segment IDs s0082 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000156 | SRC00040/t001200-001499 | covered | Timestamp interval t001200-001499; canonical segment IDs s0082 to s0082. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000157 | SRC00041/t001200-001499 | covered | Timestamp interval t001200-001499; canonical segment IDs s0082 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000158 | SRC00042/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000159 | SRC00043/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000160 | SRC00044/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000161 | SRC00045/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000162 | SRC00042/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000163 | SRC00043/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000164 | SRC00044/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000165 | SRC00045/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000166 | SRC00042/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000167 | SRC00043/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000168 | SRC00044/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000169 | SRC00045/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000170 | SRC00042/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000171 | SRC00043/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000172 | SRC00044/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0078. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000173 | SRC00045/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0060 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000174 | SRC00046/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000175 | SRC00047/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000176 | SRC00048/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000177 | SRC00049/t000000-000299 | covered | Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000178 | SRC00046/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000179 | SRC00047/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000180 | SRC00048/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000181 | SRC00049/t000300-000599 | covered | Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000182 | SRC00046/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000183 | SRC00047/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000184 | SRC00048/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000185 | SRC00049/t000600-000899 | covered | Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000186 | SRC00046/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0063 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000187 | SRC00047/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0063 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000188 | SRC00048/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0063 to s0073. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000189 | SRC00049/t000900-001199 | covered | Timestamp interval t000900-001199; canonical segment IDs s0063 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check. |
| EV0000190 | SRC00050/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000191 | SRC00052/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000192 | SRC00051/p1 | covered | Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending. |
| EV0000193 | SRC00050/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000194 | SRC00052/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000195 | SRC00051/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000196 | SRC00050/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00001 97 | SRC00052/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00001 98 | SRC00051/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00001 99 | SRC00050/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 00 | SRC00052/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 01 | SRC00051/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 02 | SRC00050/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000203 | SRC00052/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000204 | SRC00051/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000205 | SRC00050/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000206 | SRC00052/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000207 | SRC00051/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000208 | SRC00052/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000209 | SRC00052/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000210 | SRC00053/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000211 | SRC00055/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000212 | SRC00054/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000213 | SRC00053/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000214 | SRC00055/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000215 | SRC00054/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000216 | SRC00053/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000217 | SRC00055/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000218 | SRC00054/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000219 | SRC00053/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000220 | SRC00055/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000221 | SRC00054/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000222 | SRC00053/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000223 | SRC00055/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000224 | SRC00054/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000225 | SRC00053/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000226 | SRC00055/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000227 | SRC00054/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000228 | SRC00053/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000229 | SRC00055/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000230 | SRC00054/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000231 | SRC00053/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000232 | SRC00055/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000233 | SRC00054/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000234 | SRC00053/p9 | covered | Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000235 | SRC00055/p9 | covered | Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000236 | SRC00054/p9 | covered | Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000237 | SRC00053/p10 | covered | Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000238 | SRC00055/p10 | covered | Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000239 | SRC00054/p10 | covered | Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000240 | SRC00053/p11 | covered | Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000241 | SRC00055/p11 | covered | Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000242 | SRC00054/p11 | covered | Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000243 | SRC00053/p12 | covered | Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000244 | SRC00055/p12 | covered | Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000245 | SRC00054/p12 | covered | Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000246 | SRC00053/p13 | covered | Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000247 | SRC00055/p13 | covered | Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000248 | SRC00054/p13 | covered | Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000249 | SRC00053/p14 | covered | Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000250 | SRC00055/p14 | covered | Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000251 | SRC00054/p14 | covered | Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000252 | SRC00053/p15 | covered | Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000253 | SRC00055/p15 | covered | Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000254 | SRC00054/p15 | covered | Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000255 | SRC00053/p16 | covered | Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000256 | SRC00055/p16 | covered | Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000257 | SRC00054/p16 | covered | Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00002258 | SRC00053/p17 | covered | Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002259 | SRC00055/p17 | covered | Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002260 | SRC00054/p17 | covered | Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002261 | SRC00053/p18 | covered | Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002262 | SRC00055/p18 | covered | Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002263 | SRC00054/p18 | covered | Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002264 | SRC00053/p19 | covered | Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00002 65 | SRC00055/p19 | covered | Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 66 | SRC00054/p19 | covered | Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 67 | SRC00053/p20 | covered | Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 68 | SRC00055/p20 | covered | Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 69 | SRC00054/p20 | covered | Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 70 | SRC00053/p21 | covered | Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 71 | SRC00055/p21 | covered | Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000272 | SRC00054/p21 | covered | Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000273 | SRC00053/p22 | covered | Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000274 | SRC00055/p22 | covered | Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000275 | SRC00054/p22 | covered | Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000276 | SRC00053/p23 | covered | Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000277 | SRC00055/p23 | covered | Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000278 | SRC00054/p23 | covered | Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00002 79 | SRC00053/p24 | covered | Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 80 | SRC00055/p24 | covered | Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 81 | SRC00054/p24 | covered | Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 82 | SRC00053/p25 | covered | Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 83 | SRC00055/p25 | covered | Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 84 | SRC00054/p25 | covered | Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 85 | SRC00053/p26 | covered | Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00002 86 | SRC00055/p26 | covered | Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 87 | SRC00054/p26 | covered | Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 88 | SRC00053/p27 | covered | Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 89 | SRC00055/p27 | covered | Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00002 90 | SRC00054/p27 | covered | Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00002 91 | SRC00053/p28 | covered | Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00002 92 | SRC00055/p28 | covered | Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000293 | SRC00054/p28 | covered | Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000294 | SRC00053/p29 | covered | Extracted slide page 29 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000295 | SRC00055/p29 | covered | Original slide page 29 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000296 | SRC00054/p29 | covered | Alternate extracted slide text page 29; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000297 | SRC00053/p30 | covered | Extracted slide page 30 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000298 | SRC00055/p30 | covered | Original slide page 30 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000299 | SRC00054/p30 | covered | Alternate extracted slide text page 30; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000300 | SRC00053/p31 | covered | Extracted slide page 31 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000301 | SRC00055/p31 | covered | Original slide page 31 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000302 | SRC00054/p31 | covered | Alternate extracted slide text page 31; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000303 | SRC00053/p32 | covered | Extracted slide page 32 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000304 | SRC00055/p32 | covered | Original slide page 32 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000305 | SRC00054/p32 | covered | Alternate extracted slide text page 32; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000306 | SRC00053/p33 | covered | Extracted slide page 33 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000307 | SRC00055/p33 | covered | Original slide page 33 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000308 | SRC00054/p33 | covered | Alternate extracted slide text page 33; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000309 | SRC00053/p34 | covered | Extracted slide page 34 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000310 | SRC00055/p34 | covered | Original slide page 34 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000311 | SRC00054/p34 | covered | Alternate extracted slide text page 34; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000312 | SRC00053/p35 | covered | Extracted slide page 35 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000313 | SRC00055/p35 | covered | Original slide page 35 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000314 | SRC00054/p35 | covered | Alternate extracted slide text page 35; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000315 | SRC00053/p36 | covered | Extracted slide page 36 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000316 | SRC00055/p36 | covered | Original slide page 36 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000317 | SRC00054/p36 | covered | Alternate extracted slide text page 36; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000318 | SRC00053/p37 | covered | Extracted slide page 37 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000319 | SRC00055/p37 | covered | Original slide page 37 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000320 | SRC00054/p37 | covered | Alternate extracted slide text page 37; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000321 | SRC00053/p38 | covered | Extracted slide page 38 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000322 | SRC00055/p38 | covered | Original slide page 38 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000323 | SRC00054/p38 | covered | Alternate extracted slide text page 38; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000324 | SRC00053/p39 | covered | Extracted slide page 39 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000325 | SRC00055/p39 | covered | Original slide page 39 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000326 | SRC00054/p39 | covered | Alternate extracted slide text page 39; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000327 | SRC00053/p40 | covered | Extracted slide page 40 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000328 | SRC00055/p40 | covered | Original slide page 40 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000329 | SRC00054/p40 | covered | Alternate extracted slide text page 40; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000330 | SRC00053/p41 | covered | Extracted slide page 41 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000331 | SRC00055/p41 | covered | Original slide page 41 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000332 | SRC00054/p41 | covered | Alternate extracted slide text page 41; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000333 | SRC00056/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000334 | SRC00058/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000335 | SRC00057/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000336 | SRC00056/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000337 | SRC00058/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000338 | SRC00057/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000339 | SRC00056/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000340 | SRC00058/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000341 | SRC00057/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000342 | SRC00056/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000343 | SRC00058/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000344 | SRC00057/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000345 | SRC00056/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000346 | SRC00058/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000347 | SRC00057/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003 48 | SRC00056/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 49 | SRC00058/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 50 | SRC00057/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003 51 | SRC00056/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 52 | SRC00058/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 53 | SRC00057/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003 54 | SRC00059/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 55 | SRC00061/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 56 | SRC00060/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003 57 | SRC00059/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 58 | SRC00061/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 59 | SRC00060/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003360 | SRC00059/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003361 | SRC00061/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003362 | SRC00060/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003363 | SRC00059/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003364 | SRC00061/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003365 | SRC00060/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003366 | SRC00059/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003367 | SRC00061/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003368 | SRC00060/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003369 | SRC00059/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003370 | SRC00061/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003371 | SRC00060/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003772 | SRC00059/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003773 | SRC00061/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003774 | SRC00060/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003775 | SRC00062/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003776 | SRC00064/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003777 | SRC00063/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003 78 | SRC00062/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 79 | SRC00064/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 80 | SRC00063/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003 81 | SRC00062/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 82 | SRC00064/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 83 | SRC00063/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003 84 | SRC00062/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 85 | SRC00064/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 86 | SRC00063/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003 87 | SRC00062/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 88 | SRC00064/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 89 | SRC00063/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003390 | SRC00062/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003391 | SRC00064/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003392 | SRC00063/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003393 | SRC00062/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003394 | SRC00064/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003395 | SRC00063/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00003 96 | SRC00062/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00003 97 | SRC00064/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00003 98 | SRC00063/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00003 99 | SRC00062/p9 | covered | Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 00 | SRC00064/p9 | covered | Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 01 | SRC00063/p9 | covered | Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000402 | SRC00065/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000403 | SRC00067/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000404 | SRC00066/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000405 | SRC00065/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000406 | SRC00067/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000407 | SRC00066/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000408 | SRC00065/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000409 | SRC00067/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000410 | SRC00066/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000411 | SRC00065/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000412 | SRC00067/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000413 | SRC00066/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000414 | SRC00065/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000415 | SRC00067/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000416 | SRC00066/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000417 | SRC00065/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000418 | SRC00067/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000419 | SRC00066/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 20 | SRC00068/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 21 | SRC00070/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 22 | SRC00069/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 23 | SRC00068/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 24 | SRC00070/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 25 | SRC00069/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 26 | SRC00068/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 27 | SRC00070/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 28 | SRC00069/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 29 | SRC00068/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 30 | SRC00070/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 31 | SRC00069/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000432 | SRC00068/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000433 | SRC00070/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000434 | SRC00069/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000435 | SRC00068/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000436 | SRC00070/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000437 | SRC00069/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000438 | SRC00068/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000439 | SRC00070/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000440 | SRC00069/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000441 | SRC00068/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000442 | SRC00070/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000443 | SRC00069/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000444 | SRC00071/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000445 | SRC00073/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000446 | SRC00072/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000447 | SRC00071/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000448 | SRC00073/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000449 | SRC00072/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 50 | SRC00071/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 51 | SRC00073/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 52 | SRC00072/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 53 | SRC00071/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 54 | SRC00073/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 55 | SRC00072/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000456 | SRC00071/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000457 | SRC00073/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000458 | SRC00072/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000459 | SRC00071/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000460 | SRC00073/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000461 | SRC00072/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000462 | SRC00074/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000463 | SRC00076/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000464 | SRC00075/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000465 | SRC00074/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000466 | SRC00076/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000467 | SRC00075/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 68 | SRC00074/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 69 | SRC00076/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 70 | SRC00075/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 71 | SRC00074/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 72 | SRC00076/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 73 | SRC00075/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000474 | SRC00074/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000475 | SRC00076/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000476 | SRC00075/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000477 | SRC00077/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000478 | SRC00079/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000479 | SRC00078/p1 | covered | Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 80 | SRC00077/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 81 | SRC00079/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 82 | SRC00078/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 83 | SRC00077/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 84 | SRC00079/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 85 | SRC00078/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 86 | SRC00077/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 87 | SRC00079/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 88 | SRC00078/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00004 89 | SRC00077/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 90 | SRC00079/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00004 91 | SRC00078/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000492 | SRC00077/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000493 | SRC00079/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000494 | SRC00078/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000495 | SRC00077/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000496 | SRC00079/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000497 | SRC00078/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00004 98 | SRC00077/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00004 99 | SRC00079/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 00 | SRC00078/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 01 | SRC00077/p9 | covered | Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 02 | SRC00079/p9 | covered | Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 03 | SRC00078/p9 | covered | Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 04 | SRC00077/p10 | covered | Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 05 | SRC00079/p10 | covered | Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 06 | SRC00078/p10 | covered | Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 07 | SRC00077/p11 | covered | Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 08 | SRC00079/p11 | covered | Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 09 | SRC00078/p11 | covered | Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000510 | SRC00079/p12 | covered | Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000511 | SRC00079/p13 | covered | Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt; visual page included as vector figure. Human layout/content review is pending. |
| EV0000512 | SRC00080/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000513 | SRC00082/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000514 | SRC00081/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000515 | SRC00080/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000516 | SRC00082/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000517 | SRC00081/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000518 | SRC00080/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000519 | SRC00082/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000520 | SRC00081/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000521 | SRC00080/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000522 | SRC00082/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000523 | SRC00081/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000524 | SRC00080/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000525 | SRC00082/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000526 | SRC00081/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000527 | SRC00080/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 28 | SRC00082/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 29 | SRC00081/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 30 | SRC00080/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 31 | SRC00082/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 32 | SRC00081/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 33 | SRC00080/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000534 | SRC00082/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000535 | SRC00081/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000536 | SRC00080/p9 | covered | Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000537 | SRC00082/p9 | covered | Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000538 | SRC00081/p9 | covered | Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000539 | SRC00083/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000540 | SRC00085/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000541 | SRC00084/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000542 | SRC00083/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000543 | SRC00085/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000544 | SRC00084/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000545 | SRC00083/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 46 | SRC00085/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 47 | SRC00084/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 48 | SRC00083/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 49 | SRC00085/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 50 | SRC00084/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 51 | SRC00083/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000552 | SRC00085/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000553 | SRC00084/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000554 | SRC00083/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000555 | SRC00085/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000556 | SRC00084/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000557 | SRC00083/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 58 | SRC00085/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 59 | SRC00084/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 60 | SRC00083/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 61 | SRC00085/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 62 | SRC00084/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 63 | SRC00086/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000564 | SRC00088/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000565 | SRC00087/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000566 | SRC00086/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000567 | SRC00088/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000568 | SRC00087/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000569 | SRC00086/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000570 | SRC00088/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000571 | SRC00087/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000572 | SRC00086/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000573 | SRC00088/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000574 | SRC00087/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000575 | SRC00086/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 76 | SRC00088/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 77 | SRC00087/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 78 | SRC00086/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 79 | SRC00088/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 80 | SRC00087/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 81 | SRC00086/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 82 | SRC00088/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 83 | SRC00087/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 84 | SRC00086/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 85 | SRC00088/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 86 | SRC00087/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 87 | SRC00086/p9 | covered | Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00005 88 | SRC00088/p9 | covered | Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 89 | SRC00087/p9 | covered | Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 90 | SRC00086/p10 | covered | Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00005 91 | SRC00088/p10 | covered | Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00005 92 | SRC00087/p10 | covered | Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00005 93 | SRC00089/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000594 | SRC00091/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000595 | SRC00090/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000596 | SRC00089/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000597 | SRC00091/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000598 | SRC00090/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000599 | SRC00089/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000600 | SRC00091/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000601 | SRC00090/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000602 | SRC00089/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000603 | SRC00091/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000604 | SRC00090/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000605 | SRC00089/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000606 | SRC00091/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000607 | SRC00090/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000608 | SRC00089/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000609 | SRC00091/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000610 | SRC00090/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000611 | SRC00089/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000612 | SRC00091/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000613 | SRC00090/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000614 | SRC00089/p8 | covered | Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000615 | SRC00091/p8 | covered | Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000616 | SRC00090/p8 | covered | Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000617 | SRC00092/p1 | covered | Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000618 | SRC00094/p1 | covered | Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000619 | SRC00093/p1 | covered | Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000620 | SRC00092/p2 | covered | Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000621 | SRC00094/p2 | covered | Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000622 | SRC00093/p2 | covered | Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000623 | SRC00092/p3 | covered | Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00006 24 | SRC00094/p3 | covered | Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00006 25 | SRC00093/p3 | covered | Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00006 26 | SRC00092/p4 | covered | Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV00006 27 | SRC00094/p4 | covered | Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00006 28 | SRC00093/p4 | covered | Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending. |
| EV00006 29 | SRC00092/p5 | covered | Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV0000630 | SRC00094/p5 | covered | Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000631 | SRC00093/p5 | covered | Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000632 | SRC00092/p6 | covered | Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |
| EV0000633 | SRC00094/p6 | covered | Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV0000634 | SRC00093/p6 | covered | Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending. |
| EV0000635 | SRC00092/p7 | covered | Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md; original page is also embedded when the source render is available. Human subject review is pending. |

| Evidence | Source unit | Status | Decision |
|---|---|---|---|
| EV00006 36 | SRC00094/p7 | covered | Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf; visual page included as vector figure. Human layout/content review is pending. |
| EV00006 37 | SRC00093/p7 | covered | Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending. |

# C Revision and retrieval practice